Age - Sex - PA size and Mortality

This workbook walks through STATA code to replicate the CHEST article and attempt to add our analysis

Overall objectives:

1. Show that the distribution of PA diameter, Ascending Aorta Diameter, and PA:AA changes with age within this sample of patients receiving CTPAs in the emergency department. Methods: distribution visualization, quantile regression (with and without covariates)

2. Evaluate when the risk of mortality increases in our population, to see if it corresponds to when the risk increases in the population of

Note: the authors don't age or sex-adjust PA mean.

In [3]:
```
%pip install stata_setup
import stata_setup
stata_setup.config(r"/Applications/Stata/", "be")
```

```
Requirement already satisfied: stata_setup in /Users/blocke/mambaforge/envs/
mimiciv-tabular/lib/python3.11/site-packages (0.1.3)
Requirement already satisfied: numpy in /Users/blocke/mambaforge/envs/mimici
v-tabular/lib/python3.11/site-packages (from stata_setup) (1.25.2)
Requirement already satisfied: pandas in /Users/blocke/mambaforge/envs/mimic
iv-tabular/lib/python3.11/site-packages (from stata_setup) (2.0.3)
Requirement already satisfied: ipython in /Users/blocke/mambaforge/envs/mimi
civ-tabular/lib/python3.11/site-packages (from stata_setup) (9.6.0)
Requirement already satisfied: decorator in /Users/blocke/mambaforge/envs/mi
miciv-tabular/lib/python3.11/site-packages (from ipython->stata_setup) (5.2.
1)
Requirement already satisfied: ipython-pygments-lexers in /Users/blocke/mamb
aforge/envs/mimiciv-tabular/lib/python3.11/site-packages (from ipython->stat
a_setup) (1.1.1)
Requirement already satisfied: jedi>=0.16 in /Users/blocke/mambaforge/envs/m
imiciv-tabular/lib/python3.11/site-packages (from ipython->stata_setup) (0.1
9.2)
Requirement already satisfied: matplotlib-inline in /Users/blocke/mambaforg
e/envs/mimiciv-tabular/lib/python3.11/site-packages (from ipython->stata_set
up) (0.1.7)
Requirement already satisfied: pexpect>4.3 in /Users/blocke/mambaforge/envs/
mimiciv-tabular/lib/python3.11/site-packages (from ipython->stata_setup) (4.
9.0)
Requirement already satisfied: prompt_toolkit<3.1.0,>=3.0.41 in /Users/block
e/mambaforge/envs/mimiciv-tabular/lib/python3.11/site-packages (from ipython
->stata_setup) (3.0.52)
Requirement already satisfied: pygments>=2.4.0 in /Users/blocke/mambaforge/e
nvs/mimiciv-tabular/lib/python3.11/site-packages (from ipython->stata_setup)
(2.19.2)
Requirement already satisfied: stack_data in /Users/blocke/mambaforge/envs/m
imiciv-tabular/lib/python3.11/site-packages (from ipython->stata_setup) (0.
6.3)
Requirement already satisfied: traitlets>=5.13.0 in /Users/blocke/mambaforg
e/envs/mimiciv-tabular/lib/python3.11/site-packages (from ipython->stata_set
up) (5.14.3)
Requirement already satisfied: typing_extensions>=4.6 in /Users/blocke/mamba
forge/envs/mimiciv-tabular/lib/python3.11/site-packages (from ipython->stata
_setup) (4.15.0)
Requirement already satisfied: wcwidth in /Users/blocke/mambaforge/envs/mimi
civ-tabular/lib/python3.11/site-packages (from prompt_toolkit<3.1.0,>=3.0.41
->ipython->stata_setup) (0.2.14)
Requirement already satisfied: parso<0.9.0,>=0.8.4 in /Users/blocke/mambafor
ge/envs/mimiciv-tabular/lib/python3.11/site-packages (from jedi>=0.16->ipyth
on->stata_setup) (0.8.5)
Requirement already satisfied: ptyprocess>=0.5 in /Users/blocke/mambaforge/e
nvs/mimiciv-tabular/lib/python3.11/site-packages (from pexpect>4.3->ipython-
>stata_setup) (0.7.0)
Requirement already satisfied: python-dateutil>=2.8.2 in /Users/blocke/mamba
forge/envs/mimiciv-tabular/lib/python3.11/site-packages (from pandas->stata_
setup) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in /Users/blocke/mambaforge/env
s/mimiciv-tabular/lib/python3.11/site-packages (from pandas->stata_setup) (2
025.2)
Requirement already satisfied: tzdata>=2022.1 in /Users/blocke/mambaforge/en
vs/mimiciv-tabular/lib/python3.11/site-packages (from pandas->stata_setup)
(2025.2)
```

```
Requirement already satisfied: six>=1.5 in /Users/blocke/mambaforge/envs/mim
iciv-tabular/lib/python3.11/site-packages (from python-dateutil>=2.8.2->pand
as->stata_setup) (1.17.0)
Requirement already satisfied: executing>=1.2.0 in /Users/blocke/mambaforge/
envs/mimiciv-tabular/lib/python3.11/site-packages (from stack_data->ipython-
>stata_setup) (2.2.1)
Requirement already satisfied: asttokens>=2.1.0 in /Users/blocke/mambaforge/
envs/mimiciv-tabular/lib/python3.11/site-packages (from stack_data->ipython-
>stata_setup) (3.0.0)
Requirement already satisfied: pure_eval in /Users/blocke/mambaforge/envs/mi
miciv-tabular/lib/python3.11/site-packages (from stack_data->ipython->stata_
setup) (0.2.3)
Note: you may need to restart the kernel to use updated packages.
```

Data Cleaning Code

In [4]:
```
%%stata
cd "/Users/blocke/Box Sync/Residency Personal Files/Scholarly Work/Locke Res

capture mkdir "Results and Figures"
capture mkdir "Results and Figures/$S_DATE/" //make new folder for figure ou
capture mkdir "Results and Figures/$S_DATE/Logs/" //new folder for stata log
local a1=substr(c(current_time),1,2)
local a2=substr(c(current_time),4,2)
local a3=substr(c(current_time),7,2)
local b = "BL CombofilewithoutEMPI.do" // do file name
copy "`b'" "Results and Figures/$S_DATE/Logs/(`a1'_`a2'_`a3')`b'"

use final_noempi, clear

set scheme cleanplots //cleanplots white_tableau white_w3d //3 options I usu

//missings dropobs, force //drop all rows with no observations. None
//missings dropvars, force //drop all columns with no observations
//missings report
//mdesc
//codebook
count // n=990
count if lastfollowupday==0 // n=78 without any follow-up data

/* -----
Data Cleaning
--------*/

label variable age "Age (years)"

label define male_lab 0 "Female" 1 "Male"
label variable male "Male Sex?"
label values male male_lab

bysort obesity: sum bmi, detail
label variable obesity "Obesity (ICD only)"

generate obesity_calc = .
replace obesity_calc = 0 if !missing(obesity) & obesity == 0
replace obesity_calc = 1 if !missing(obesity) & obesity == 1
```

```
replace obesity_calc = 1 if !missing(bmi) & bmi >= 30
replace obesity_calc = 0 if !missing(bmi) & bmi < 30
label define obesity_calc_lab 0 "Not Obese" 1 "Obese"
label variable obesity_calc "Obesity"
label values obesity_calc obesity_calc_lab
tab obesity obesity_calc, missing
tab obesity_calc, missing
tab obesity, missing

//traditional cutpoints
label variable enlargedratio "PA:AA Ratio"
label define enlargedratio_lab 0 "Normal PA:AA (<0.9)" 1 "Increased PA:AA (0
label values enlargedratio enlargedratio_lab

label variable enlargedpa "PA diameter"
label define enlargedpa_lab 0 "Normal PAd" 1 "Enlarged PAd"
label values enlargedpa enlargedpa_lab

gen pa_confusion_matrix = .
replace pa_confusion_matrix = 0 if (enlargedpa == 0) & (enlargedratio == 0)
replace pa_confusion_matrix = 1 if (enlargedpa == 1) & (enlargedratio == 0)
replace pa_confusion_matrix = 2 if (enlargedpa == 0) & (enlargedratio == 1)
replace pa_confusion_matrix = 3 if (enlargedpa == 1) & (enlargedratio == 1)
label variable pa_confusion_matrix "PAd high, PA:AA high, neither, or both?"
label define pa_confusion_lab 0 "Neither" 1 "Only Enlarged PAd" 2 "Only High
label values pa_confusion_matrix pa_confusion_lab
tab enlargedpa pa_confusion_matrix, col //sanity checks
tab enlargedratio pa_confusion_matrix


//label comorbdities
label variable pulmdisease "Pulmonary Disease"
label variable chf "Congestive Heart Failure"
label variable diabetes3 "Diabetes"
label define diabetes_val 0 "No DM" 1 "Uncomplicated DM" 2 "DM with Complica
label values diabetes3 diabetes_val
label variable hypertension "Hypertension"
label variable pulmonarycircdisorder "Pulm Circ. Disorder"
label variable peripheralvascdisorders "Periph Vasc. Disease"
label variable renalfailure "Kidney Disease"

// [ ] remove?
gen known_assoc_comorb = 0
replace known_assoc_comorb = 1 if (pulmdisease == 1 | pulmonarycircdisorders
tab chf known_assoc_comorb, missing //sanity checks
tab pulmdisease known_assoc_comorb, missing
tab pulmonarycircdisorders known_assoc_comorb, missing
label variable known_assoc_comorb "Known Comorbidity w PA->Mortality Associa
label define known_assoc_comorb_lab 0 "No Pulm Dz/CHF/Pulm Vasc Dz" 1 "Any c
label values known_assoc_comorb known_assoc_comorb_lab

// Split diabetes into complicated vs not.
tab diabetes3
capture drop diabetes1
gen diabetes1=cond(diabetes3==1,1,0)
tab diabetes3 diabetes1, missing
```

```
capture drop diabetes2
gen diabetes2=cond(diabetes3==2,1,0)
tab diabetes3 diabetes2, missing
label variable diabetes1 "Uncomplicated Diabetes"
label variable diabetes2 "Diabetes with Complication(s)"

// Outcomes
label variable death "Death"
label define death_label 0 "Alive or Censored" 1 "Died"
label values death death_label
label variable lastfollowupyear "Follow-up or death(years)"

label variable anyemergency "Any Emergency Visits in Follow-up?"
label variable anyadmission "Any Hospital Admission in Follow-up?"

gen time_of_death = lastfollowupyear if death == 1
label variable time_of_death "Time of Death"
gen time_of_censoring = lastfollowupyear if death != 1
label variable time_of_censoring "Time of longest follow-up alive"


//Categorizations of continuous variables; for graphs

recode age min/30=0 30/40=1 40/50=2 50/60=3 60/70=4 70/max=5, gen(age_decade
label define age_dec_lab 0 "<30 years" 1 "30-40 years" 2 "40-50 years" 3 "50
label variable age_decade "Age (by decade)"
label values age_decade age_dec_lab

gen sex_norm_mpad = .
replace sex_norm_mpad = mpad + 1 if male == 0
replace sex_norm_mpad = mpad - 1 if male == 1
sum sex_norm_mpad, detail
recode sex_norm_mpad min/25=0 25/28=1 28/31=2 31/max=3, gen(mpad_cat)
label define mpad_cat_lab 0 "F:<24 mm, M:<26 mm" 1 "F:24-27 mm, M:26-29 mm"
//TODO: Cutpoints for mPA were defined with ≤27 mm(F) and ≤29 mm(M) as the r
label variable mpad_cat "PA_d strata"
label values mpad_cat mpad_cat_lab
bysort mpad_cat: sum sex_norm_mpad, detail
bysort mpad_cat: sum mpad if male == 0, detail  //sanity check
bysort mpad_cat: sum mpad if male == 1, detail  //sanity check

gen sex_norm_aa = .
replace sex_norm_aa = ascendingaorta + 1.5 if male == 0 //men 3mm larger; mo
replace sex_norm_aa = ascendingaorta - 1.5 if male == 1
sum sex_norm_aa, detail
recode sex_norm_aa min/30=0 30/33.5=1 33.5/37=2 37/max=3, gen(aa_cat)
label define aa_cat_lab 0 "F:<28.5 mm, M:<31.5 mm" 1 "F:28.5-32 mm, M:31.5-3
label variable aa_cat "AA_d strata"
label values aa_cat aa_cat_lab
bysort aa_cat: sum sex_norm_aa, detail
bysort aa_cat: sum ascendingaorta if male == 0, detail
bysort aa_cat: sum ascendingaorta if male == 1, detail

save cleaned_noempi, replace
```

```
. cd "/Users/blocke/Box Sync/Residency Personal Files/Scholarly Work/Locke R
ese
> arch Projects/Pulm Artery Stuff w Scarps/Local Analysis"
/Users/blocke/Box Sync/Residency Personal Files/Scholarly Work/Locke Researc
h P
> rojects/Pulm Artery Stuff w Scarps/Local Analysis

.
. capture mkdir "Results and Figures"

. capture mkdir "Results and Figures/$S_DATE/" //make new folder for figure
out
> put if needed

. capture mkdir "Results and Figures/$S_DATE/Logs/" //new folder for stata l
ogs

. local a1=substr(c(current_time),1,2)

. local a2=substr(c(current_time),4,2)

. local a3=substr(c(current_time),7,2)

. local b = "BL CombofilewithoutEMPI.do" // do file name

. copy "`b'" "Results and Figures/$S_DATE/Logs/(`a1'_`a2'_`a3')`b'"

.
. use final_noempi, clear

.
. set scheme cleanplots //cleanplots white_tableau white_w3d //3 options I u
sua
> lly use

.
. //missings dropobs, force //drop all rows with no observations. None
. //missings dropvars, force //drop all columns with no observations
. //missings report
. //mdesc
. //codebook
. count // n=990
  990

. count if lastfollowupday==0 // n=78 without any follow-up data
  78

.
. /* -----
> Data Cleaning
> --------*/
.
. label variable age "Age (years)"

.
. label define male_lab 0 "Female" 1 "Male"
```

```
. label variable male "Male Sex?"

. label values male male_lab

.
. bysort obesity: sum bmi, detail

---------------------------------------------------------------------------------
---
-> obesity = 0

                             bmi
-------------------------------------------------------------
      Percentiles      Smallest
 1%        16.01           13.6
 5%         19.2          15.24
10%         21.1          15.39       Obs                 391
25%        23.63          16.01       Sum of wgt.         391

50%        26.88                      Mean           27.81962
                        Largest       Std. dev.      6.329926
75%        31.65          48.52
90%        35.78          50.01       Variance       40.06796
95%        38.43          51.05       Skewness       1.239868
99%        48.52          69.18       Kurtosis       7.622813

---------------------------------------------------------------------------------
---
-> obesity = 1

                             bmi
-------------------------------------------------------------
      Percentiles      Smallest
 1%         19.6          19.55
 5%        24.09           19.6
10%         28.7          21.53       Obs                 166
25%        32.73          21.99       Sum of wgt.         166

50%       37.745                      Mean           39.03705
                        Largest       Std. dev.      9.527408
75%        44.28          66.22
90%        50.28          68.09       Variance       90.77149
95%        54.88          71.92       Skewness       .8388844
99%        71.92             72       Kurtosis       4.528189

---------------------------------------------------------------------------------
---
-> obesity = .

                             bmi
-------------------------------------------------------------
      Percentiles      Smallest
 1%        17.16          17.16
 5%        17.16          22.61
10%        17.16          28.65       Obs                   7
```

```
25%        22.61          29.62       Sum of wgt.           7

50%        29.62                      Mean           27.51286
                        Largest       Std. dev.      5.588985
75%        30.99          29.62
90%        32.77          30.79       Variance       31.23676
95%        32.77          30.99       Skewness      -1.025105
99%        32.77          32.77       Kurtosis       2.613054


. label variable obesity "Obesity (ICD only)"

.
. generate obesity_calc = .
(990 missing values generated)

. replace obesity_calc = 0 if !missing(obesity) & obesity == 0
(719 real changes made)

. replace obesity_calc = 1 if !missing(obesity) & obesity == 1
(228 real changes made)

. replace obesity_calc = 1 if !missing(bmi) & bmi >= 30
(125 real changes made)

. replace obesity_calc = 0 if !missing(bmi) & bmi < 30
(27 real changes made)

. label define obesity_calc_lab 0 "Not Obese" 1 "Obese"

. label variable obesity_calc "Obesity"

. label values obesity_calc obesity_calc_lab

. tab obesity obesity_calc, missing

   Obesity |             Obesity
(ICD only) | Not Obese      Obese          . |     Total
-----------+---------------------------------+----------
         0 |       597        122          0 |       719
         1 |        23        205          0 |       228
         . |         4          3         36 |        43
-----------+---------------------------------+----------
     Total |       624        330         36 |       990

. tab obesity_calc, missing

    Obesity |      Freq.     Percent        Cum.
------------+-----------------------------------
  Not Obese |        624       63.03       63.03
      Obese |        330       33.33       96.36
          . |         36        3.64      100.00
------------+-----------------------------------
      Total |        990      100.00

. tab obesity, missing
```

```
     Obesity |
  (ICD only) |      Freq.     Percent        Cum.
-------------+-----------------------------------
           0 |        719       72.63       72.63
           1 |        228       23.03       95.66
           . |         43        4.34      100.00
-------------+-----------------------------------
       Total |        990      100.00

. 
. //traditional cutpoints
. label variable enlargedratio "PA:AA Ratio"

. label define enlargedratio_lab 0 "Normal PA:AA (<0.9)" 1 "Increased PA:AA
(0.
> 9+)"

. label values enlargedratio enlargedratio_lab

. 
. label variable enlargedpa "PA diameter"

. label define enlargedpa_lab 0 "Normal PAd" 1 "Enlarged PAd"

. label values enlargedpa enlargedpa_lab

. 
. gen pa_confusion_matrix = .
(990 missing values generated)

. replace pa_confusion_matrix = 0 if (enlargedpa == 0) & (enlargedratio ==
0)
(542 real changes made)

. replace pa_confusion_matrix = 1 if (enlargedpa == 1) & (enlargedratio ==
0)
(109 real changes made)

. replace pa_confusion_matrix = 2 if (enlargedpa == 0) & (enlargedratio ==
1)
(172 real changes made)

. replace pa_confusion_matrix = 3 if (enlargedpa == 1) & (enlargedratio ==
1)
(167 real changes made)

. label variable pa_confusion_matrix "PAd high, PA:AA high, neither, or bot
h?"

. label define pa_confusion_lab 0 "Neither" 1 "Only Enlarged PAd" 2 "Only Hi
gh
> PA:AA" 3 "Both"

. label values pa_confusion_matrix pa_confusion_lab
```

```
. tab enlargedpa pa_confusion_matrix, col //sanity checks

+-------------------+
| Key               |
|-------------------|
|     frequency     |
| column percentage |
+-------------------+

             |   PAd high, PA:AA high, neither, or both?
 PA diameter |   Neither  Only Enla  Only High       Both |     Total
-------------+--------------------------------------------+----------
  Normal PAd |       542          0        172          0 |       714
             |    100.00       0.00     100.00       0.00 |     72.12
-------------+--------------------------------------------+----------
Enlarged PAd |         0        109          0        167 |       276
             |      0.00     100.00       0.00     100.00 |     27.88
-------------+--------------------------------------------+----------
       Total |       542        109        172        167 |       990
             |    100.00     100.00     100.00     100.00 |    100.00

. tab enlargedratio pa_confusion_matrix

                      |   PAd high, PA:AA high, neither, or both?
          PA:AA Ratio |   Neither  Only Enla  Only High       Both |     Tot
al
----------------------+--------------------------------------------+--------
--
  Normal PA:AA (<0.9) |       542        109          0          0 |       6
51
Increased PA:AA (0.9+ |         0          0        172        167 |       3
39
----------------------+--------------------------------------------+--------
--
                Total |       542        109        172        167 |       9
90

.
.
. //label comorbdities
. label variable pulmdisease "Pulmonary Disease"

. label variable chf "Congestive Heart Failure"

. label variable diabetes3 "Diabetes"

. label define diabetes_val 0 "No DM" 1 "Uncomplicated DM" 2 "DM with Compli
cat
> ion(s)"

. label values diabetes3 diabetes_val

. label variable hypertension "Hypertension"

. label variable pulmonarycircdisorder "Pulm Circ. Disorder"
```

```
. label variable peripheralvascdisorders "Periph Vasc. Disease"

. label variable renalfailure "Kidney Disease"

.
. // [ ] remove?
. gen known_assoc_comorb = 0

. replace known_assoc_comorb = 1 if (pulmdisease == 1 | pulmonarycircdisorde
rs
> == 1 | chf == 1)
(571 real changes made)

. tab chf known_assoc_comorb, missing //sanity checks

Congestive |
     Heart |  known_assoc_comorb
   Failure |         0          1 |     Total
-----------+----------------------+----------
         0 |       376        401 |       777
         1 |         0        170 |       170
         . |        43          0 |        43
-----------+----------------------+----------
     Total |       419        571 |       990

. tab pulmdisease known_assoc_comorb, missing

 Pulmonary |  known_assoc_comorb
   Disease |         0          1 |     Total
-----------+----------------------+----------
         0 |       376         88 |       464
         1 |         0        483 |       483
         . |        43          0 |        43
-----------+----------------------+----------
     Total |       419        571 |       990

. tab pulmonarycircdisorders known_assoc_comorb, missing

Pulm Circ. |  known_assoc_comorb
  Disorder |         0          1 |     Total
-----------+----------------------+----------
         0 |       376        428 |       804
         1 |         0        143 |       143
         . |        43          0 |        43
-----------+----------------------+----------
     Total |       419        571 |       990

. label variable known_assoc_comorb "Known Comorbidity w PA->Mortality Assoc
iat
> ion?"

. label define known_assoc_comorb_lab 0 "No Pulm Dz/CHF/Pulm Vasc Dz" 1 "Any
of
>  Pulm Dz/CHF/Pulm Vasc Dz"

. label values known_assoc_comorb known_assoc_comorb_lab
```

```
.
. // Split diabetes into complicated vs not.
. tab diabetes3

               Diabetes |      Freq.     Percent        Cum.
------------------------+-----------------------------------
                  No DM |        718       75.82       75.82
        Uncomplicated DM |        154       16.26       92.08
DM with Complication(s) |         75        7.92      100.00
------------------------+-----------------------------------
                  Total |        947      100.00

. capture drop diabetes1

. gen diabetes1=cond(diabetes3==1,1,0)

. tab diabetes3 diabetes1, missing

                      |       diabetes1
             Diabetes |         0          1 |     Total
----------------------+----------------------+----------
                No DM |       718          0 |       718
     Uncomplicated DM |         0        154 |       154
DM with Complication( |        75          0 |        75
                    . |        43          0 |        43
----------------------+----------------------+----------
                Total |       836        154 |       990

. capture drop diabetes2

. gen diabetes2=cond(diabetes3==2,1,0)

. tab diabetes3 diabetes2, missing

                      |       diabetes2
             Diabetes |         0          1 |     Total
----------------------+----------------------+----------
                No DM |       718          0 |       718
     Uncomplicated DM |       154          0 |       154
DM with Complication( |         0         75 |        75
                    . |        43          0 |        43
----------------------+----------------------+----------
                Total |       915         75 |       990

. label variable diabetes1 "Uncomplicated Diabetes"

. label variable diabetes2 "Diabetes with Complication(s)"

.
. // Outcomes
. label variable death "Death"

. label define death_label 0 "Alive or Censored" 1 "Died"

. label values death death_label
```

```
. label variable lastfollowupyear "Follow-up or death(years)"

.
. label variable anyemergency "Any Emergency Visits in Follow-up?"

. label variable anyadmission "Any Hospital Admission in Follow-up?"

.
. gen time_of_death = lastfollowupyear if death == 1
(727 missing values generated)

. label variable time_of_death "Time of Death"

. gen time_of_censoring = lastfollowupyear if death != 1
(263 missing values generated)

. label variable time_of_censoring "Time of longest follow-up alive"

.
.
. //Categorizations of continuous variables; for graphs
.
. recode age min/30=0 30/40=1 40/50=2 50/60=3 60/70=4 70/max=5, gen(age_deca
de)
(990 differences between age and age_decade)

. label define age_dec_lab 0 "<30 years" 1 "30-40 years" 2 "40-50 years" 3
"50-
> 60 years" 4 "60-70 years" 5 "70+ years"

. label variable age_decade "Age (by decade)"

. label values age_decade age_dec_lab

.
. gen sex_norm_mpad = .
(990 missing values generated)

. replace sex_norm_mpad = mpad + 1 if male == 0
(620 real changes made)

. replace sex_norm_mpad = mpad - 1 if male == 1
(370 real changes made)

. sum sex_norm_mpad, detail

                        sex_norm_mpad
-------------------------------------------------------------
      Percentiles      Smallest
 1%         18.2           16.4
 5%         19.5           16.9
10%         20.8           17.2       Obs                 990
25%           23           17.5       Sum of wgt.         990

50%         25.5                      Mean           25.87222
```

```
                        Largest       Std. dev.      4.209439
75%         28.4           40.5
90%        31.25           40.6       Variance       17.71938
95%         33.4           41.2       Skewness         .64384
99%         38.3           41.3       Kurtosis       3.598582

. recode sex_norm_mpad min/25=0 25/28=1 28/31=2 31/max=3, gen(mpad_cat)
(990 differences between sex_norm_mpad and mpad_cat)

. label define mpad_cat_lab 0 "F:<24 mm, M:<26 mm" 1 "F:24-27 mm, M:26-29 m
m" 2
>  "F:27-30 mm, M:29-32 mm" 3 "F:30+ mm, M32+ mm"

. //TODO: Cutpoints for mPA were defined with ≤27 mm(F) and ≤29 mm(M) as the
no
> rmal reference range; mild as >27 to <31 mm(F) and >29 to <31 mm(M); moder
ate
> ≥31–34 mm; and severe>34 mm.  from -  Truong QA, Bhatia HS, Szymonifka J,
et
> al. A four-tier classification system of pulmonary artery metrics on compu
ted
>  tomography for the diagnosis and prognosis of pulmonary hypertension. J C
ard
> iovasc Comput Tomogr 2018;12(1):60-66.
. label variable mpad_cat "PA_d strata"

. label values mpad_cat mpad_cat_lab

. bysort mpad_cat: sum sex_norm_mpad, detail

-----------------------------------------------------------------------------
---
-> mpad_cat = F:<24 mm, M:<26 mm

                        sex_norm_mpad
-------------------------------------------------------------
      Percentiles      Smallest
 1%         17.6           16.4
 5%         18.8           16.9
10%         19.5           17.2       Obs                 452
25%           21           17.5       Sum of wgt.         452

50%         22.6                      Mean           22.33584
                        Largest       Std. dev.      1.904982
75%           24             25
90%         24.6             25       Variance       3.628956
95%         24.8             25       Skewness      -.5804249
99%           25             25       Kurtosis       2.527295

-----------------------------------------------------------------------------
---
-> mpad_cat = F:24-27 mm, M:26-29 mm

                        sex_norm_mpad
-------------------------------------------------------------
      Percentiles      Smallest
```

```
 1%         25.1           25.1
 5%         25.2           25.1
10%         25.3           25.1       Obs                 269
25%         25.7           25.1       Sum of wgt.         269

50%         26.4                      Mean           26.46543
                        Largest       Std. dev.      .8501723
75%         27.2             28
90%         27.7             28       Variance       .7227929
95%         27.8             28       Skewness       .1066503
99%           28             28       Kurtosis       1.846716

--------------------------------------------------------------------------------------
---
-> mpad_cat = F:27-30 mm, M:29-32 mm

                        sex_norm_mpad
-------------------------------------------------------------
      Percentiles      Smallest
 1%         28.2           28.1
 5%         28.2           28.2
10%         28.3           28.2       Obs                 166
25%         28.7           28.2       Sum of wgt.         166

50%         29.3                      Mean           29.38253
                        Largest       Std. dev.       .801133
75%           30             31
90%         30.6             31       Variance       .6418141
95%         30.8             31       Skewness        .305048
99%           31             31       Kurtosis       2.134404

--------------------------------------------------------------------------------------
---
-> mpad_cat = F:30+ mm, M32+ mm

                        sex_norm_mpad
-------------------------------------------------------------
      Percentiles      Smallest
 1%         31.2           31.2
 5%         31.4           31.2
10%         31.6           31.2       Obs                 103
25%         32.1           31.2       Sum of wgt.         103

50%         33.3                      Mean           34.18447
                        Largest       Std. dev.      2.579548
75%         35.5           40.5
90%         37.8           40.6       Variance       6.654069
95%         39.6           41.2       Skewness       1.040311
99%         41.2           41.3       Kurtosis       3.273034


. bysort mpad_cat: sum mpad if male == 0, detail  //sanity check

--------------------------------------------------------------------------------------
---
-> mpad_cat = F:<24 mm, M:<26 mm
```

```
                            MPAD
-------------------------------------------------------------
      Percentiles      Smallest
 1%         16.6           15.9
 5%         17.9           16.5
10%         18.7           16.6       Obs                 277
25%         20.3           17.2       Sum of wgt.         277

50%         21.7                      Mean           21.42238
                        Largest       Std. dev.      1.813437
75%         22.9             24
90%         23.6             24       Variance       3.288555
95%         23.8             24       Skewness      -.6016113
99%           24             24       Kurtosis       2.600962

----------------------------------------------------------------------------
---
-> mpad_cat = F:24-27 mm, M:26-29 mm

                            MPAD
-------------------------------------------------------------
      Percentiles      Smallest
 1%         24.1           24.1
 5%         24.1           24.1
10%         24.3           24.1       Obs                 175
25%         24.7           24.1       Sum of wgt.         175

50%         25.4                      Mean           25.43429
                        Largest       Std. dev.       .859077
75%         26.1             27
90%         26.7             27       Variance       .7380132
95%         26.8             27       Skewness        .135076
99%           27             27       Kurtosis        1.81374

----------------------------------------------------------------------------
---
-> mpad_cat = F:27-30 mm, M:29-32 mm

                            MPAD
-------------------------------------------------------------
      Percentiles      Smallest
 1%        27.15           27.1
 5%         27.2           27.2
10%         27.3           27.2       Obs                 100
25%        27.85           27.2       Sum of wgt.         100

50%         28.3                      Mean              28.42
                        Largest       Std. dev.      .7902179
75%           29             30
90%         29.5             30       Variance       .6244443
95%         29.9             30       Skewness       .2389809
99%           30             30       Kurtosis       2.239303

----------------------------------------------------------------------------
---
```

```
-> mpad_cat = F:30+ mm, M32+ mm

                            MPAD
-------------------------------------------------------------
      Percentiles      Smallest
 1%         30.2           30.2
 5%         30.5           30.2
10%         30.5           30.3       Obs                  68
25%         30.9           30.5       Sum of wgt.          68

50%        32.05                      Mean           32.85441
                        Largest       Std. dev.      2.403072
75%         34.2           37.3
90%         36.4           39.1       Variance       5.774755
95%         37.3           39.6       Skewness        1.19754
99%         40.3           40.3       Kurtosis       3.913179


. bysort mpad_cat: sum mpad if male == 1, detail  //sanity check

----------------------------------------------------------------------------
---
-> mpad_cat = F:<24 mm, M:<26 mm

                            MPAD
-------------------------------------------------------------
      Percentiles      Smallest
 1%         18.2           17.4
 5%         19.6           18.2
10%         20.2           18.6       Obs                 175
25%         21.8           18.8       Sum of wgt.         175

50%         23.4                      Mean           23.19886
                        Largest       Std. dev.      2.039354
75%           25             26
90%         25.6             26       Variance       4.158964
95%         25.9             26       Skewness      -.5145537
99%           26             26       Kurtosis       2.354854

----------------------------------------------------------------------------
---
-> mpad_cat = F:24-27 mm, M:26-29 mm

                            MPAD
-------------------------------------------------------------
      Percentiles      Smallest
 1%         26.1           26.1
 5%         26.2           26.1
10%         26.4           26.1       Obs                  94
25%         26.9           26.2       Sum of wgt.          94

50%         27.5                      Mean            27.5234
                        Largest       Std. dev.      .8347848
75%         28.2             29
90%         28.7             29       Variance       .6968657
95%         28.9             29       Skewness       .0628926
```

```
99%           29             29       Kurtosis       1.918017

-----------------------------------------------------------------------------
---
-> mpad_cat = F:27-30 mm, M:29-32 mm

                            MPAD
-------------------------------------------------------------
      Percentiles      Smallest
 1%         29.2           29.2
 5%         29.2           29.2
10%         29.3           29.2       Obs                  66
25%         29.6           29.2       Sum of wgt.          66

50%         30.1                      Mean           30.32576
                        Largest       Std. dev.      .8201896
75%         30.8           31.7
90%         31.7           31.8       Variance       .6727109
95%         31.7           31.9       Skewness       .4127733
99%         31.9           31.9       Kurtosis       2.031273

-----------------------------------------------------------------------------
---
-> mpad_cat = F:30+ mm, M32+ mm

                            MPAD
-------------------------------------------------------------
      Percentiles      Smallest
 1%         32.2           32.2
 5%         32.2           32.2
10%           33           32.4       Obs                  35
25%         33.9             33       Sum of wgt.          35

50%           35                      Mean           35.82571
                        Largest       Std. dev.      2.818045
75%         37.5           40.5
90%         40.5           40.6       Variance       7.941377
95%         41.5           41.5       Skewness       .7522805
99%         42.2           42.2       Kurtosis        2.45316


.
. gen sex_norm_aa = .
(990 missing values generated)

. replace sex_norm_aa = ascendingaorta + 1.5 if male == 0 //men 3mm larger;
mor
> e dispersion (not accounted for)
(620 real changes made)

. replace sex_norm_aa = ascendingaorta - 1.5 if male == 1
(370 real changes made)

. sum sex_norm_aa, detail

                         sex_norm_aa
```

```
-------------------------------------------------------------
      Percentiles      Smallest
 1%         21.4           16.5
 5%         23.4           17.5
10%         25.1           17.7       Obs                 990
25%         27.5           19.3       Sum of wgt.         990

50%         30.7                      Mean           30.72808
                        Largest       Std. dev.      4.593335
75%         33.6           46.6
90%        36.45           46.7       Variance       21.09873
95%         38.6           46.9       Skewness       .3447436
99%         43.1           49.9       Kurtosis       3.550049

. recode sex_norm_aa min/30=0 30/33.5=1 33.5/37=2 37/max=3, gen(aa_cat)
(990 differences between sex_norm_aa and aa_cat)

. label define aa_cat_lab 0 "F:<28.5 mm, M:<31.5 mm" 1 "F:28.5-32 mm, M:31.5
-35
>  mm" 2 "F:32-35.5 mm, M:35-38.5mm" 3 "F:35.5+mm, M:38.5+mm" //35 and 32 as
th
> resholds (similar percentil as for MPAd) 33.5 as threshold for abn, roughl
y

. label variable aa_cat "AA_d strata"

. label values aa_cat aa_cat_lab

. bysort aa_cat: sum sex_norm_aa, detail

----------------------------------------------------------------------------
---
-> aa_cat = F:<28.5 mm, M:<31.5 mm

                         sex_norm_aa
-------------------------------------------------------------
      Percentiles      Smallest
 1%           20           16.5
 5%         22.1           17.5
10%         23.2           17.7       Obs                 439
25%         25.3           19.3       Sum of wgt.         439

50%         27.1                      Mean           26.71549
                        Largest       Std. dev.      2.416522
75%         28.7             30
90%         29.4             30       Variance       5.839577
95%         29.8             30       Skewness      -.8962911
99%           30             30       Kurtosis       3.759939

----------------------------------------------------------------------------
---
-> aa_cat = F:28.5-32 mm, M:31.5-35 mm

                         sex_norm_aa
-------------------------------------------------------------
      Percentiles      Smallest
```

```
 1%         30.1           30.1
 5%         30.2           30.1
10%         30.3           30.1       Obs                 293
25%         30.9           30.1       Sum of wgt.         293

50%         31.7                      Mean           31.66758
                        Largest       Std. dev.      .9372467
75%         32.4           33.4
90%         33.1           33.5       Variance       .8784313
95%         33.3           33.5       Skewness       .0904055
99%         33.5           33.5       Kurtosis       2.044953

----------------------------------------------------------------------------
---
-> aa_cat = F:32-35.5 mm, M:35-38.5mm

                        sex_norm_aa
-------------------------------------------------------------
      Percentiles      Smallest
 1%         33.6           33.6
 5%         33.6           33.6
10%         33.7           33.6       Obs                 175
25%           34           33.6       Sum of wgt.         175

50%         34.7                      Mean           34.86743
                        Largest       Std. dev.      .9798673
75%         35.6           36.9
90%         36.3           36.9       Variance       .9601398
95%         36.7           36.9       Skewness       .5101286
99%         36.9             37       Kurtosis       2.085036

----------------------------------------------------------------------------
---
-> aa_cat = F:35.5+mm, M:38.5+mm

                        sex_norm_aa
-------------------------------------------------------------
      Percentiles      Smallest
 1%         37.1           37.1
 5%         37.3           37.1
10%         37.5           37.2       Obs                  83
25%           38           37.2       Sum of wgt.          83

50%         39.2                      Mean           39.90723
                        Largest       Std. dev.      2.628724
75%         40.8           46.6
90%         44.1           46.7       Variance       6.910191
95%           46           46.9       Skewness       1.592395
99%         49.9           49.9       Kurtosis       5.422965


. bysort aa_cat: sum ascendingaorta if male == 0, detail

----------------------------------------------------------------------------
---
-> aa_cat = F:<28.5 mm, M:<31.5 mm
```

```
                       Ascending Aorta
-------------------------------------------------------------
      Percentiles      Smallest
 1%         19.5           18.5
 5%         21.1             19
10%         21.8           19.5       Obs                 266
25%         23.9           19.8       Sum of wgt.         266

50%         25.7                      Mean           25.36617
                        Largest       Std. dev.      2.273801
75%         27.4           28.4
90%           28           28.5       Variance       5.170172
95%         28.3           28.5       Skewness      -.6430157
99%         28.5           28.5       Kurtosis       2.642303

----------------------------------------------------------------------------
---
-> aa_cat = F:28.5-32 mm, M:31.5-35 mm

                       Ascending Aorta
-------------------------------------------------------------
      Percentiles      Smallest
 1%         28.6           28.6
 5%         28.7           28.6
10%         28.8           28.6       Obs                 200
25%         29.4           28.6       Sum of wgt.         200

50%         30.1                      Mean            30.1565
                        Largest       Std. dev.       .938007
75%         30.9           31.9
90%         31.5             32       Variance       .8798571
95%         31.7             32       Skewness       .0801883
99%           32             32       Kurtosis        2.02428

----------------------------------------------------------------------------
---
-> aa_cat = F:32-35.5 mm, M:35-38.5mm

                       Ascending Aorta
-------------------------------------------------------------
      Percentiles      Smallest
 1%         32.1           32.1
 5%         32.1           32.1
10%         32.2           32.1       Obs                 108
25%        32.55           32.1       Sum of wgt.         108

50%        33.15                      Mean           33.40648
                        Largest       Std. dev.      .9998385
75%        34.15           35.3
90%         34.9           35.3       Variance        .999677
95%         35.1           35.4       Skewness        .433915
99%         35.4           35.4       Kurtosis       1.915829

----------------------------------------------------------------------------
---
```

```
-> aa_cat = F:35.5+mm, M:38.5+mm

                       Ascending Aorta
-------------------------------------------------------------
      Percentiles      Smallest
 1%         35.6           35.6
 5%         35.7           35.6
10%         35.8           35.7       Obs                  46
25%         36.7           35.8       Sum of wgt.          46

50%        37.85                      Mean           38.56087
                        Largest       Std. dev.      2.823275
75%         39.3           43.4
90%         43.4           44.5       Variance        7.97088
95%         44.5           45.2       Skewness       1.586845
99%         48.4           48.4       Kurtosis       5.303856


. bysort aa_cat: sum ascendingaorta if male == 1, detail

----------------------------------------------------------------------------
---
-> aa_cat = F:<28.5 mm, M:<31.5 mm

                       Ascending Aorta
-------------------------------------------------------------
      Percentiles      Smallest
 1%           19             18
 5%         23.3             19
10%         24.5           19.2       Obs                 173
25%         26.6           20.8       Sum of wgt.         173

50%         28.5                      Mean           27.98382
                        Largest       Std. dev.      2.610706
75%         30.1           31.4
90%         30.8           31.5       Variance       6.815783
95%         31.1           31.5       Skewness      -1.093603
99%         31.5           31.5       Kurtosis       4.397207

----------------------------------------------------------------------------
---
-> aa_cat = F:28.5-32 mm, M:31.5-35 mm

                       Ascending Aorta
-------------------------------------------------------------
      Percentiles      Smallest
 1%         31.6           31.6
 5%         31.7           31.6
10%         31.9           31.6       Obs                  93
25%         32.4           31.6       Sum of wgt.          93

50%         33.3                      Mean            33.1914
                        Largest       Std. dev.      .9402422
75%         33.8           34.8
90%         34.6           34.9       Variance       .8840555
95%         34.8           34.9       Skewness       .1126076
```

```
99%          34.9           34.9       Kurtosis       2.084397

-----------------------------------------------------------------------------
---
-> aa_cat = F:32-35.5 mm, M:35-38.5mm

                       Ascending Aorta
-------------------------------------------------------------
      Percentiles      Smallest
 1%         35.1           35.1
 5%         35.1           35.1
10%         35.2           35.1       Obs                  67
25%         35.5           35.1       Sum of wgt.          67

50%         36.2                      Mean           36.30448
                        Largest       Std. dev.      .9508264
75%         36.8           38.2
90%         37.8           38.3       Variance       .9040708
95%         38.2           38.4       Skewness        .636223
99%         38.5           38.5       Kurtosis       2.427223

-----------------------------------------------------------------------------
---
-> aa_cat = F:35.5+mm, M:38.5+mm

                       Ascending Aorta
-------------------------------------------------------------
      Percentiles      Smallest
 1%         38.7           38.7
 5%         38.9           38.9
10%           39             39       Obs                  37
25%         39.4             39       Sum of wgt.          37

50%         40.5                      Mean           41.21622
                        Largest       Std. dev.      2.389271
75%         42.2           43.8
90%         43.8           45.6       Variance       5.708617
95%         48.1           48.1       Skewness       1.488033
99%         48.4           48.4       Kurtosis       4.924933


.
. save cleaned_noempi, replace
file cleaned_noempi.dta saved

.
```

Data Analysis

In [5]:
```
%%stata
clear
use cleaned_noempi

/* Analysis */

/*
```

```
Table 1
*/

table1_mc, by (age_decade) ///
vars( ///
male bin %4.1f \ ///
obesity_calc bin %4.1f \ ///
pulmdisease bin %4.1f \ ///
chf bin %4.1f \ ///
diabetes3 cat %4.1f \ ///
hypertension bin %4.1f \ ///
pulmonarycircdisorders bin %4.1f \ ///
peripheralvascdisorders bin %4.1f \ ///
renalfailure bin %4.1f \ ///
mpad conts %4.1f \ ///
enlargedpa cat %4.1f \ ///
ascendingaorta conts %4.1f \ ///
mpaaa conts %4.2f \ ///
enlargedratio cat %4.1f \ ///
anyemergency bin %4.1f \ ///
anyadmission bin %4.1f \ ///
death bin %4.1f \ ///
time_of_death conts %4.1f \ ///
time_of_death conts %4.1f \ ///
) ///
percent_n percsign("%") iqrmiddle(",") sdleft(" (±") sdright(")") onecol tot
saving("Results and Figures/$S_DATE/Table 1 PA enlargement by Age.xlsx", rep
```

```
. clear

. use cleaned_noempi

.
. /* Analysis */
.
. /*
> Table 1
> */
.
. table1_mc, by (age_decade) ///
> vars( ///
> male bin %4.1f \ ///
> obesity_calc bin %4.1f \ ///
> pulmdisease bin %4.1f \ ///
> chf bin %4.1f \ ///
> diabetes3 cat %4.1f \ ///
> hypertension bin %4.1f \ ///
> pulmonarycircdisorders bin %4.1f \ ///
> peripheralvascdisorders bin %4.1f \ ///
> renalfailure bin %4.1f \ ///
> mpad conts %4.1f \ ///
> enlargedpa cat %4.1f \ ///
> ascendingaorta conts %4.1f \ ///
> mpaaa conts %4.2f \ ///
> enlargedratio cat %4.1f \ ///
> anyemergency bin %4.1f \ ///
> anyadmission bin %4.1f \ ///
> death bin %4.1f \ ///
> time_of_death conts %4.1f \ ///
> time_of_death conts %4.1f \ ///
> ) ///
> percent_n percsign("%") iqrmiddle(",") sdleft(" (±") sdright(")") onecol t
ota
> l(before) ///
> saving("Results and Figures/$S_DATE/Table 1 PA enlargement by Age.xlsx", r
epl
> ace)

  +-------------------------------------------------------------------------
---
> ---------------------------------------------+
  | factor                                N_T   N_0   N_1   N_2   N_3   N_4
> N_5   m_T   m_0   m_1   m_2   m_3   m_4   m_5 |
  |-------------------------------------------------------------------------
---
> ---------------------------------------------|
  | Male Sex?                             990   147   165   169   159   145
> 205     0     0     0     0     0     0     0 |
  |-------------------------------------------------------------------------
---
> ---------------------------------------------|
  | Obesity                               954   134   156   160   158   142
> 204    36    13     9     9     1     3     1 |
  |-------------------------------------------------------------------------
```

```
---
> ----------------------------------------------|
  | Pulmonary Disease                      947   129   154   160   158   142
> 204    43    18    11     9     1     3     1 |
  |--------------------------------------------------------------------------
---
> ----------------------------------------------|
  | Congestive Heart Failure               947   129   154   160   158   142
> 204    43    18    11     9     1     3     1 |
  |--------------------------------------------------------------------------
---
> ----------------------------------------------|
  | Diabetes                               947   129   154   160   158   142
> 204    43    18    11     9     1     3     1 |
  |--------------------------------------------------------------------------
---
> ----------------------------------------------|
  | Hypertension                           947   129   154   160   158   142
> 204    43    18    11     9     1     3     1 |
  |--------------------------------------------------------------------------
---
> ----------------------------------------------|
  | Pulm Circ. Disorder                    947   129   154   160   158   142
> 204    43    18    11     9     1     3     1 |
  |--------------------------------------------------------------------------
---
> ----------------------------------------------|
  | Periph Vasc. Disease                   947   129   154   160   158   142
> 204    43    18    11     9     1     3     1 |
  |--------------------------------------------------------------------------
---
> ----------------------------------------------|
  | Kidney Disease                         947   129   154   160   158   142
> 204    43    18    11     9     1     3     1 |
  |--------------------------------------------------------------------------
---
> ----------------------------------------------|
  | MPAD                                   990   147   165   169   159   145
> 205     0     0     0     0     0     0     0 |
  |--------------------------------------------------------------------------
---
> ----------------------------------------------|
  | PA diameter                            990   147   165   169   159   145
> 205     0     0     0     0     0     0     0 |
  |--------------------------------------------------------------------------
---
> ----------------------------------------------|
  | Ascending Aorta                        990   147   165   169   159   145
> 205     0     0     0     0     0     0     0 |
  |--------------------------------------------------------------------------
---
> ----------------------------------------------|
  | MPA/AA                                 990   147   165   169   159   145
> 205     0     0     0     0     0     0     0 |
  |--------------------------------------------------------------------------
---
```

```
> ----------------------------------------------|
  | PA:AA Ratio                           990   147   165   169   159   145
> 205     0     0     0     0     0     0     0 |
  |---------------------------------------------------------------------------
---
> ----------------------------------------------|
  | Any Emergency Visits in Follow-up?    990   147   165   169   159   145
> 205     0     0     0     0     0     0     0 |
  |---------------------------------------------------------------------------
---
> ----------------------------------------------|
  | Any Hospital Admission in Follow-up?  990   147   165   169   159   145
> 205     0     0     0     0     0     0     0 |
  |---------------------------------------------------------------------------
---
> ----------------------------------------------|
  | Death                                 990   147   165   169   159   145
> 205     0     0     0     0     0     0     0 |
  |---------------------------------------------------------------------------
---
> ----------------------------------------------|
  | Time of Death                         263     2    19    29    37    42
> 134   727   145   146   140   122   103    71 |
  | Time of Death                         263     2    19    29    37    42
> 134   727   145   146   140   122   103    71 |
  +---------------------------------------------------------------------------
---
> ----------------------------------------------+
   N_ ... #records used below,   m_ ... #records not used

  +---------------------------------------------------------------------------
---
> ---------------------------------------------------------------------------
---
> ----------------------------+
  |                                       Total              <30 years
>   30-40 years        40-50 years       50-60 years       60-70 years
>  70+ years          p-value |
  |---------------------------------------------------------------------------
---
> ---------------------------------------------------------------------------
---
> ----------------------------|
  |                                       N=990              N=147
>   N=165              N=169             N=159             N=145
>  N=205                      |
  |---------------------------------------------------------------------------
---
> ---------------------------------------------------------------------------
---
> ----------------------------|
  | Male Sex?                             37.4% (370)        30.6% (45)
>   37.0% (61)         40.8% (69)        46.5% (74)        36.6% (53)
>  33.2% (68)          0.051  |
  |---------------------------------------------------------------------------
---
```

```
> ----------------------------------------------------------------------------
---
> ----------------------------|
  | Obesity                               34.6% (330)        25.4% (34)
>   41.0% (64)         37.5% (60)         45.6% (72)         38.0% (54)
>  22.5% (46)        <0.001  |
  |---------------------------------------------------------------------------
---
> ----------------------------------------------------------------------------
---
> ----------------------------|
  | Pulmonary Disease                     51.0% (483)        34.9% (45)
>   50.6% (78)         49.4% (79)         52.5% (83)         56.3% (80)
>  57.8% (118)         0.002  |
  |---------------------------------------------------------------------------
---
> ----------------------------------------------------------------------------
---
> ----------------------------|
  | Congestive Heart Failure              18.0% (170)         3.1% (4)
>    7.1% (11)          6.9% (11)         22.2% (35)         16.9% (24)
>  41.7% (85)        <0.001  |
  |---------------------------------------------------------------------------
---
> ----------------------------------------------------------------------------
---
> ----------------------------|
  | Diabetes
>
>                    <0.001  |
  |    No DM                              75.8% (718)        96.9% (125)
>   82.5% (127)        81.9% (131)        62.7% (99)         69.7% (99)
>  67.2% (137)                 |
  |    Uncomplicated DM                   16.3% (154)         2.3% (3)
>   12.3% (19)         13.1% (21)         22.8% (36)         21.1% (30)
>  22.1% (45)                  |
  |    DM with Complication(s)             7.9% (75)          0.8% (1)
>    5.2% (8)           5.0% (8)          14.6% (23)          9.2% (13)
>  10.8% (22)                  |
  |---------------------------------------------------------------------------
---
> ----------------------------------------------------------------------------
---
> ----------------------------|
  | Hypertension                          54.7% (518)        13.2% (17)
>   31.8% (49)         44.4% (71)         65.2% (103)        73.9% (105)
>  84.8% (173)       <0.001  |
  |---------------------------------------------------------------------------
---
> ----------------------------------------------------------------------------
---
> ----------------------------|
  | Pulm Circ. Disorder                   15.1% (143)        11.6% (15)
>   13.6% (21)         16.2% (26)         13.9% (22)         16.2% (23)
>  17.6% (36)          0.70   |
  |---------------------------------------------------------------------------
```

```
---
> ---------------------------------------------------------------------------
---
> ----------------------------|
  | Periph Vasc. Disease                     10.9% (103)        2.3% (3)
>    1.9% (3)          4.4% (7)         12.7% (20)         9.9% (14)
>  27.5% (56)        <0.001  |
  |--------------------------------------------------------------------------
---
> ---------------------------------------------------------------------------
---
> ----------------------------|
  | Kidney Disease                            6.4% (61)         2.3% (3)
>    2.6% (4)          3.1% (5)          8.2% (13)         5.6% (8)
>  13.7% (28)        <0.001  |
  |--------------------------------------------------------------------------
---
> ---------------------------------------------------------------------------
---
> ----------------------------|
  | MPAD                                     25.2 (22.5,28.2)   23.4 (21.1,25.
8)
>   24.1 (22.2,26.6)   24.7 (22.0,27.9)   25.8 (23.2,28.4)   25.9 (23.2,29.
2)
>  26.7 (24.1,29.7)   <0.001  |
  |--------------------------------------------------------------------------
---
> ---------------------------------------------------------------------------
---
> ----------------------------|
  | PA diameter
>
>                    <0.001  |
  |   Normal PAd                             72.1% (714)       87.1% (128)
>   81.8% (135)       74.6% (126)       73.0% (116)       66.2% (96)
>  55.1% (113)                 |
  |   Enlarged PAd                           27.9% (276)       12.9% (19)
>   18.2% (30)        25.4% (43)        27.0% (43)        33.8% (49)
>  44.9% (92)                  |
  |--------------------------------------------------------------------------
---
> ---------------------------------------------------------------------------
---
> ----------------------------|
  | Ascending Aorta                          30.1 (27.2,33.3)   25.1 (23.3,27.
9)
>   27.9 (25.3,30.2)   29.2 (27.2,32.3)   31.7 (29.4,34.8)   31.8 (29.7,34.
8)
>  32.9 (30.2,35.8)   <0.001  |
  |--------------------------------------------------------------------------
---
> ---------------------------------------------------------------------------
---
> ----------------------------|
  | MPA/AA                                   0.85 (0.76,0.93)   0.93 (0.85,1.0
2)
```

```
>   0.89 (0.83,0.96)   0.84 (0.78,0.92)   0.79 (0.73,0.91)   0.81 (0.73,0.8
9)
>  0.81 (0.72,0.90)   <0.001  |
  |--------------------------------------------------------------------------
---
> --------------------------------------------------------------------------
---
> ----------------------------|
  | PA:AA Ratio
>
>                     <0.001  |
  |    Normal PA:AA (<0.9)                65.8% (651)        39.5% (58)
>   53.9% (89)         71.6% (121)        74.2% (118)        78.6% (114)
>  73.7% (151)                |
  |    Increased PA:AA (0.9+)             34.2% (339)        60.5% (89)
>   46.1% (76)         28.4% (48)         25.8% (41)         21.4% (31)
>  26.3% (54)                 |
  |--------------------------------------------------------------------------
---
> --------------------------------------------------------------------------
---
> ----------------------------|
  | Any Emergency Visits in Follow-up?    76.9% (761)        84.4% (124)
>   83.6% (138)        81.1% (137)        75.5% (120)        66.2% (96)
>  71.2% (146)        <0.001  |
  |--------------------------------------------------------------------------
---
> --------------------------------------------------------------------------
---
> ----------------------------|
  | Any Hospital Admission in Follow-up?  52.7% (522)        51.0% (75)
>   44.2% (73)         46.2% (78)         54.1% (86)         59.3% (86)
>  60.5% (124)        0.009   |
  |--------------------------------------------------------------------------
---
> --------------------------------------------------------------------------
---
> ----------------------------|
  | Death                                 26.6% (263)         1.4% (2)
>   11.5% (19)         17.2% (29)         23.3% (37)         29.0% (42)
>  65.4% (134)        <0.001  |
  |--------------------------------------------------------------------------
---
> --------------------------------------------------------------------------
---
> ----------------------------|
  | Time of Death                         3.5 (1.2,6.6)      4.3 (3.7,4.9)
>   2.6 (0.8,10.7)     2.4 (1.4,5.6)      3.3 (0.8,5.1)      3.4 (1.2,6.1)
>  4.1 (1.3,7.3)      0.65    |
  | Time of Death                         3.5 (1.2,6.6)      4.3 (3.7,4.9)
>   2.6 (0.8,10.7)     2.4 (1.4,5.6)      3.3 (0.8,5.1)      3.4 (1.2,6.1)
>  4.1 (1.3,7.3)      0.65    |
  +--------------------------------------------------------------------------
---
> --------------------------------------------------------------------------
---
```

```
> ----------------------------+
Data are presented as median (IQR) for continuous measures, and % (n) for ca
teg
> orical measures.

file Results and Figures/25 Oct 2025/Table 1 PA enlargement by Age.xlsx save
d
```

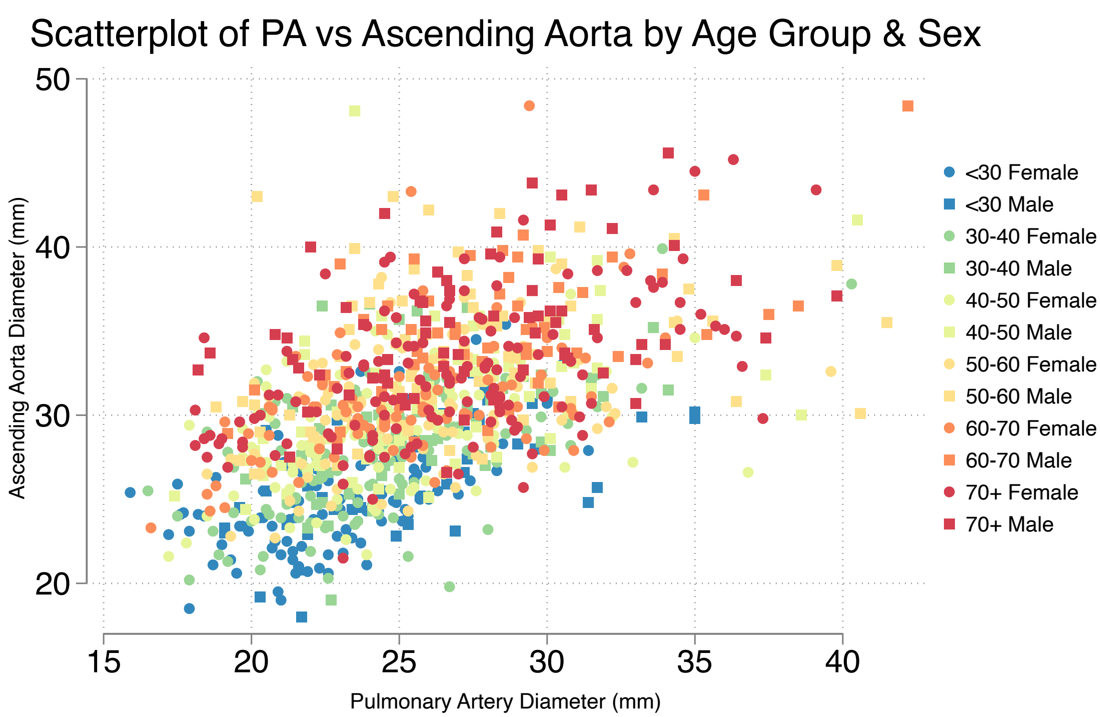
.

Table 1 Details - all the pertinent info is easier to see in images below

In [6]:
```
%%stata
colorpalette Spectral, n(6) nograph  // [ ] TODO: use same CET C6 pallette s
local color6 `"`r(p1)'"' // Reverse order
local color5 `"`r(p2)'"'
local color4 `"`r(p3)'"'
local color3 `"`r(p4)'"'
local color2 `"`r(p5)'"'
local color1 `"`r(p6)'"'

twoway ///
    (scatter ascendingaorta mpad if age_decade == 0 & male == 0, mcolor("`co
    (scatter ascendingaorta mpad if age_decade == 0 & male == 1, mcolor("`co
    (scatter ascendingaorta mpad if age_decade == 1 & male == 0, mcolor("`co
    (scatter ascendingaorta mpad if age_decade == 1 & male == 1, mcolor("`co
    (scatter ascendingaorta mpad if age_decade == 2 & male == 0, mcolor("`co
    (scatter ascendingaorta mpad if age_decade == 2 & male == 1, mcolor("`co
    (scatter ascendingaorta mpad if age_decade == 3 & male == 0, mcolor("`co
    (scatter ascendingaorta mpad if age_decade == 3 & male == 1, mcolor("`co
    (scatter ascendingaorta mpad if age_decade == 4 & male == 0, mcolor("`co
    (scatter ascendingaorta mpad if age_decade == 4 & male == 1, mcolor("`co
    (scatter ascendingaorta mpad if age_decade == 5 & male == 0, mcolor("`co
    (scatter ascendingaorta mpad if age_decade == 5 & male == 1, mcolor("`co
    ///
    legend(order(1 "<30 Female" 2 "<30 Male" 3 "30-40 Female" 4 "30-40 Male"
                 5 "40-50 Female" 6 "40-50 Male" 7 "50-60 Female" 8 "50-60 M
                 9 "60-70 Female" 10 "60-70 Male" 11 "70+ Female" 12 "70+ Ma
    xtitle("Pulmonary Artery Diameter (mm)") ytitle("Ascending Aorta Diamete
    title("Scatterplot of PA vs Ascending Aorta by Age Group & Sex") ///
    xlabel(, labsize(medlarge)) ylabel(, labsize(medlarge)) ///
    scheme(white_w3d)
```

```
. colorpalette Spectral, n(6) nograph  // [ ] TODO: use same CET C6 pallette
so
> mehow?

. local color6 `"`r(p1)'"' // Reverse order

. local color5 `"`r(p2)'"'

. local color4 `"`r(p3)'"'

. local color3 `"`r(p4)'"'

. local color2 `"`r(p5)'"'

. local color1 `"`r(p6)'"'

.
. twoway ///
>     (scatter ascendingaorta mpad if age_decade == 0 & male == 0, mcolor("`
col
> or1'") msymbol(O)) || /// <30 years, Female
>     (scatter ascendingaorta mpad if age_decade == 0 & male == 1, mcolor("`
col
> or1'") msymbol(S)) || /// <30 years, Male
>     (scatter ascendingaorta mpad if age_decade == 1 & male == 0, mcolor("`
col
> or2'") msymbol(O)) || /// 30-40 years, Female
>     (scatter ascendingaorta mpad if age_decade == 1 & male == 1, mcolor("`
col
> or2'") msymbol(S)) || /// 30-40 years, Male
>     (scatter ascendingaorta mpad if age_decade == 2 & male == 0, mcolor("`
col
> or3'") msymbol(O)) || /// 40-50 years, Female
>     (scatter ascendingaorta mpad if age_decade == 2 & male == 1, mcolor("`
col
> or3'") msymbol(S)) || /// 40-50 years, Male
>     (scatter ascendingaorta mpad if age_decade == 3 & male == 0, mcolor("`
col
> or4'") msymbol(O)) || /// 50-60 years, Female
>     (scatter ascendingaorta mpad if age_decade == 3 & male == 1, mcolor("`
col
> or4'") msymbol(S)) || /// 50-60 years, Male
>     (scatter ascendingaorta mpad if age_decade == 4 & male == 0, mcolor("`
col
> or5'") msymbol(O)) || /// 60-70 years, Female
>     (scatter ascendingaorta mpad if age_decade == 4 & male == 1, mcolor("`
col
> or5'") msymbol(S)) || /// 60-70 years, Male
>     (scatter ascendingaorta mpad if age_decade == 5 & male == 0, mcolor("`
col
> or6'") msymbol(O)) || /// 70+ years, Female
>     (scatter ascendingaorta mpad if age_decade == 5 & male == 1, mcolor("`
col
> or6'") msymbol(S)), /// 70+ years, Male
>     ///
>     legend(order(1 "<30 Female" 2 "<30 Male" 3 "30-40 Female" 4 "30-40 Mal
```

```
e"
> ///
>                     5 "40-50 Female" 6 "40-50 Male" 7 "50-60 Female" 8 "50-60
Ma
> le" ///
>                     9 "60-70 Female" 10 "60-70 Male" 11 "70+ Female" 12 "70+
Mal
> e")) ///
>       xtitle("Pulmonary Artery Diameter (mm)") ytitle("Ascending Aorta Diame
ter
>    (mm)") ///
>       title("Scatterplot of PA vs Ascending Aorta by Age Group & Sex") ///
>       xlabel(, labsize(medlarge)) ylabel(, labsize(medlarge)) ///
>       scheme(white_w3d)

.
```

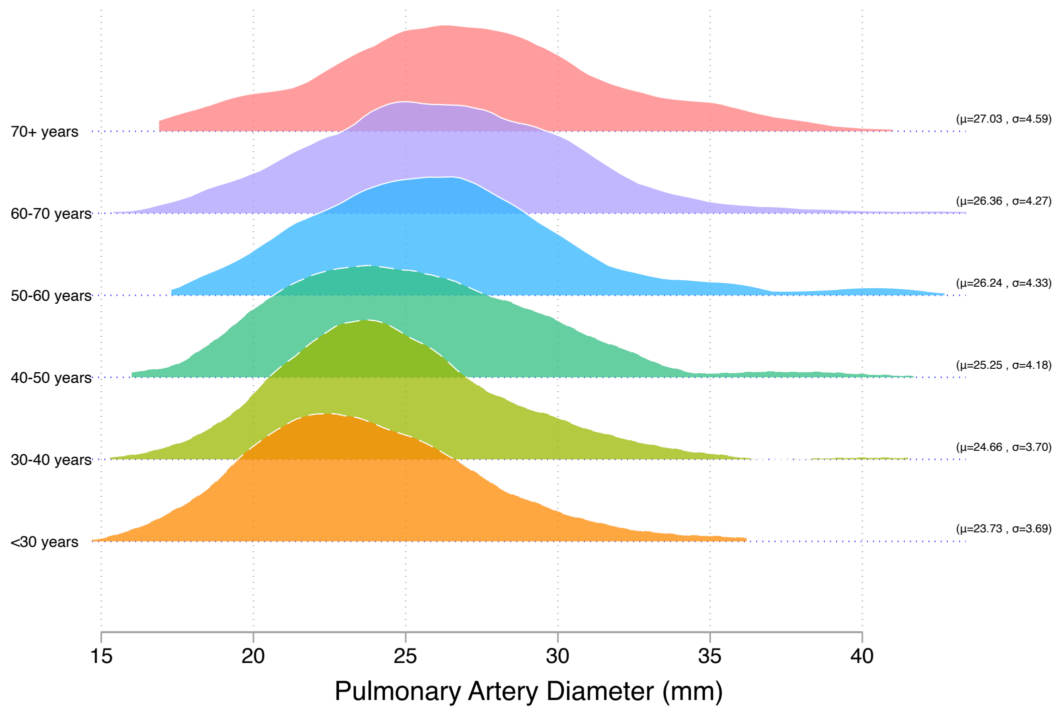


this one shows how sex, age, PA and AA size relate. (no correlation to outcomes yet). Clearly an upward trend as people get older, and somewhat shifted distributions by sex

Assessing the relationship between Age, Sex and PA size

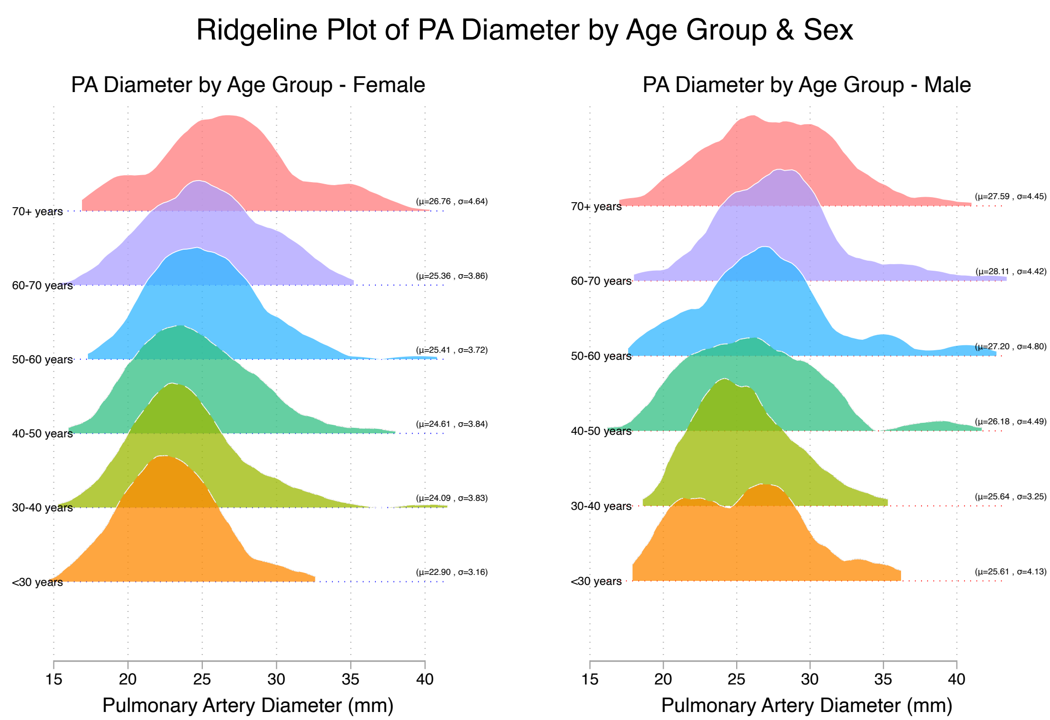
In [7]:
```
%%stata
/* PAd */
ridgeline mpad, by(age_decade) yline ylw(0.2) overlap(1.7) ylc(blue) ylp(dot
        labpos(right) bwid(1.2) laboffset(-3) showstats xlabel(15(5)40) ///
        palette(CET C6) alpha(75) xtitle("Pulmonary Artery Diameter (mm)")
```

```
. /* PAd */
. ridgeline mpad, by(age_decade) yline ylw(0.2) overlap(1.7) ylc(blue) ylp(d
ot)
>  ////
>         labpos(right) bwid(1.2) laboffset(-3) showstats xlabel(15(5)40)
///
>         palette(CET C6) alpha(75) xtitle("Pulmonary Artery Diameter (mm)")
(note:  named style i not found in class symbol, default attributes used)

.
```



In [8]:
```
%%stata
* Ridgeline plot for Females (male == 0)
ridgeline mpad if male == 0, by(age_decade) yline ylw(0.2) overlap(1.7) ///
    ylc(blue) ylp(dot) labpos(right) bwid(1.2) laboffset(-3) ///
    showstats xlabel(15(5)40) palette(CET C6) alpha(75) ///
    xtitle("Pulmonary Artery Diameter (mm)") ///
    title("PA Diameter by Age Group - Female") ///
    saving(ridgeline_female, replace)

* Ridgeline plot for Males (male == 1)
ridgeline mpad if male == 1, by(age_decade) yline ylw(0.2) overlap(1.7) ///
    ylc(red) ylp(dot) labpos(right) bwid(1.2) laboffset(-3) ///
    showstats xlabel(15(5)40) palette(CET C6) alpha(75) ///
    xtitle("Pulmonary Artery Diameter (mm)") ///
    title("PA Diameter by Age Group - Male") ///
    saving(ridgeline_male, replace)

graph combine ridgeline_female.gph ridgeline_male.gph, ///
    title("Ridgeline Plot of PA Diameter by Age Group & Sex")
```

```
. * Ridgeline plot for Females (male == 0)
. ridgeline mpad if male == 0, by(age_decade) yline ylw(0.2) overlap(1.7)
///
>     ylc(blue) ylp(dot) labpos(right) bwid(1.2) laboffset(-3) ///
>     showstats xlabel(15(5)40) palette(CET C6) alpha(75) ///
>     xtitle("Pulmonary Artery Diameter (mm)") ///
>     title("PA Diameter by Age Group - Female") ///
>     saving(ridgeline_female, replace)

.
. * Ridgeline plot for Males (male == 1)
. ridgeline mpad if male == 1, by(age_decade) yline ylw(0.2) overlap(1.7)
///
>     ylc(red) ylp(dot) labpos(right) bwid(1.2) laboffset(-3) ///
>     showstats xlabel(15(5)40) palette(CET C6) alpha(75) ///
>     xtitle("Pulmonary Artery Diameter (mm)") ///
>     title("PA Diameter by Age Group - Male") ///
>     saving(ridgeline_male, replace)

.
. graph combine ridgeline_female.gph ridgeline_male.gph, ///
>     title("Ridgeline Plot of PA Diameter by Age Group & Sex")
(note:  named style i not found in class symbol, default attributes used)
(note:  named style i not found in class symbol, default attributes used)

.
```



larger PAs as you get older; true of both sexes.

In [9]:
```
%%stata
/* Run Bootstrap Quantile Regressions for the Full Population */
bsqreg mpad c.age, quantile(50) reps(500) // 50th Percentile
```

```
. /* Run Bootstrap Quantile Regressions for the Full Population */
. bsqreg mpad c.age, quantile(50) reps(500) // 50th Percentile
(fitting base model)

Bootstrap replications (500): .........10.........20.........30.........4
0.....
> ....50.........60.........70.........80.........90.........100.........11
0...
> ......120.........130.........140.........150.........160.........17
0........
> .180.........190.........200.........210.........220.........230.........2
40.
> ........250.........260.........270.........280.........290.........30
0......
> ...310.........320.........330.........340.........350.........36
0.........37
> 0.........380.........390.........400.........410.........420.........43
0....
> .....440.........450.........460.........470.........480.........49
0.........
> 500 done

Median regression, bootstrap(500) SEs              Number of obs =        9
90
  Raw sum of deviations  1665.45 (about 25.200001)
  Min sum of deviations 1591.973                   Pseudo R2     =     0.04
41

----------------------------------------------------------------------------
--
        mpad | Coefficient  Std. err.      t    P>|t|     [95% conf. interva
l]
-------------+--------------------------------------------------------------
--
         age |   .0636364   .0071022     8.96   0.000     .0496993    .07757
35
       _cons |   21.91818   .3954829    55.42   0.000     21.14209    22.694
26
----------------------------------------------------------------------------
--

.
```

In [10]:

```
%%stata

bsqreg mpad c.age, quantile(90) reps(500) // 90th Percentile
```

```
.
. bsqreg mpad c.age, quantile(90) reps(500) // 90th Percentile
(fitting base model)

Bootstrap replications (500): .........10.........20.........30.........4
0.....
> ....50.........60.........70.........80.........90.........100.........11
0...
> ......120.........130.........140.........150.........160.........17
0........
> .180.........190.........200.........210.........220.........230.........2
40.
> ........250.........260.........270.........280.........290.........30
0......
> ...310.........320.........330.........340.........350.........36
0.........37
> 0.........380.........390.........400.........410.........420.........43
0....
> .....440.........450.........460.........470.........480.........49
0.........
> 500 done

.9 Quantile regression, bootstrap(500) SEs          Number of obs =        9
90
  Raw sum of deviations   842.75 (about 30.9)
  Min sum of deviations 805.1974                    Pseudo R2     =     0.04
46

----------------------------------------------------------------------------
--
        mpad | Coefficient  Std. err.      t    P>|t|     [95% conf. interva
l]
-------------+--------------------------------------------------------------
--
         age |   .0913044   .0162558     5.62   0.000     .0594046    .12320
42
       _cons |    26.3087   .7865813    33.45   0.000     24.76513    27.852
26
----------------------------------------------------------------------------
--

.
```

In [11]:

```
%%stata

bsqreg mpad c.age, quantile(95) reps(500) // 95th Percentile
```

```
.
. bsqreg mpad c.age, quantile(95) reps(500) // 95th Percentile
(fitting base model)

Bootstrap replications (500): .........10.........20.........30.........4
0.....
> ....50.........60.........70.........80.........90.........100.........11
0...
> ......120.........130.........140.........150.........160.........17
0........
> .180.........190.........200.........210.........220.........230.........2
40.
> ........250.........260.........270.........280.........290.........30
0......
> ...310.........320.........330.........340.........350.........36
0.........37
> 0.........380.........390.........400.........410.........420.........43
0....
> .....440.........450.........460.........470.........480.........49
0.........
> 500 done

.95 Quantile regression, bootstrap(500) SEs         Number of obs =         9
90
  Raw sum of deviations  526.025 (about 33.599998)
  Min sum of deviations 497.4948                    Pseudo R2     =     0.05
42

------------------------------------------------------------------------------
--
        mpad | Coefficient  Std. err.      t    P>|t|     [95% conf. interva
l]
-------------+----------------------------------------------------------------
--
         age |   .0955224   .0200925     4.75   0.000     .0560935    .13495
13
       _cons |   27.89851   .9993441    27.92   0.000     25.93743    29.859
59
------------------------------------------------------------------------------
--

.
```

The preceding three quantile regressions ask how the 50th, 90th, and 95th percentile are expected to change as one gets older.

The 95th percentile is expected to get 0.096 mm bigger each year. The fact that this is more than the median suggests that the distribution widens with age.

In [12]:

```
%%stata

/* Extract Colors from Spectral Palette */
colorpalette Spectral, n(6) nograph
local color6 `"`r(p1)'"' // Reverse order
local color5 `"`r(p2)'"'
```

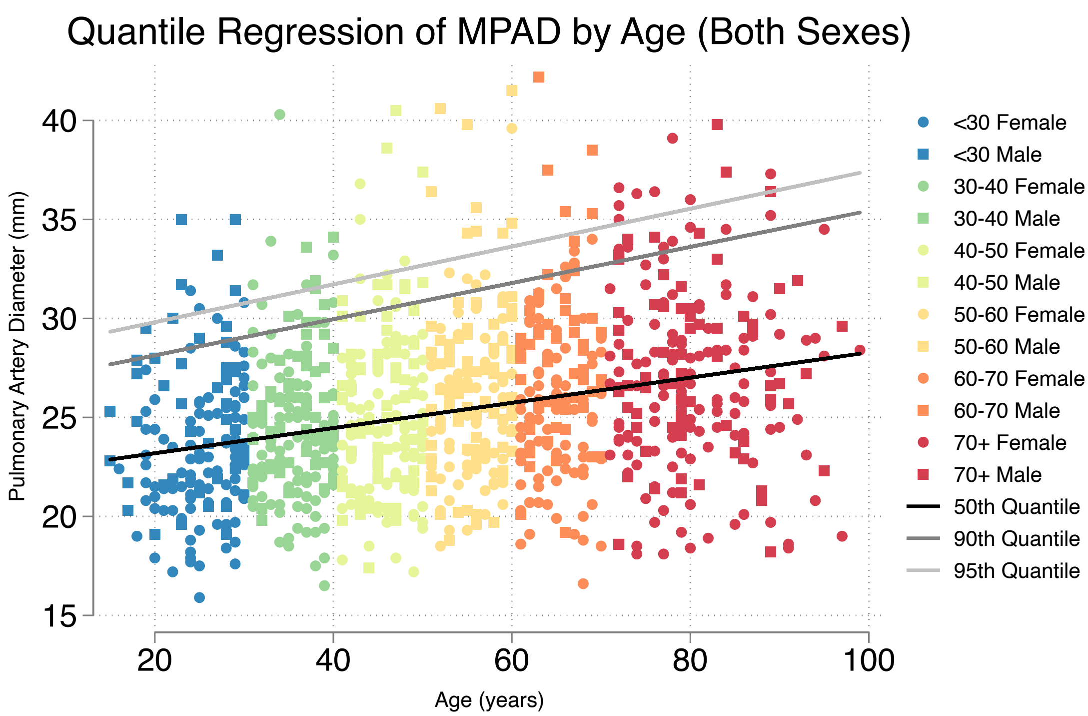
```
local color4 `"`r(p3)'"'
local color3 `"`r(p4)'"'
local color2 `"`r(p5)'"'
local color1 `"`r(p6)'"'

/* Run Quantile Regressions for the Full Population */
qreg mpad c.age, quantile(50) // 50th Percentile
predict mpad_q50_all, xb

qreg mpad c.age, quantile(90) // 90th Percentile
predict mpad_q90_all, xb

qreg mpad c.age, quantile(95) // 95th Percentile
predict mpad_q95_all, xb

/* Scatter Plot with Both Genders + Quantile Regression Lines */
twoway ///
    (scatter mpad age if age_decade == 0 & male == 0, mcolor("`color1'") msy
    (scatter mpad age if age_decade == 0 & male == 1, mcolor("`color1'") msy
    (scatter mpad age if age_decade == 1 & male == 0, mcolor("`color2'") msy
    (scatter mpad age if age_decade == 1 & male == 1, mcolor("`color2'") msy
    (scatter mpad age if age_decade == 2 & male == 0, mcolor("`color3'") msy
    (scatter mpad age if age_decade == 2 & male == 1, mcolor("`color3'") msy
    (scatter mpad age if age_decade == 3 & male == 0, mcolor("`color4'") msy
    (scatter mpad age if age_decade == 3 & male == 1, mcolor("`color4'") msy
    (scatter mpad age if age_decade == 4 & male == 0, mcolor("`color5'") msy
    (scatter mpad age if age_decade == 4 & male == 1, mcolor("`color5'") msy
    (scatter mpad age if age_decade == 5 & male == 0, mcolor("`color6'") msy
    (scatter mpad age if age_decade == 5 & male == 1, mcolor("`color6'") msy
    (line mpad_q50_all age, lcolor(black) lpattern(solid) lwidth(medthick))
    (line mpad_q90_all age, lcolor(gs8) lpattern(solid) lwidth(medthick)) ||
    (line mpad_q95_all age, lcolor(gs12) lpattern(solid) lwidth(medthick)),
    legend(order(1 "<30 Female" 2 "<30 Male" 3 "30-40 Female" 4 "30-40 Male"
                 5 "40-50 Female" 6 "40-50 Male" 7 "50-60 Female" 8 "50-60 M
                 9 "60-70 Female" 10 "60-70 Male" 11 "70+ Female" 12 "70+ Ma
                 13 "50th Quantile" 14 "90th Quantile" 15 "95th Quantile"))
    xtitle("Age (years)") ytitle("Pulmonary Artery Diameter (mm)") ///
    title("Quantile Regression of MPAD by Age (Both Sexes)") ///
    xlabel(, labsize(medlarge)) ylabel(, labsize(medlarge)) ///
    scheme(white_w3d)
```

```
.
. /* Extract Colors from Spectral Palette */
. colorpalette Spectral, n(6) nograph

. local color6 `"`r(p1)'"' // Reverse order

. local color5 `"`r(p2)'"'

. local color4 `"`r(p3)'"'

. local color3 `"`r(p4)'"'

. local color2 `"`r(p5)'"'

. local color1 `"`r(p6)'"'

.
. /* Run Quantile Regressions for the Full Population */
. qreg mpad c.age, quantile(50) // 50th Percentile

Iteration 1:  WLS sum of weighted deviations =  1598.5126

Iteration 1:  Sum of abs. weighted deviations =  1598.1065
Iteration 2:  Sum of abs. weighted deviations =  1592.1619
Iteration 3:  Sum of abs. weighted deviations =  1592.0304
Iteration 4:  Sum of abs. weighted deviations =  1591.9727

Median regression                                  Number of obs =       9
90
  Raw sum of deviations  1665.45 (about 25.200001)
  Min sum of deviations 1591.973                   Pseudo R2     =     0.04
41

----------------------------------------------------------------------------
--
        mpad | Coefficient  Std. err.      t    P>|t|     [95% conf. interva
l]
-------------+--------------------------------------------------------------
--
         age |   .0636364   .0079815     7.97   0.000     .0479737    .07929
91
       _cons |   21.91818    .446773    49.06   0.000     21.04144    22.794
91
----------------------------------------------------------------------------
--

. predict mpad_q50_all, xb

.
. qreg mpad c.age, quantile(90) // 90th Percentile

Iteration 1:  WLS sum of weighted deviations =  1418.7219

Iteration 1:  Sum of abs. weighted deviations =  1429.4705
Iteration 2:  Sum of abs. weighted deviations =  932.06999
Iteration 3:  Sum of abs. weighted deviations =  819.51355
```

```
Iteration 4:  Sum of abs. weighted deviations =  815.08121
Iteration 5:  Sum of abs. weighted deviations =     805.21
Iteration 6:  Sum of abs. weighted deviations =  805.20325
Iteration 7:  Sum of abs. weighted deviations =  805.19739

.9 Quantile regression                              Number of obs =        9
90
  Raw sum of deviations   842.75 (about 30.9)
  Min sum of deviations 805.1974                    Pseudo R2     =     0.04
46

----------------------------------------------------------------------------
--
        mpad | Coefficient  Std. err.      t    P>|t|     [95% conf. interva
l]
-------------+--------------------------------------------------------------
--
         age |   .0913044   .0148248     6.16   0.000     .0622126    .12039
62
       _cons |    26.3087   .8298327    31.70   0.000     24.68026    27.937
13
----------------------------------------------------------------------------
--

. predict mpad_q90_all, xb

.
. qreg mpad c.age, quantile(95) // 95th Percentile

Iteration 1:  WLS sum of weighted deviations =  1363.1293

Iteration 1:  Sum of abs. weighted deviations =  1366.4641
Iteration 2:  Sum of abs. weighted deviations =  859.05065
Iteration 3:  Sum of abs. weighted deviations =  513.28842
Iteration 4:  Sum of abs. weighted deviations =     498.22
Iteration 5:  Sum of abs. weighted deviations =  497.49485

.95 Quantile regression                             Number of obs =        9
90
  Raw sum of deviations  526.025 (about 33.599998)
  Min sum of deviations 497.4948                    Pseudo R2     =     0.05
42

----------------------------------------------------------------------------
--
        mpad | Coefficient  Std. err.      t    P>|t|     [95% conf. interva
l]
-------------+--------------------------------------------------------------
--
         age |   .0955224     .02062     4.63   0.000     .0550583    .13598
65
       _cons |   27.89851   1.154223    24.17   0.000      25.6335    30.163
52
----------------------------------------------------------------------------
--
```

```
. predict mpad_q95_all, xb

.
. /* Scatter Plot with Both Genders + Quantile Regression Lines */
. twoway ///
>     (scatter mpad age if age_decade == 0 & male == 0, mcolor("`color1'") m
sym
> bol(O)) || /// <30 years, Female
>     (scatter mpad age if age_decade == 0 & male == 1, mcolor("`color1'") m
sym
> bol(S)) || /// <30 years, Male
>     (scatter mpad age if age_decade == 1 & male == 0, mcolor("`color2'") m
sym
> bol(O)) || /// 30-40 years, Female
>     (scatter mpad age if age_decade == 1 & male == 1, mcolor("`color2'") m
sym
> bol(S)) || /// 30-40 years, Male
>     (scatter mpad age if age_decade == 2 & male == 0, mcolor("`color3'") m
sym
> bol(O)) || /// 40-50 years, Female
>     (scatter mpad age if age_decade == 2 & male == 1, mcolor("`color3'") m
sym
> bol(S)) || /// 40-50 years, Male
>     (scatter mpad age if age_decade == 3 & male == 0, mcolor("`color4'") m
sym
> bol(O)) || /// 50-60 years, Female
>     (scatter mpad age if age_decade == 3 & male == 1, mcolor("`color4'") m
sym
> bol(S)) || /// 50-60 years, Male
>     (scatter mpad age if age_decade == 4 & male == 0, mcolor("`color5'") m
sym
> bol(O)) || /// 60-70 years, Female
>     (scatter mpad age if age_decade == 4 & male == 1, mcolor("`color5'") m
sym
> bol(S)) || /// 60-70 years, Male
>     (scatter mpad age if age_decade == 5 & male == 0, mcolor("`color6'") m
sym
> bol(O)) || /// 70+ years, Female
>     (scatter mpad age if age_decade == 5 & male == 1, mcolor("`color6'") m
sym
> bol(S)) || /// 70+ years, Male
>     (line mpad_q50_all age, lcolor(black) lpattern(solid) lwidth(medthic
k)) |
> | /// 50th quantile
>     (line mpad_q90_all age, lcolor(gs8) lpattern(solid) lwidth(medthick))
||
> /// 90th quantile
>     (line mpad_q95_all age, lcolor(gs12) lpattern(solid) lwidth(medthic
k)), /
> // 95th quantile
>     legend(order(1 "<30 Female" 2 "<30 Male" 3 "30-40 Female" 4 "30-40 Mal
e"
> ///
>                 5 "40-50 Female" 6 "40-50 Male" 7 "50-60 Female" 8 "50-60
Ma
> le" ///
```

```
>                  9 "60-70 Female" 10 "60-70 Male" 11 "70+ Female" 12 "70+
Mal
> e" ///
>                  13 "50th Quantile" 14 "90th Quantile" 15 "95th Quantil
e")) /
> //
>     xtitle("Age (years)") ytitle("Pulmonary Artery Diameter (mm)") ///
>     title("Quantile Regression of MPAD by Age (Both Sexes)") ///
>     xlabel(, labsize(medlarge)) ylabel(, labsize(medlarge)) ///
>     scheme(white_w3d)

.
```



confirmed visually.

In [13]:

```
%%stata

//this code gives confidence intervals - all signficant

/* Run Bootstrap Quantile Regressions for the Men */
bsqreg mpad c.age if male == 1, quantile(50) reps(500) // 50th Percentile
bsqreg mpad c.age if male == 1, quantile(90) reps(500) // 90th Percentile
bsqreg mpad c.age if male == 1, quantile(95) reps(500) // 95th Percentile

/* Run Bootstrap Quantile Regressions for the Women */
bsqreg mpad c.age if male == 0, quantile(50) reps(500) // 50th Percentile
bsqreg mpad c.age if male == 0, quantile(90) reps(500) // 90th Percentile
bsqreg mpad c.age if male == 0, quantile(95) reps(500) // 95th Percentile
```

```
.
. //this code gives confidence intervals - all signficant
.
. /* Run Bootstrap Quantile Regressions for the Men */
. bsqreg mpad c.age if male == 1, quantile(50) reps(500) // 50th Percentile
(fitting base model)

Bootstrap replications (500): .........10.........20.........30.........4
0.....
> ....50.........60.........70.........80.........90.........100.........11
0...
> ......120.........130.........140.........150.........160.........17
0.......
> .180.........190.........200.........210.........220.........230.........2
40.
> ........250.........260.........270.........280.........290.........30
0......
> ...310.........320.........330.........340.........350.........36
0.........37
> 0.........380.........390.........400.........410.........420.........43
0....
> .....440.........450.........460.........470.........480.........49
0.........
> 500 done

Median regression, bootstrap(500) SEs              Number of obs =         3
70
  Raw sum of deviations      629 (about 26.4)
  Min sum of deviations   615.95                   Pseudo R2     =     0.02
07

----------------------------------------------------------------------------
--
        mpad | Coefficient  Std. err.      t    P>|t|     [95% conf. interva
l]
-------------+--------------------------------------------------------------
--
         age |        .05   .0185639     2.69   0.007     .0134954    .08650
47
       _cons |      23.85   .9334349    25.55   0.000     22.01446    25.685
53
----------------------------------------------------------------------------
--

. bsqreg mpad c.age if male == 1, quantile(90) reps(500) // 90th Percentile
(fitting base model)

Bootstrap replications (500): .........10.........20.........30.........4
0.....
> ....50.........60.........70.........80.........90.........100.........11
0...
> ......120.........130.........140.........150.........160.........17
0.......
> .180.........190.........200.........210.........220.........230.........2
40.
> ........250.........260.........270.........280.........290......x..30
```

```
0......
> ...310........320........330........340........350........36
0........37
> 0........380........390........400........410........420........43
0....
> .....440........450........460........470........x........49
0........50
> 0 done
..x: Error occurred when bsreg executed qreg.

.9 Quantile regression, bootstrap(500) SEs          Number of obs =        3
70
  Raw sum of deviations   327.46 (about 31.799999)
  Min sum of deviations 317.3634                    Pseudo R2     =     0.03
08

------------------------------------------------------------------------------
--
        mpad | Coefficient  Std. err.      t    P>|t|     [95% conf. interva
l]
-------------+----------------------------------------------------------------
--
         age |   .0758621   .0274477     2.76   0.006     .0218882     .1298
36
       _cons |   28.05862    1.34574    20.85   0.000     25.41231    30.704
93
------------------------------------------------------------------------------
--

. bsqreg mpad c.age if male == 1, quantile(95) reps(500) // 95th Percentile
(fitting base model)

Bootstrap replications (500): .........10.........20.........30.........4
0.....
> ....50.........60.........70.........80.........90.........100.........11
0...
> ......120........130........140........150........160........17
0.......
> .180........190........200........210........220........230........2
40.
> ........250........260........270........280........290........30
0......
> ...310........320........330........340........350........36
0........37
> 0........380........390........400........410........420........43
0....
> .....440........450........460........470........480........49
0........
> 500 done

.95 Quantile regression, bootstrap(500) SEs         Number of obs =        3
70
  Raw sum of deviations   205.18 (about 34.799999)
  Min sum of deviations 199.6834                    Pseudo R2     =     0.02
68
```

```
------------------------------------------------------------------------------
--
        mpad | Coefficient  Std. err.      t    P>|t|     [95% conf. interva
l]
-------------+----------------------------------------------------------------
--
         age |   .0620691    .041402     1.50   0.135    -.0193451    .14348
33
       _cons |   31.30344   2.288345    13.68   0.000     26.80357    35.803
32
------------------------------------------------------------------------------
--

.
. /* Run Bootstrap Quantile Regressions for the Women */
. bsqreg mpad c.age if male == 0, quantile(50) reps(500) // 50th Percentile
(fitting base model)

Bootstrap replications (500): .........10.........20.........30.........4
0.....
> ....50.........60.........70.........80.........90.........100.........11
0...
> ......120.........130.........140.........150.........160.........17
0........
> .180.........190.........200.........210.........220.........230.........2
40.
> ........250.........260.........270.........280.........290.........30
0......
> ...310.........320.........330.........340.........350.........36
0.........37
> 0.........380.........390.........400.........410.........420.........43
0....
> .....440.........450.........460.........470.........480.........49
0........
> 500 done

Median regression, bootstrap(500) SEs              Number of obs =       6
20
  Raw sum of deviations   995.25 (about 24.5)
  Min sum of deviations 940.5036                   Pseudo R2     =    0.05
50

------------------------------------------------------------------------------
--
        mpad | Coefficient  Std. err.      t    P>|t|     [95% conf. interva
l]
-------------+----------------------------------------------------------------
--
         age |   .0690477   .0100627     6.86   0.000     .0492865    .08880
89
       _cons |   21.17619   .5143339    41.17   0.000     20.16613    22.186
24
------------------------------------------------------------------------------
--

. bsqreg mpad c.age if male == 0, quantile(90) reps(500) // 90th Percentile
```

```
(fitting base model)

Bootstrap replications (500): .........10.........20.........30.........4
0.....
> ....50.........60.........70.........80.........90.........100.........11
0...
> ......120.........130.........140.........150.........160.........17
0........
> .180.........190.........200.........210.........220.........230.........2
40.
> ........250.........260.........270.........280.........290.........30
0......
> ...310.........320.........330.........340.........350.........36
0.........37
> 0.........380.........390.........400.........410.........420.........43
0....
> .....440.........450.........460.........470.........480.........49
0.........
> 500 done

.9 Quantile regression, bootstrap(500) SEs          Number of obs =        6
20
  Raw sum of deviations   505.79 (about 30.5)
  Min sum of deviations 472.1886                    Pseudo R2     =     0.06
64

------------------------------------------------------------------------------
--
        mpad | Coefficient  Std. err.      t    P>|t|     [95% conf. interva
l]
-------------+----------------------------------------------------------------
--
         age |   .1095238   .0243154     4.50   0.000     .0617729    .15727
46
       _cons |   24.38572   1.290648    18.89   0.000     21.85113     26.92
03
------------------------------------------------------------------------------
--

. bsqreg mpad c.age if male == 0, quantile(95) reps(500) // 95th Percentile
(fitting base model)

Bootstrap replications (500): .........10.........20.........30.........4
0.....
> ....50.........60.........70.........80.........90.........100.........11
0...
> ......120.........130.........140.........150.........160.........17
0........
> .180.........190.........200.........210.........220.........230.........2
40.
> ........250.........260.........270.........280.........290.........30
0......
> ...310.........320.........330.........340.........350.........36
0.........37
> 0.........380.........390.........400.........410.........420.........43
0....
```

```
> .....440.........450.........460.........470.........480.........49
0.........
> 500 done

.95 Quantile regression, bootstrap(500) SEs        Number of obs =         6
20
  Raw sum of deviations  309.645 (about 32.200001)
  Min sum of deviations 283.3593                    Pseudo R2     =     0.08
49

----------------------------------------------------------------------------
--
        mpad | Coefficient  Std. err.      t    P>|t|     [95% conf. interva
l]
-------------+--------------------------------------------------------------
--
         age |   .0952381   .0224581     4.24   0.000     .0511346    .13934
16
       _cons |   26.77143   1.316143    20.34   0.000     24.18677    29.356
08
----------------------------------------------------------------------------
--

.
```

In [14]:
```
%%stata

/* -----
Sex-Stratified Quantile Regression Scatter Plots
-----*/

/* Extract Colors from Spectral Palette */
colorpalette Spectral, n(6) nograph
local color6 `"`r(p1)'"' // Reverse order
local color5 `"`r(p2)'"'
local color4 `"`r(p3)'"'
local color3 `"`r(p4)'"'
local color2 `"`r(p5)'"'
local color1 `"`r(p6)'"'

/* -----
Generate Quantile Regression Predictions for Males
-----*/
qreg mpad c.age if male == 1, quantile(50) // Male
predict mpad_q50_male if male == 1, xb
qreg mpad c.age if male == 1, quantile(90) // Male
predict mpad_q90_male if male == 1, xb
qreg mpad c.age if male == 1, quantile(95) // Male
predict mpad_q95_male if male == 1, xb

/* -----
Male Scatter Plot
-----*/
twoway ///
    (scatter mpad age if age_decade == 0 & male == 1, mcolor("`color1'") msy
    (scatter mpad age if age_decade == 1 & male == 1, mcolor("`color2'") msy
```

```
    (scatter mpad age if age_decade == 2 & male == 1, mcolor("`color3'") msy
    (scatter mpad age if age_decade == 3 & male == 1, mcolor("`color4'") msy
    (scatter mpad age if age_decade == 4 & male == 1, mcolor("`color5'") msy
    (scatter mpad age if age_decade == 5 & male == 1, mcolor("`color6'") msy
    (line mpad_q50_male age, lcolor(black) lpattern(solid) lwidth(medthick))
    (line mpad_q90_male age, lcolor(gs8) lpattern(solid) lwidth(medthick)) |
    (line mpad_q95_male age, lcolor(gs12) lpattern(solid) lwidth(medthick)),
    ///
    legend(order(7 "50th Quantile" 8 "90th Quantile" 9 "95th Quantile") pos(
    xtitle("Age (years)") ytitle("Pulmonary Artery Diameter (mm)", margin(me
    title("Men") ///
    xlabel(, labsize(medlarge)) ylabel(15(5)40, nogrid labsize(medlarge)) //
    scheme(white_w3d) yscale(range(15 40)) ///
    saving(male_qreg_plot, replace)

/* -----
Generate Quantile Regression Predictions for Females
-----*/
qreg mpad c.age if male == 0, quantile(50) // Female
predict mpad_q50_female if male == 0, xb
qreg mpad c.age if male == 0, quantile(90) // Female
predict mpad_q90_female if male == 0, xb
qreg mpad c.age if male == 0, quantile(95) // Female
predict mpad_q95_female if male == 0, xb

/* -----
Female Scatter Plot
-----*/
twoway ///
    (scatter mpad age if age_decade == 0 & male == 0, mcolor("`color1'") msy
    (scatter mpad age if age_decade == 1 & male == 0, mcolor("`color2'") msy
    (scatter mpad age if age_decade == 2 & male == 0, mcolor("`color3'") msy
    (scatter mpad age if age_decade == 3 & male == 0, mcolor("`color4'") msy
    (scatter mpad age if age_decade == 4 & male == 0, mcolor("`color5'") msy
    (scatter mpad age if age_decade == 5 & male == 0, mcolor("`color6'") msy
    (line mpad_q50_female age, lcolor(black) lpattern(solid) lwidth(medthick
    (line mpad_q90_female age, lcolor(gs8) lpattern(solid) lwidth(medthick))
    (line mpad_q95_female age, lcolor(gs12) lpattern(solid) lwidth(medthick)
    ///
    legend(order(7 "50th Quantile" 8 "90th Quantile" 9 "95th Quantile") pos(
    xtitle("Age (years)") ytitle("Pulmonary Artery Diameter (mm)", margin(me
    title("Women") ///
    xlabel(, labsize(medlarge)) ylabel(15(5)40, nogrid labsize(medlarge)) //
    scheme(white_w3d) yscale(range(15 40)) ///
    saving(female_qreg_plot, replace)

/* -----
Combine the Two Plots into a Single Figure
-----*/
graph combine male_qreg_plot.gph female_qreg_plot.gph, ///
    title("Sex-Stratified Quantile Regression of MPAD by Age") ///
    ycommon xcommon ///
    iscale(*1.2) ///
    rows(1) ///
    graphregion(margin(2 2 2 2)) ///
```

```
    xsize(9) ysize(5)
```

```
.
. /* -----
> Sex-Stratified Quantile Regression Scatter Plots
> -----*/
.
. /* Extract Colors from Spectral Palette */
. colorpalette Spectral, n(6) nograph

. local color6 `"`r(p1)'"' // Reverse order

. local color5 `"`r(p2)'"'

. local color4 `"`r(p3)'"'

. local color3 `"`r(p4)'"'

. local color2 `"`r(p5)'"'

. local color1 `"`r(p6)'"'

.
. /* -----
> Generate Quantile Regression Predictions for Males
> -----*/
. qreg mpad c.age if male == 1, quantile(50) // Male

Iteration 1:  WLS sum of weighted deviations =  616.82051

Iteration 1:  Sum of abs. weighted deviations =  616.81601
Iteration 2:  Sum of abs. weighted deviations =  616.59395
Iteration 3:  Sum of abs. weighted deviations =  615.95001

Median regression                                  Number of obs =        3
70
  Raw sum of deviations      629 (about 26.4)
  Min sum of deviations   615.95                   Pseudo R2     =     0.02
07

----------------------------------------------------------------------------
--
        mpad | Coefficient  Std. err.      t    P>|t|     [95% conf. interva
l]
-------------+--------------------------------------------------------------
--
         age |        .05   .0156922     3.19   0.002     .0191425    .08085
76
       _cons |      23.85   .8761326    27.22   0.000     22.12714    25.572
85
----------------------------------------------------------------------------
--

. predict mpad_q50_male if male == 1, xb
(620 missing values generated)

. qreg mpad c.age if male == 1, quantile(90) // Male
```

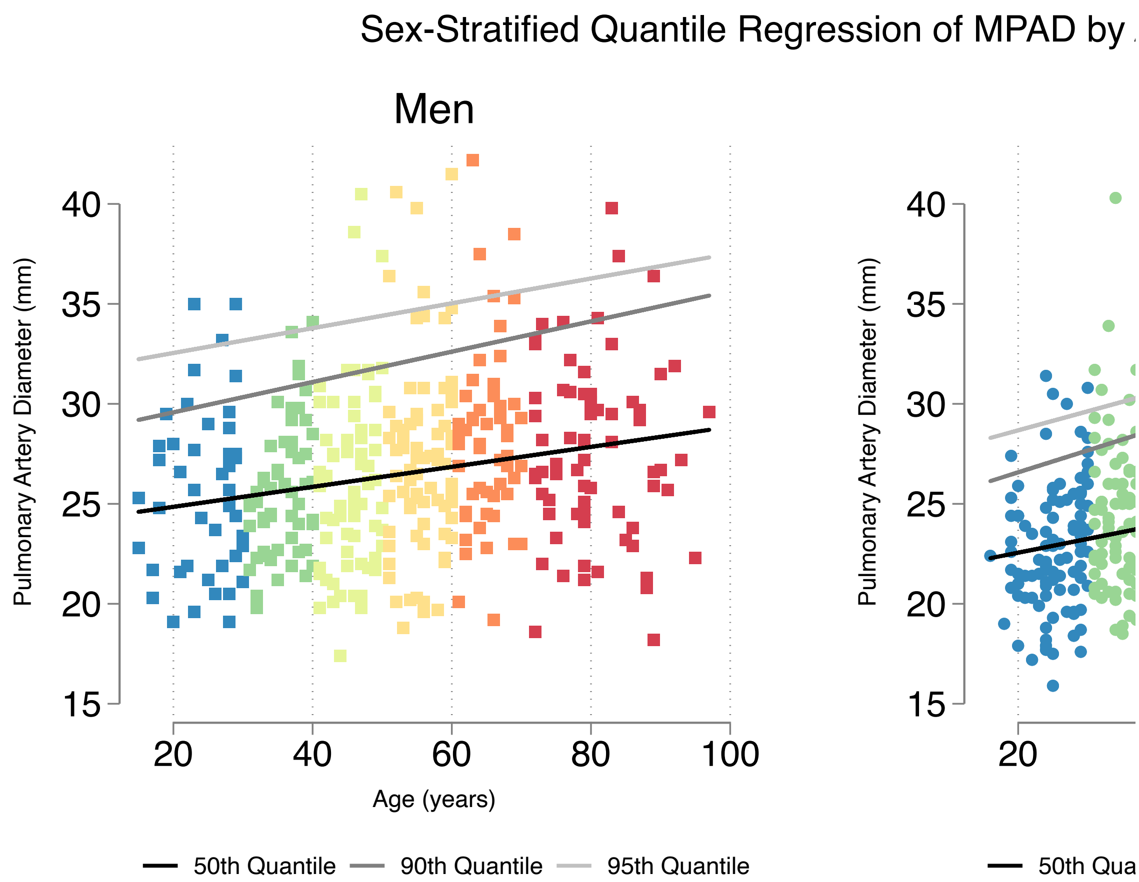
```
Iteration 1:  WLS sum of weighted deviations =  516.95839

Iteration 1:  Sum of abs. weighted deviations =   521.0177
Iteration 2:  Sum of abs. weighted deviations =    418.688
Iteration 3:  Sum of abs. weighted deviations =  327.67707
Iteration 4:  Sum of abs. weighted deviations =  317.36345

.9 Quantile regression                             Number of obs =        3
70
  Raw sum of deviations   327.46 (about 31.799999)
  Min sum of deviations 317.3634                   Pseudo R2     =     0.03
08

----------------------------------------------------------------------------
--
        mpad | Coefficient  Std. err.      t    P>|t|     [95% conf. interva
l]
-------------+--------------------------------------------------------------
--
         age |   .0758621   .0306207     2.48   0.014     .0156486    .13607
56
       _cons |   28.05862   1.709631    16.41   0.000     24.69675    31.420
49
----------------------------------------------------------------------------
--

. predict mpad_q90_male if male == 1, xb
(620 missing values generated)

. qreg mpad c.age if male == 1, quantile(95) // Male

Iteration 1:  WLS sum of weighted deviations =  488.35735

Iteration 1:  Sum of abs. weighted deviations =  488.41762
Iteration 2:  Sum of abs. weighted deviations =  252.46043
Iteration 3:  Sum of abs. weighted deviations =  226.43999
Iteration 4:  Sum of abs. weighted deviations =  201.09818
Iteration 5:  Sum of abs. weighted deviations =  200.04639
Iteration 6:  Sum of abs. weighted deviations =   199.8393
Iteration 7:  Sum of abs. weighted deviations =  199.68345

.95 Quantile regression                            Number of obs =        3
70
  Raw sum of deviations   205.18 (about 34.799999)
  Min sum of deviations 199.6834                   Pseudo R2     =     0.02
68

----------------------------------------------------------------------------
--
        mpad | Coefficient  Std. err.      t    P>|t|     [95% conf. interva
l]
-------------+--------------------------------------------------------------
--
         age |   .0620691   .0486485     1.28   0.203    -.033595    .15773
31
       _cons |   31.30344   2.716171    11.52   0.000     25.96228    36.644
```

```
61
----------------------------------------------------------------------------
--

. predict mpad_q95_male if male == 1, xb
(620 missing values generated)

.
. /* -----
> Male Scatter Plot
> -----*/
. twoway ///
>     (scatter mpad age if age_decade == 0 & male == 1, mcolor("`color1'") m
sym
> bol(S)) || /// <30 years
>     (scatter mpad age if age_decade == 1 & male == 1, mcolor("`color2'") m
sym
> bol(S)) || /// 30-40 years
>     (scatter mpad age if age_decade == 2 & male == 1, mcolor("`color3'") m
sym
> bol(S)) || /// 40-50 years
>     (scatter mpad age if age_decade == 3 & male == 1, mcolor("`color4'") m
sym
> bol(S)) || /// 50-60 years
>     (scatter mpad age if age_decade == 4 & male == 1, mcolor("`color5'") m
sym
> bol(S)) || /// 60-70 years
>     (scatter mpad age if age_decade == 5 & male == 1, mcolor("`color6'") m
sym
> bol(S)) || /// 70+ years
>     (line mpad_q50_male age, lcolor(black) lpattern(solid) lwidth(medthic
k))
> || /// 50th percentile line
>     (line mpad_q90_male age, lcolor(gs8) lpattern(solid) lwidth(medthick))
||
>  /// 90th percentile line
>     (line mpad_q95_male age, lcolor(gs12) lpattern(solid) lwidth(medthic
k)),
> /// 95th percentile line
>     ///
>     legend(order(7 "50th Quantile" 8 "90th Quantile" 9 "95th Quantile") po
s(6
> ) row(1)) /// Only quantile labels in legend
>     xtitle("Age (years)") ytitle("Pulmonary Artery Diameter (mm)", margin
(med
> large)) ///
>     title("Men") ///
>     xlabel(, labsize(medlarge)) ylabel(15(5)40, nogrid labsize(medlarge))
///
>  Adjusted Y-axis range
>     scheme(white_w3d) yscale(range(15 40)) ///
>     saving(male_qreg_plot, replace)
file male_qreg_plot.gph saved

.
. /* -----
```

```
> Generate Quantile Regression Predictions for Females
> -----*/
. qreg mpad c.age if male == 0, quantile(50) // Female

Iteration 1:  WLS sum of weighted deviations =  942.48547

Iteration 1:  Sum of abs. weighted deviations =  942.91875
Iteration 2:  Sum of abs. weighted deviations =  940.53838
Iteration 3:  Sum of abs. weighted deviations =  940.50455
Iteration 4:  Sum of abs. weighted deviations =  940.50357

Median regression                                 Number of obs =        6
20
  Raw sum of deviations   995.25 (about 24.5)
  Min sum of deviations 940.5036                  Pseudo R2     =     0.05
50

----------------------------------------------------------------------------
--
        mpad | Coefficient  Std. err.      t    P>|t|     [95% conf. interva
l]
-------------+--------------------------------------------------------------
--
         age |   .0690477   .0098963     6.98   0.000     .0496132    .08848
22
       _cons |   21.17619   .5547992    38.17   0.000     20.08667    22.265
71
----------------------------------------------------------------------------
--

. predict mpad_q50_female if male == 0, xb
(370 missing values generated)

. qreg mpad c.age if male == 0, quantile(90) // Female

Iteration 1:  WLS sum of weighted deviations =  861.62076

Iteration 1:  Sum of abs. weighted deviations =   862.6474
Iteration 2:  Sum of abs. weighted deviations =  631.42063
Iteration 3:  Sum of abs. weighted deviations =  478.09688
Iteration 4:  Sum of abs. weighted deviations =  472.40414
Iteration 5:  Sum of abs. weighted deviations =  472.18857

.9 Quantile regression                            Number of obs =        6
20
  Raw sum of deviations   505.79 (about 30.5)
  Min sum of deviations 472.1886                  Pseudo R2     =     0.06
64

----------------------------------------------------------------------------
--
        mpad | Coefficient  Std. err.      t    P>|t|     [95% conf. interva
l]
-------------+--------------------------------------------------------------
--
         age |   .1095238     .01843     5.94   0.000     .0733307    .14571
```

```
69
      _cons |   24.38572   1.033209    23.60   0.000     22.35669    26.414
74
----------------------------------------------------------------------------
--

. predict mpad_q90_female if male == 0, xb
(370 missing values generated)

. qreg mpad c.age if male == 0, quantile(95) // Female

Iteration 1:  WLS sum of weighted deviations =  838.61546

Iteration 1:  Sum of abs. weighted deviations =  839.25782
Iteration 2:  Sum of abs. weighted deviations =  374.71483
Iteration 3:  Sum of abs. weighted deviations =  317.24902
Iteration 4:  Sum of abs. weighted deviations =  285.19244
Iteration 5:  Sum of abs. weighted deviations =     283.44
Iteration 6:  Sum of abs. weighted deviations =  283.35929

.95 Quantile regression                            Number of obs =       6
20
  Raw sum of deviations  309.645 (about 32.200001)
  Min sum of deviations 283.3593                   Pseudo R2     =    0.08
49

----------------------------------------------------------------------------
--
        mpad | Coefficient  Std. err.      t    P>|t|     [95% conf. interva
l]
-------------+--------------------------------------------------------------
--
         age |   .0952381   .0199511     4.77   0.000     .0560579    .13441
83
       _cons |   26.77143   1.118483    23.94   0.000     24.57494    28.967
92
----------------------------------------------------------------------------
--

. predict mpad_q95_female if male == 0, xb
(370 missing values generated)

.
. /* -----
> Female Scatter Plot
> -----*/
. twoway ///
>     (scatter mpad age if age_decade == 0 & male == 0, mcolor("`color1'") m
sym
> bol(O)) || /// <30 years
>     (scatter mpad age if age_decade == 1 & male == 0, mcolor("`color2'") m
sym
> bol(O)) || /// 30-40 years
>     (scatter mpad age if age_decade == 2 & male == 0, mcolor("`color3'") m
sym
> bol(O)) || /// 40-50 years
```

```
>     (scatter mpad age if age_decade == 3 & male == 0, mcolor("`color4'") m
sym
> bol(O)) || /// 50-60 years
>     (scatter mpad age if age_decade == 4 & male == 0, mcolor("`color5'") m
sym
> bol(O)) || /// 60-70 years
>     (scatter mpad age if age_decade == 5 & male == 0, mcolor("`color6'") m
sym
> bol(O)) || /// 70+ years
>     (line mpad_q50_female age, lcolor(black) lpattern(solid) lwidth(medthi
ck)
> ) || /// 50th percentile line
>     (line mpad_q90_female age, lcolor(gs8) lpattern(solid) lwidth(medthic
k))
> || /// 90th percentile line
>     (line mpad_q95_female age, lcolor(gs12) lpattern(solid) lwidth(medthic
k))
> , /// 95th percentile line
>     ///
>     legend(order(7 "50th Quantile" 8 "90th Quantile" 9 "95th Quantile") po
s(6
> ) row(1)) /// Only quantile labels in legend
>     xtitle("Age (years)") ytitle("Pulmonary Artery Diameter (mm)", margin
(med
> large)) ///
>     title("Women") ///
>     xlabel(, labsize(medlarge)) ylabel(15(5)40, nogrid labsize(medlarge))
///
>  Adjusted Y-axis range
>     scheme(white_w3d) yscale(range(15 40)) ///
>     saving(female_qreg_plot, replace)
file female_qreg_plot.gph saved

.
. /* -----
> Combine the Two Plots into a Single Figure
> -----*/
. graph combine male_qreg_plot.gph female_qreg_plot.gph, ///
>     title("Sex-Stratified Quantile Regression of MPAD by Age") ///
>     ycommon xcommon ///
>     iscale(*1.2) ///
>     rows(1) ///
>     graphregion(margin(2 2 2 2)) ///
>     xsize(9) ysize(5)

.
.
```



And true in both sexes

Survival analysis

Now we move to assessing the relation between PA size and mortality.

In [15]:
```
%%stata

stset lastfollowupyear, failure(death==1)
```

```
.
. stset lastfollowupyear, failure(death==1)

Survival-time data settings

         Failure event: death==1
Observed time interval: (0, lastfollowupyear]
     Exit on or before: failure

--------------------------------------------------------------------------
        990  total observations
         78  observations end on or before enter()
--------------------------------------------------------------------------
        912  observations remaining, representing
        263  failures in single-record/single-failure data
  6,981.359  total analysis time at risk and under observation
                                                At risk from t =         0
                                     Earliest observed entry t =         0
                                          Last observed exit t =   12.4819

.
```

In [ ]:

```

```

Non-age,sex standardized approach:

mirrors the CHEST article

In [16]:

```
%%stata

* Generate tertile cutoffs
xtile mpad_tertile = mpad, n(3)  // Divides into 3 equal-sized groups

* Label the tertile groups
label define mpad_tertile_lbl 1 "MPAD Tertile 1" 2 "MPAD Tertile 2" 3 "MPAD
label values mpad_tertile mpad_tertile_lbl

* Verify the tertile distribution
tab mpad_tertile

sts test mpad_tertile, logrank
```

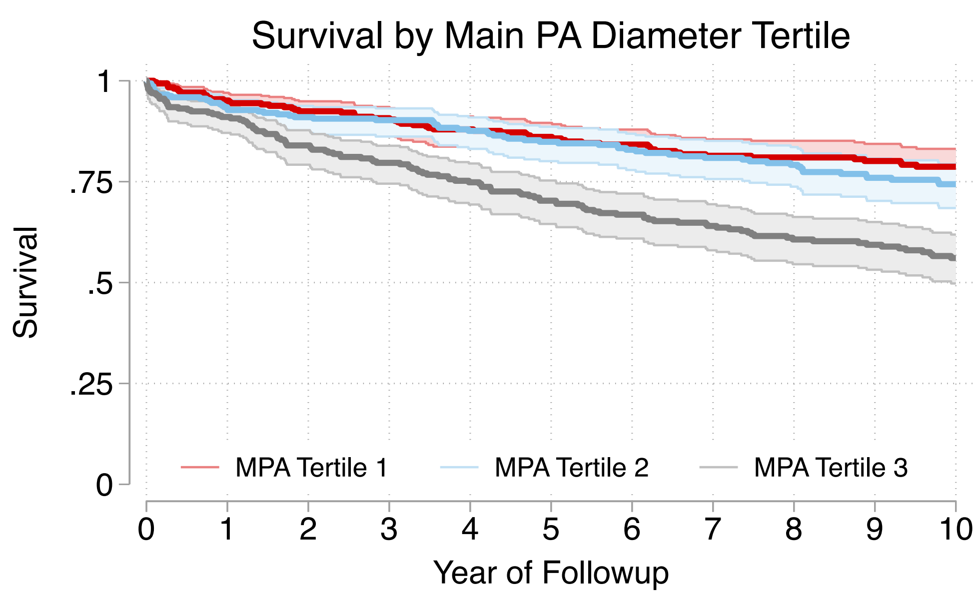
```
.
. * Generate tertile cutoffs
. xtile mpad_tertile = mpad, n(3)  // Divides into 3 equal-sized groups

.
. * Label the tertile groups
. label define mpad_tertile_lbl 1 "MPAD Tertile 1" 2 "MPAD Tertile 2" 3 "MPA
D T
> ertile 3"

. label values mpad_tertile mpad_tertile_lbl

.
. * Verify the tertile distribution
. tab mpad_tertile

3 quantiles of |
          mpad |      Freq.     Percent        Cum.
---------------+-----------------------------------
MPAD Tertile 1 |        336       33.94       33.94
MPAD Tertile 2 |        336       33.94       67.88
MPAD Tertile 3 |        318       32.12      100.00
---------------+-----------------------------------
         Total |        990      100.00

.
. sts test mpad_tertile, logrank

        Failure _d: death==1
  Analysis time _t: lastfollowupyear

Equality of survivor functions
Log-rank test

               |  Observed       Expected
mpad_tertile   |    events         events
---------------+-------------------------
MPAD Tertile 1 |        63          94.96
MPAD Tertile 2 |        75          89.19
MPAD Tertile 3 |       125          78.85
---------------+-------------------------
         Total |       263         263.00

                         chi2(2) =  40.07
                         Pr>chi2 = 0.0000

.
```

Unadjusted size predicts mortality

In [17]:
```
%%stata

sts graph, by(mpad_tertile) tmax(10) ci surv ///
 plotopts(lwidth(thick)) ///
 risktable(0(1)10, order(1 "MPA Tertile 1" 2 "MPA Tertile 2" 3 "MPA Tertile
 xlabel(0(1)10, labsize(medlarge)) ///
```

```
ylabel(,labsize(medlarge)) ///
xtitle("Year of Followup", size(medlarge)) ///
ytitle("Survival", size(medlarge)) ///
legend(order(2 "MPA Tertile 1" 4 "MPA Tertile 2" 6 "MPA Tertile 3") positic
title("Survival by Main PA Diameter Tertile", size(large))
```

```
.
. sts graph, by(mpad_tertile) tmax(10) ci surv ///
>  plotopts(lwidth(thick)) ///
>  risktable(0(1)10, order(1 "MPA Tertile 1" 2 "MPA Tertile 2" 3 "MPA Tertil
e 3
> ") title("Number Eligible", size(medium)) size(medsmall)) ///
>  xlabel(0(1)10, labsize(medlarge)) ///
>  ylabel(,labsize(medlarge)) ///
>  xtitle("Year of Followup", size(medlarge)) ///
>  ytitle("Survival", size(medlarge)) ///
>  legend(order(2 "MPA Tertile 1" 4 "MPA Tertile 2" 6 "MPA Tertile 3") posit
ion
> (6) ring(0) rows(1) size(medsmall)) ///
>  title("Survival by Main PA Diameter Tertile", size(large))

        Failure _d: death==1
  Analysis time _t: lastfollowupyear
(note:  named style i not found in class symbol, default attributes used)

.
```



Number Eligible

	0	1	2	3	4	5	6	7	8	9	10
MPA Tertile 1	313	290	274	263	240	228	213	203	194	181	148
MPA Tertile 2	302	265	254	243	228	219	208	195	178	159	133
MPA Tertile 3	297	260	237	216	199	183	170	157	141	133	111

mostly highest tertile PAs that die. (suspect age is confounding this)

In [18]:

```
%%stata

stcox i.mpad_tertile
```

```
estat concordance
stbrier i.mpad_tertile, bt(12.4819)
```

```
.
. stcox i.mpad_tertile

        Failure _d: death==1
  Analysis time _t: lastfollowupyear

Iteration 0:  Log likelihood = -1693.3646
Iteration 1:  Log likelihood = -1674.8791
Iteration 2:  Log likelihood = -1674.5942
Iteration 3:  Log likelihood = -1674.5942
Refining estimates:
Iteration 0:  Log likelihood = -1674.5942

Cox regression with Breslow method for ties

No. of subjects =        912                           Number of obs =    9
12
No. of failures =        263
Time at risk    = 6,981.3588
                                                        LR chi2(2)    =  37.
54
Log likelihood = -1674.5942                             Prob > chi2   = 0.00
00

-----------------------------------------------------------------------------
--
          _t | Haz. ratio   Std. err.      z    P>|z|     [95% conf. interva
l]
-------------+---------------------------------------------------------------
--
mpad_tertile |
MPAD Tert..  |    1.26738   .2166131     1.39   0.166     .9066184    1.7716
96
MPAD Tert..  |   2.391676   .3696904     5.64   0.000     1.766562    3.2379
92
-----------------------------------------------------------------------------
--

. estat concordance

        Failure _d: death==1
  Analysis time _t: lastfollowupyear

Harrell's C concordance statistic

  Number of subjects (N)             =      912
  Number of comparison pairs (P)     =   174623
  Number of orderings as expected (E) =    75942
  Number of tied predictions (T)     =    56982

          Harrell's C = (E + T/2) / P =   0.5980
                            Somers' D =   0.1961

. stbrier i.mpad_tertile, bt(12.4819)

Mean estimation                      Number of obs   =       990
```

```
------------------------------------------------------------------
                |       Mean    Std. Err.    [95% Conf. Interval]
----------------+-------------------------------------------------
    Brier score |   4.696547     .5729393    3.572231    5.820863
------------------------------------------------------------------

.
```

Harrell's C = 0.5980 .... not great prediction accuracy using just tertile

Adding age and sex as predictors to the model:

In [19]:
```
%%stata

stcox i.mpad_tertile c.age i.male
estat concordance
stbrier i.mpad_tertile, bt(12.4819)
```

```
.
. stcox i.mpad_tertile c.age i.male

        Failure _d: death==1
  Analysis time _t: _t

Iteration 0:  Log likelihood = -1693.3646
Iteration 1:  Log likelihood =  -1570.436
Iteration 2:  Log likelihood = -1569.7309
Iteration 3:  Log likelihood = -1569.7305
Refining estimates:
Iteration 0:  Log likelihood = -1569.7305

Cox regression with Breslow method for ties

No. of subjects =        912                           Number of obs =    9
12
No. of failures =        263
Time at risk    = 6,981.3588
                                                        LR chi2(4)    = 247.
27
Log likelihood = -1569.7305                             Prob > chi2   = 0.00
00

----------------------------------------------------------------------------
--
          _t | Haz. ratio   Std. err.      z    P>|z|     [95% conf. interva
l]
-------------+--------------------------------------------------------------
--
mpad_tertile |
MPAD Tert..  |   1.102178   .1889749     0.57   0.570     .7876044    1.5423
94
MPAD Tert..  |   1.574439   .2465408     2.90   0.004     1.158341    2.1400
07
             |
         age |   1.052754   .0040015    13.53   0.000      1.04494    1.0606
26
             |
        male |
       Male  |   1.386117   .1756037     2.58   0.010     1.081343    1.7767
92
----------------------------------------------------------------------------
--

. estat concordance

        Failure _d: death==1
  Analysis time _t: _t

Harrell's C concordance statistic

  Number of subjects (N)             =      912
  Number of comparison pairs (P)     =   174623
  Number of orderings as expected (E) =   131708
  Number of tied predictions (T)     =      392
```

```
          Harrell's C = (E + T/2) / P =   0.7554
                            Somers' D =   0.5107

. stbrier i.mpad_tertile, bt(12.4819)

Mean estimation                       Number of obs   =       990

---------------------------------------------------------------
               |       Mean    Std. Err.     [95% Conf. Interval]
---------------+-----------------------------------------------
   Brier score |   4.696547     .5729393    3.572231    5.820863
---------------------------------------------------------------

.
```

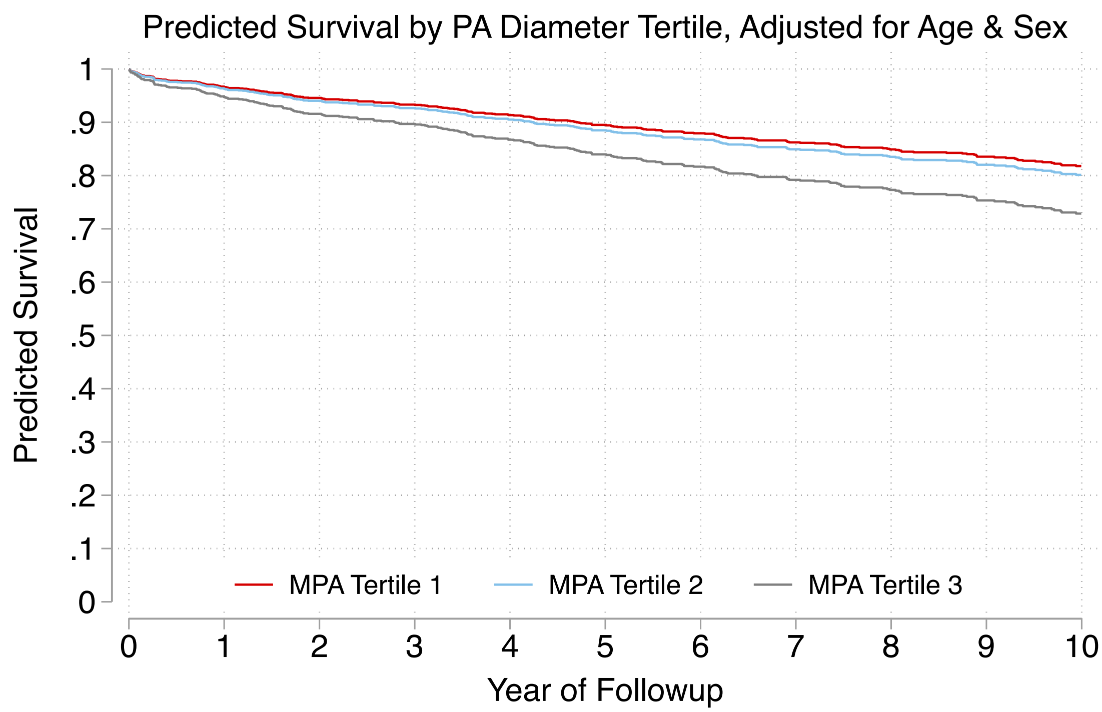
quite a bit better prediction: Harrell C 0.7539

In [20]:

```
%%stata

stcox i.mpad_tertile c.age male
stcurve, surv at(mpad_tertile=1 male=0.3737 age=52) /// at mean values
          at(mpad_tertile=2 male=0.3737 age=52) ///
          at(mpad_tertile=3 male=0.3737 age=52) ///
          range(0 10) ///
          xlabel(0(1)10, labsize(medlarge)) ///
          ylabel(0(.1)1,labsize(medlarge)) ///
          xtitle("Year of Followup", size(medlarge)) ///
          ytitle("Predicted Survival", size(medlarge)) ///
          title("Predicted Survival by PA Diameter Tertile, Adjusted for Age
          legend(order(1 "MPA Tertile 1" 2 "MPA Tertile 2" 3 "MPA Tertile 3"
                 position(6) ring(0) rows(1) size(medsmall))
```

```
.
. stcox i.mpad_tertile c.age male

        Failure _d: death==1
  Analysis time _t: _t

Iteration 0:  Log likelihood = -1693.3646
Iteration 1:  Log likelihood =  -1570.436
Iteration 2:  Log likelihood = -1569.7309
Iteration 3:  Log likelihood = -1569.7305
Refining estimates:
Iteration 0:  Log likelihood = -1569.7305

Cox regression with Breslow method for ties

No. of subjects =        912                           Number of obs =    9
12
No. of failures =        263
Time at risk    = 6,981.3588
                                                        LR chi2(4)    = 247.
27
Log likelihood = -1569.7305                             Prob > chi2   = 0.00
00

----------------------------------------------------------------------------
--
          _t | Haz. ratio   Std. err.      z    P>|z|     [95% conf. interva
l]
-------------+--------------------------------------------------------------
--
mpad_tertile |
MPAD Tert..  |   1.102178   .1889749     0.57   0.570     .7876044    1.5423
94
MPAD Tert..  |   1.574439   .2465408     2.90   0.004     1.158341    2.1400
07
             |
         age |   1.052754   .0040015    13.53   0.000      1.04494    1.0606
26
        male |   1.386117   .1756037     2.58   0.010     1.081343    1.7767
92
----------------------------------------------------------------------------
--

. stcurve, surv at(mpad_tertile=1 male=0.3737 age=52) /// at mean values
>           at(mpad_tertile=2 male=0.3737 age=52) ///
>           at(mpad_tertile=3 male=0.3737 age=52) ///
>           range(0 10) ///
>           xlabel(0(1)10, labsize(medlarge)) ///
>           ylabel(0(.1)1,labsize(medlarge)) ///
>           xtitle("Year of Followup", size(medlarge)) ///
>           ytitle("Predicted Survival", size(medlarge)) ///
>           title("Predicted Survival by PA Diameter Tertile, Adjusted for A
ge
> & Sex", size(medlarge)) ///
>           legend(order(1 "MPA Tertile 1" 2 "MPA Tertile 2" 3 "MPA Tertile
3")
```

```
>  ///
>                    position(6) ring(0) rows(1) size(medsmall))
note: function evaluated at specified values of selected covariates and
      overall means of other covariates (if any).

.
```



note: predicted death rates are lower because this displays at the mean age of 52; hazard ratios are the same throughout tho because of proportional hazards assumption

Age, Sex Specific Tertile Approach MPA diameter

In [21]:
```
%%stata

/* Method = use conditional quantile regression to generate age-sex specific
quantile estimates, then split into tertiles based on that and repeat */

/* -----
Step 1: Compute 33.3rd and 66.6th Percentiles Using Quantile Regression
-----*/
* Run Quantile Regression Separately for Males
qreg mpad c.age if male == 1, quantile(0.333)
predict mpad_q33_male if male == 1, xb

qreg mpad c.age if male == 1, quantile(0.666)
predict mpad_q66_male if male == 1, xb

* Run Quantile Regression Separately for Females
qreg mpad c.age if male == 0, quantile(0.333)
predict mpad_q33_female if male == 0, xb

qreg mpad c.age if male == 0, quantile(0.666)
```

```
predict mpad_q66_female if male == 0, xb

/* -----
Step 2: Categorize Individuals Into Predicted Tertiles
-----*/

* Create an empty variable for tertile classification
gen mpad_pred_tertile = .

* Assign individuals into tertiles based on their predicted quantile cutoffs
replace mpad_pred_tertile = 1 if male == 1 & mpad < mpad_q33_male
replace mpad_pred_tertile = 2 if male == 1 & mpad >= mpad_q33_male & mpad <
replace mpad_pred_tertile = 3 if male == 1 & mpad >= mpad_q66_male

replace mpad_pred_tertile = 1 if male == 0 & mpad < mpad_q33_female
replace mpad_pred_tertile = 2 if male == 0 & mpad >= mpad_q33_female & mpad
replace mpad_pred_tertile = 3 if male == 0 & mpad >= mpad_q66_female

/* -----
Step 3: Label the Categories
-----*/

label define mpad_pred_tertile_lbl 1 "MPA Pred Tertile 1" 2 "MPAD Pred Terti
label values mpad_pred_tertile mpad_pred_tertile_lbl

* Verify the tertile distribution
tab mpad_pred_tertile mpad_tertile
```

```
.
. /* Method = use conditional quantile regression to generate age-sex specif
ic
> quantile estimates, then split into tertiles based on that and repeat */
.
. /* -----
> Step 1: Compute 33.3rd and 66.6th Percentiles Using Quantile Regression
> -----*/
. * Run Quantile Regression Separately for Males
. qreg mpad c.age if male == 1, quantile(0.333)

Iteration 1:  WLS sum of weighted deviations =  590.66637

Iteration 1:  Sum of abs. weighted deviations =  591.93094
Iteration 2:  Sum of abs. weighted deviations =   551.7607
Iteration 3:  Sum of abs. weighted deviations =  550.28064
Iteration 4:  Sum of abs. weighted deviations =  541.46338
Iteration 5:  Sum of abs. weighted deviations =  540.80858
Iteration 6:  Sum of abs. weighted deviations =  540.80635

.333 Quantile regression                           Number of obs =        3
70
  Raw sum of deviations 554.4912 (about 24.799999)
  Min sum of deviations 540.8063                   Pseudo R2     =     0.02
47

----------------------------------------------------------------------------
--
        mpad | Coefficient  Std. err.      t    P>|t|     [95% conf. interva
l]
-------------+--------------------------------------------------------------
--
         age |   .0489796     .01428     3.43   0.001     .0208989    .07706
03
       _cons |   22.24082    .797291    27.90   0.000       20.673    23.808
64
----------------------------------------------------------------------------
--

. predict mpad_q33_male if male == 1, xb
(620 missing values generated)

.
. qreg mpad c.age if male == 1, quantile(0.666)

Iteration 1:  WLS sum of weighted deviations =  603.07386

Iteration 1:  Sum of abs. weighted deviations =  605.13012
Iteration 2:  Sum of abs. weighted deviations =  584.63494
Iteration 3:  Sum of abs. weighted deviations =  575.40786
Iteration 4:  Sum of abs. weighted deviations =  575.40425

.666 Quantile regression                           Number of obs =        3
70
  Raw sum of deviations 594.4544 (about 28.200001)
  Min sum of deviations 575.4043                   Pseudo R2     =     0.03
```

```
20

------------------------------------------------------------------------------
--
        mpad | Coefficient  Std. err.      t    P>|t|     [95% conf. interva
l]
-------------+----------------------------------------------------------------
--
         age |   .0538462    .013823     3.90   0.000     .0266643     .0810
28
       _cons |   25.46923   .7717703    33.00   0.000      23.9516    26.986
86
------------------------------------------------------------------------------
--

. predict mpad_q66_male if male == 1, xb
(620 missing values generated)

.
. * Run Quantile Regression Separately for Females
. qreg mpad c.age if male == 0, quantile(0.333)

Iteration 1:  WLS sum of weighted deviations =  921.71448

Iteration 1:  Sum of abs. weighted deviations =  926.34884
Iteration 2:  Sum of abs. weighted deviations =  903.78933
Iteration 3:  Sum of abs. weighted deviations =  842.65138
Iteration 4:  Sum of abs. weighted deviations =  830.14194
Iteration 5:  Sum of abs. weighted deviations =  829.60329
Iteration 6:  Sum of abs. weighted deviations =  829.58113

.333 Quantile regression                           Number of obs =        6
20
  Raw sum of deviations 870.6123 (about 22.9)
  Min sum of deviations 829.5811                   Pseudo R2     =     0.04
71

------------------------------------------------------------------------------
--
        mpad | Coefficient  Std. err.      t    P>|t|     [95% conf. interva
l]
-------------+----------------------------------------------------------------
--
         age |       .052   .0087723     5.93   0.000      .034773     .0692
27
       _cons |     20.304   .4917824    41.29   0.000     19.33823    21.269
77
------------------------------------------------------------------------------
--

. predict mpad_q33_female if male == 0, xb
(370 missing values generated)

.
. qreg mpad c.age if male == 0, quantile(0.666)
```

```
Iteration 1:  WLS sum of weighted deviations =  931.25557

Iteration 1:  Sum of abs. weighted deviations =  930.79749
Iteration 2:  Sum of abs. weighted deviations =  910.54622
Iteration 3:  Sum of abs. weighted deviations =  889.95718
Iteration 4:  Sum of abs. weighted deviations =  889.81602
Iteration 5:  Sum of abs. weighted deviations =  889.77739

.666 Quantile regression                            Number of obs =        6
20
  Raw sum of deviations 952.9886 (about 26.200001)
  Min sum of deviations 889.7774                    Pseudo R2     =     0.06
63

----------------------------------------------------------------------------
--
        mpad | Coefficient  Std. err.      t    P>|t|     [95% conf. interva
l]
-------------+--------------------------------------------------------------
--
         age |   .0694915    .010144     6.85   0.000     .0495707    .08941
24
       _cons |   22.56271   .5686828    39.68   0.000     21.44593     23.67
95
----------------------------------------------------------------------------
--

. predict mpad_q66_female if male == 0, xb
(370 missing values generated)

.
. /* -----
> Step 2: Categorize Individuals Into Predicted Tertiles
> -----*/
.
. * Create an empty variable for tertile classification
. gen mpad_pred_tertile = .
(990 missing values generated)

.
. * Assign individuals into tertiles based on their predicted quantile cutof
fs
. replace mpad_pred_tertile = 1 if male == 1 & mpad < mpad_q33_male
(122 real changes made)

. replace mpad_pred_tertile = 2 if male == 1 & mpad >= mpad_q33_male & mpad
< m
> pad_q66_male
(123 real changes made)

. replace mpad_pred_tertile = 3 if male == 1 & mpad >= mpad_q66_male
(125 real changes made)

.
. replace mpad_pred_tertile = 1 if male == 0 & mpad < mpad_q33_female
(206 real changes made)
```

```
. replace mpad_pred_tertile = 2 if male == 0 & mpad >= mpad_q33_female & mpa
d <
>  mpad_q66_female
(205 real changes made)

. replace mpad_pred_tertile = 3 if male == 0 & mpad >= mpad_q66_female
(209 real changes made)

.
. /* -----
> Step 3: Label the Categories
> -----*/
.
. label define mpad_pred_tertile_lbl 1 "MPA Pred Tertile 1" 2 "MPAD Pred Ter
til
> e 2" 3 "MPAD Pred Tertile 3"

. label values mpad_pred_tertile mpad_pred_tertile_lbl

.
. * Verify the tertile distribution
. tab mpad_pred_tertile mpad_tertile

                    |       3 quantiles of mpad
  mpad_pred_tertile | MPAD Tert  MPAD Tert  MPAD Tert |     Total
--------------------+---------------------------------+----------
 MPA Pred Tertile 1 |       281         47          0 |       328
MPAD Pred Tertile 2 |        55        224         49 |       328
MPAD Pred Tertile 3 |         0         65        269 |       334
--------------------+---------------------------------+----------
              Total |       336        336        318 |       990

.
```

In [22]:

```
%%stata

tab mpad_pred_tertile mpad_tertile

kappaetc mpad_pred_tertile mpad_tertile
```

```
.
. tab mpad_pred_tertile mpad_tertile

                    |      3 quantiles of mpad
  mpad_pred_tertile | MPAD Tert  MPAD Tert  MPAD Tert |     Total
--------------------+---------------------------------+----------
 MPA Pred Tertile 1 |       281         47          0 |       328
MPAD Pred Tertile 2 |        55        224         49 |       328
MPAD Pred Tertile 3 |         0         65        269 |       334
--------------------+---------------------------------+----------
              Total |       336        336        318 |       990

.
. kappaetc mpad_pred_tertile mpad_tertile

Interrater agreement                             Number of subjects =     99
0
                                                Ratings per subject =      
2
                                        Number of rating categories =      
3
----------------------------------------------------------------------------
--
                    |   Coef.  Std. Err.    t    P>|t|   [95% Conf. Interva
l]
--------------------+-------------------------------------------------------
--
   Percent Agreement |  0.7818    0.0131  59.53   0.000     0.7560     0.807
6
Brennan and Prediger |  0.6727    0.0197  34.15   0.000     0.6341     0.711
4
Cohen/Conger's Kappa |  0.6728    0.0197  34.15   0.000     0.6341     0.711
4
    Scott/Fleiss' Pi |  0.6727    0.0197  34.14   0.000     0.6340     0.711
4
           Gwet's AC |  0.6727    0.0197  34.16   0.000     0.6341     0.711
4
Krippendorff's Alpha |  0.6729    0.0197  34.14   0.000     0.6342     0.711
6
----------------------------------------------------------------------------
--

.
```

In [23]:

```
%%stata

tab3way mpad_pred_tertile mpad_tertile age_decade
```

```
.
. tab3way mpad_pred_tertile mpad_tertile age_decade
```

Table entries are cell frequencies
Missing categories ignored

	Age (by decade) and 3 quantiles of mpad		
	<30 years		
mpad_pred_tertile	MPAD Tertile 1	MPAD Tertile 2	MPAD Tertile 3
MPA Pred Tertile 1	52		
MPAD Pred Tertile 2	23	20	
MPAD Pred Tertile 3		27	25

	Age (by decade) and 3 quantiles of mpad		
	30-40 years		
mpad_pred_tertile	MPAD Tertile 1	MPAD Tertile 2	MPAD Tertile 3
MPA Pred Tertile 1	49	2	
MPAD Pred Tertile 2	18	43	3
MPAD Pred Tertile 3		16	34

	Age (by decade) and 3 quantiles of mpad		
	40-50 years		
mpad_pred_tertile	MPAD Tertile 1	MPAD Tertile 2	MPAD Tertile 3
MPA Pred Tertile 1	54	4	
MPAD Pred Tertile 2	10	39	5
MPAD Pred Tertile 3		10	47

	Age (by decade) and 3 quantiles of mpad		
	50-60 years		
mpad_pred_tertile	MPAD Tertile 1	MPAD Tertile 2	MPAD Tertile 3
MPA Pred Tertile 1	43	6	
MPAD Pred Tertile 2	4	39	10
MPAD Pred Tertile 3		11	46

	Age (by decade) and 3 quantiles of mpad		
	60-70 years		
mpad_pred_tertile	MPAD Tertile 1	MPAD Tertile 2	MPAD Tertile 3
MPA Pred Tertile 1	39	9	
MPAD Pred Tertile 2		40	8
MPAD Pred Tertile 3		1	48

```
-------------------------------------------------------------------
                    |    Age (by decade) and 3 quantiles of mpad
                    | ------------------ 70+ years -----------------
  mpad_pred_tertile | MPAD Tertile 1  MPAD Tertile 2  MPAD Tertile 3
--------------------+----------------------------------------------
 MPA Pred Tertile 1 |            44              26
MPAD Pred Tertile 2 |                            43              23
MPAD Pred Tertile 3 |                                            69
-------------------------------------------------------------------
```

.

actual tertile to age-sex-specific tertile shifts definitely change by age group

In [24]:

```
%%stata

/* -----
Define Colors for MPAD Predicted Tertiles
-----*/

colorpalette Spectral, n(3) nograph
local color3 `"`r(p1)'"'  // MPAD Tertile 3 (High)
local color2 `"`r(p2)'"'  // MPAD Tertile 2 (Middle)
local color1 `"`r(p3)'"'  // MPAD Tertile 1 (Low)

/* -----
Generate Quantile Regression Predictions for Males
-----*/

qreg mpad c.age if male == 1, quantile(0.333) // Male
predict mpad_33_male if male == 1, xb
qreg mpad c.age if male == 1, quantile(0.666) // Male
predict mpad_66_male if male == 1, xb
qreg mpad c.age if male == 1, quantile(95) // Male
predict mpad_95_male if male == 1, xb

/* -----
Male Scatter Plot (Color by MPAD Predicted Tertile)
-----*/
twoway ///
    (scatter mpad age if male == 1 & mpad_tertile == 1, mcolor("`color1'") m
    (scatter mpad age if male == 1 & mpad_tertile == 2, mcolor("`color2'") m
    (scatter mpad age if male == 1 & mpad_tertile == 3, mcolor("`color3'") m
    (line mpad_33_male age, lcolor(black) lpattern(solid) lwidth(medthick))
    (line mpad_66_male age, lcolor(gs8) lpattern(dash) lwidth(medthick)) ||
    (line mpad_95_male age, lcolor(gs12) lpattern(longdash) lwidth(medthick)
    ///
    legend(order(4 "1st-2nd Tertile" 5 "2nd-3rd Tertile" 6 "95th Percentile"
                 1 "MPAD Pred Tertile 1" 2 "MPAD Pred Tertile 2" 3 "MPAD Pre
           pos(6) row(2)) ///
    xtitle("Age (years)") ytitle("Pulmonary Artery Diameter (mm)", margin(me
    title("Men") ///
    xlabel(, labsize(medlarge)) ylabel(15(5)40, nogrid labsize(medlarge)) //
    scheme(white_w3d) yscale(range(15 40)) ///
    saving(male_qreg_plot, replace)
```

```
/* -----
Generate Quantile Regression Predictions for Females
-----*/

qreg mpad c.age if male == 0, quantile(0.333) // Female
predict mpad_33_female if male == 0, xb
qreg mpad c.age if male == 0, quantile(0.666) // Female
predict mpad_66_female if male == 0, xb
qreg mpad c.age if male == 0, quantile(95) // Female
predict mpad_95_female if male == 0, xb

/* -----
Female Scatter Plot (Color by MPAD Predicted Tertile)
-----*/
twoway ///
    (scatter mpad age if male == 0 & mpad_tertile == 1, mcolor("`color1'") m
    (scatter mpad age if male == 0 & mpad_tertile == 2, mcolor("`color2'") m
    (scatter mpad age if male == 0 & mpad_tertile == 3, mcolor("`color3'") m
    (line mpad_33_female age, lcolor(black) lpattern(solid) lwidth(medthick)
    (line mpad_66_female age, lcolor(gs8) lpattern(dash) lwidth(medthick)) |
    (line mpad_95_female age, lcolor(gs12) lpattern(longdash) lwidth(medthic
    ///
    legend(order(4 "1st-2nd Tertile" 5 "2nd-3rd Tertile" 6 "95th Quantile" /
                 1 "MPAD Tertile 1" 2 "MPAD Tertile 2" 3 "MPAD Tertile 3") /
           pos(6) row(2)) ///
    xtitle("Age (years)") ytitle("Pulmonary Artery Diameter (mm)", margin(me
    title("Women") ///
    xlabel(, labsize(medlarge)) ylabel(15(5)40, nogrid labsize(medlarge)) //
    scheme(white_w3d) yscale(range(15 40)) ///
    saving(female_qreg_plot, replace)


/* -----
Combine the Two Plots into a Single Figure
-----*/
graph combine male_qreg_plot.gph female_qreg_plot.gph, ///
    title("Sex-Stratified Quantile Regression of MPAD by Predicted Tertile")
    ycommon xcommon ///
    iscale(*1.2) ///
    rows(1) ///
    graphregion(margin(2 2 2 2)) ///
    xsize(9) ysize(5)
```

```
.
. /* -----
> Define Colors for MPAD Predicted Tertiles
> -----*/
.
. colorpalette Spectral, n(3) nograph

. local color3 `"`r(p1)'"'  // MPAD Tertile 3 (High)

. local color2 `"`r(p2)'"'  // MPAD Tertile 2 (Middle)

. local color1 `"`r(p3)'"'  // MPAD Tertile 1 (Low)

.
. /* -----
> Generate Quantile Regression Predictions for Males
> -----*/
.
. qreg mpad c.age if male == 1, quantile(0.333) // Male

Iteration 1:  WLS sum of weighted deviations =  590.66637

Iteration 1:  Sum of abs. weighted deviations =  591.93094
Iteration 2:  Sum of abs. weighted deviations =   551.7607
Iteration 3:  Sum of abs. weighted deviations =  550.28064
Iteration 4:  Sum of abs. weighted deviations =  541.46338
Iteration 5:  Sum of abs. weighted deviations =  540.80858
Iteration 6:  Sum of abs. weighted deviations =  540.80635

.333 Quantile regression                           Number of obs =        3
70
  Raw sum of deviations 554.4912 (about 24.799999)
  Min sum of deviations 540.8063                   Pseudo R2     =     0.02
47

------------------------------------------------------------------------------
--
        mpad | Coefficient  Std. err.      t    P>|t|     [95% conf. interva
l]
-------------+----------------------------------------------------------------
--
         age |   .0489796     .01428     3.43   0.001     .0208989    .07706
03
       _cons |   22.24082    .797291    27.90   0.000       20.673    23.808
64
------------------------------------------------------------------------------
--

. predict mpad_33_male if male == 1, xb
(620 missing values generated)

. qreg mpad c.age if male == 1, quantile(0.666) // Male

Iteration 1:  WLS sum of weighted deviations =  603.07386

Iteration 1:  Sum of abs. weighted deviations =  605.13012
```

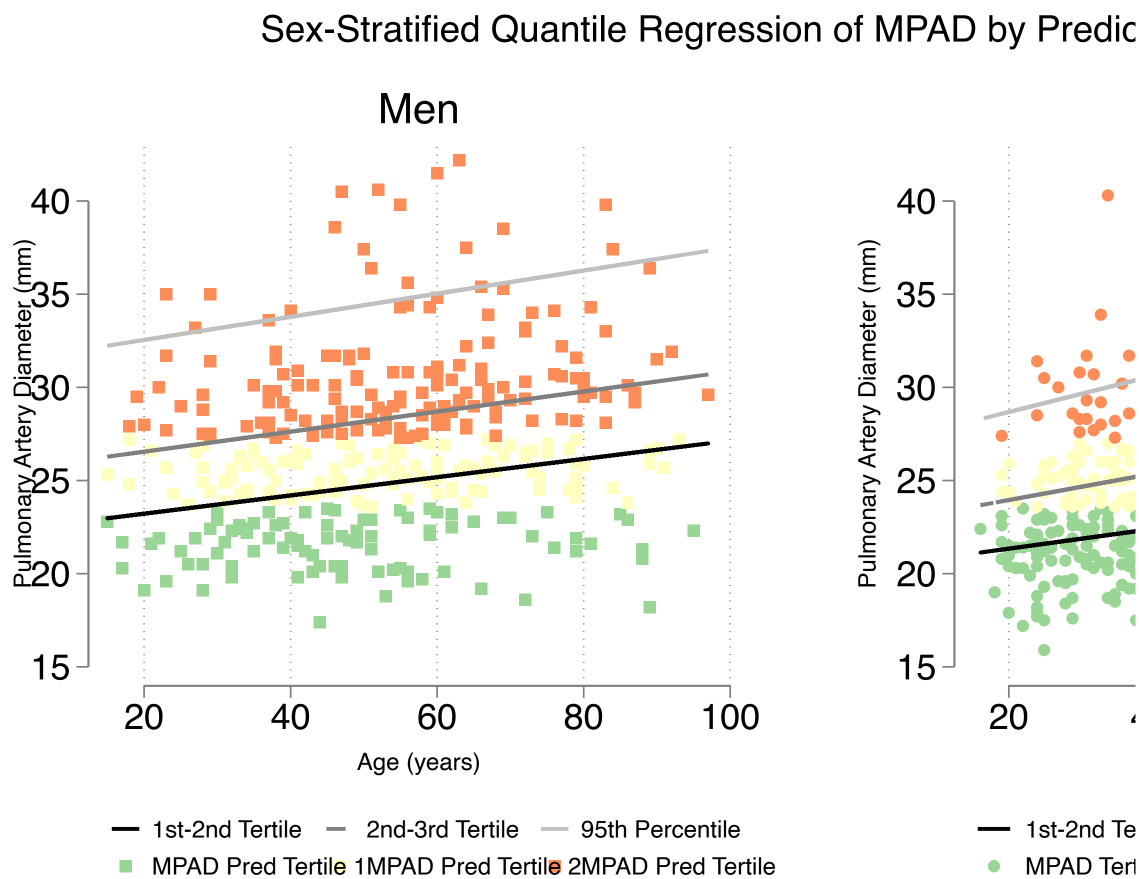
```
Iteration 2:  Sum of abs. weighted deviations =  584.63494
Iteration 3:  Sum of abs. weighted deviations =  575.40786
Iteration 4:  Sum of abs. weighted deviations =  575.40425

.666 Quantile regression                           Number of obs =       3
70
  Raw sum of deviations 594.4544 (about 28.200001)
  Min sum of deviations 575.4043                   Pseudo R2     =    0.03
20

----------------------------------------------------------------------------
--
        mpad | Coefficient  Std. err.      t    P>|t|     [95% conf. interva
l]
-------------+--------------------------------------------------------------
--
         age |   .0538462    .013823     3.90   0.000     .0266643     .0810
28
       _cons |   25.46923   .7717703    33.00   0.000      23.9516    26.986
86
----------------------------------------------------------------------------
--

. predict mpad_66_male if male == 1, xb
(620 missing values generated)

. qreg mpad c.age if male == 1, quantile(95) // Male

Iteration 1:  WLS sum of weighted deviations =  488.35735

Iteration 1:  Sum of abs. weighted deviations =  488.41762
Iteration 2:  Sum of abs. weighted deviations =  252.46043
Iteration 3:  Sum of abs. weighted deviations =  226.43999
Iteration 4:  Sum of abs. weighted deviations =  201.09818
Iteration 5:  Sum of abs. weighted deviations =  200.04639
Iteration 6:  Sum of abs. weighted deviations =   199.8393
Iteration 7:  Sum of abs. weighted deviations =  199.68345

.95 Quantile regression                            Number of obs =       3
70
  Raw sum of deviations   205.18 (about 34.799999)
  Min sum of deviations 199.6834                   Pseudo R2     =    0.02
68

----------------------------------------------------------------------------
--
        mpad | Coefficient  Std. err.      t    P>|t|     [95% conf. interva
l]
-------------+--------------------------------------------------------------
--
         age |   .0620691   .0486485     1.28   0.203    -.033595    .15773
31
       _cons |   31.30344   2.716171    11.52   0.000     25.96228    36.644
61
----------------------------------------------------------------------------
--
```

```
. predict mpad_95_male if male == 1, xb
(620 missing values generated)

.
. /* -----
> Male Scatter Plot (Color by MPAD Predicted Tertile)
> -----*/
. twoway ///
>     (scatter mpad age if male == 1 & mpad_tertile == 1, mcolor("`color1'")
ms
> ymbol(S)) || /// MPAD Tertile 1 (Low)
>     (scatter mpad age if male == 1 & mpad_tertile == 2, mcolor("`color2'")
ms
> ymbol(S)) || /// MPAD Tertile 2 (Middle)
>     (scatter mpad age if male == 1 & mpad_tertile == 3, mcolor("`color3'")
ms
> ymbol(S)) || /// MPAD Tertile 3 (High)
>     (line mpad_33_male age, lcolor(black) lpattern(solid) lwidth(medthic
k)) |
> | /// 50th percentile line
>     (line mpad_66_male age, lcolor(gs8) lpattern(dash) lwidth(medthick)) |
| /
> // 90th percentile line
>     (line mpad_95_male age, lcolor(gs12) lpattern(longdash) lwidth(medthic
k))
> , /// 95th percentile line
>     ///
>     legend(order(4 "1st-2nd Tertile" 5 "2nd-3rd Tertile" 6 "95th Percentil
e"
> ///
>                  1 "MPAD Pred Tertile 1" 2 "MPAD Pred Tertile 2" 3 "MPAD P
red
>  Tertile 3") ///
>           pos(6) row(2)) ///
>     xtitle("Age (years)") ytitle("Pulmonary Artery Diameter (mm)", margin
(med
> large)) ///
>     title("Men") ///
>     xlabel(, labsize(medlarge)) ylabel(15(5)40, nogrid labsize(medlarge))
///
>  Adjusted Y-axis range
>     scheme(white_w3d) yscale(range(15 40)) ///
>     saving(male_qreg_plot, replace)
file male_qreg_plot.gph saved

.
.
. /* -----
> Generate Quantile Regression Predictions for Females
> -----*/
.
. qreg mpad c.age if male == 0, quantile(0.333) // Female

Iteration 1:  WLS sum of weighted deviations =  921.71448
```

```
Iteration 1:  Sum of abs. weighted deviations =  926.34884
Iteration 2:  Sum of abs. weighted deviations =  903.78933
Iteration 3:  Sum of abs. weighted deviations =  842.65138
Iteration 4:  Sum of abs. weighted deviations =  830.14194
Iteration 5:  Sum of abs. weighted deviations =  829.60329
Iteration 6:  Sum of abs. weighted deviations =  829.58113

.333 Quantile regression                           Number of obs =        6
20
  Raw sum of deviations 870.6123 (about 22.9)
  Min sum of deviations 829.5811                   Pseudo R2     =     0.04
71

---------------------------------------------------------------------------
--
        mpad | Coefficient  Std. err.      t    P>|t|     [95% conf. interva
l]
-------------+-------------------------------------------------------------
--
         age |       .052   .0087723     5.93   0.000      .034773     .0692
27
       _cons |     20.304   .4917824    41.29   0.000     19.33823    21.269
77
---------------------------------------------------------------------------
--

. predict mpad_33_female if male == 0, xb
(370 missing values generated)

. qreg mpad c.age if male == 0, quantile(0.666) // Female

Iteration 1:  WLS sum of weighted deviations =  931.25557

Iteration 1:  Sum of abs. weighted deviations =  930.79749
Iteration 2:  Sum of abs. weighted deviations =  910.54622
Iteration 3:  Sum of abs. weighted deviations =  889.95718
Iteration 4:  Sum of abs. weighted deviations =  889.81602
Iteration 5:  Sum of abs. weighted deviations =  889.77739

.666 Quantile regression                           Number of obs =        6
20
  Raw sum of deviations 952.9886 (about 26.200001)
  Min sum of deviations 889.7774                   Pseudo R2     =     0.06
63

---------------------------------------------------------------------------
--
        mpad | Coefficient  Std. err.      t    P>|t|     [95% conf. interva
l]
-------------+-------------------------------------------------------------
--
         age |   .0694915    .010144     6.85   0.000     .0495707    .08941
24
       _cons |   22.56271   .5686828    39.68   0.000     21.44593     23.67
95
---------------------------------------------------------------------------
```

```
--

. predict mpad_66_female if male == 0, xb
(370 missing values generated)

. qreg mpad c.age if male == 0, quantile(95) // Female

Iteration 1:  WLS sum of weighted deviations =  838.61546

Iteration 1:  Sum of abs. weighted deviations =  839.25782
Iteration 2:  Sum of abs. weighted deviations =  374.71483
Iteration 3:  Sum of abs. weighted deviations =  317.24902
Iteration 4:  Sum of abs. weighted deviations =  285.19244
Iteration 5:  Sum of abs. weighted deviations =     283.44
Iteration 6:  Sum of abs. weighted deviations =  283.35929

.95 Quantile regression                             Number of obs =       6
20
  Raw sum of deviations  309.645 (about 32.200001)
  Min sum of deviations 283.3593                    Pseudo R2     =    0.08
49

----------------------------------------------------------------------------
--
        mpad | Coefficient  Std. err.      t    P>|t|     [95% conf. interva
l]
-------------+--------------------------------------------------------------
--
         age |   .0952381   .0199511     4.77   0.000     .0560579    .13441
83
       _cons |   26.77143   1.118483    23.94   0.000     24.57494    28.967
92
----------------------------------------------------------------------------
--

. predict mpad_95_female if male == 0, xb
(370 missing values generated)

.
. /* -----
> Female Scatter Plot (Color by MPAD Predicted Tertile)
> -----*/
. twoway ///
>     (scatter mpad age if male == 0 & mpad_tertile == 1, mcolor("`color1'")
ms
> ymbol(O)) || /// MPAD Tertile 1 (Low)
>     (scatter mpad age if male == 0 & mpad_tertile == 2, mcolor("`color2'")
ms
> ymbol(O)) || /// MPAD Tertile 2 (Middle)
>     (scatter mpad age if male == 0 & mpad_tertile == 3, mcolor("`color3'")
ms
> ymbol(O)) || /// MPAD Tertile 3 (High)
>     (line mpad_33_female age, lcolor(black) lpattern(solid) lwidth(medthic
k))
>  || /// 50th percentile line
>     (line mpad_66_female age, lcolor(gs8) lpattern(dash) lwidth(medthick))
```

```
||
>  /// 90th percentile line
>     (line mpad_95_female age, lcolor(gs12) lpattern(longdash) lwidth(medth
ick
> )), /// 95th percentile line
>     ///
>     legend(order(4 "1st-2nd Tertile" 5 "2nd-3rd Tertile" 6 "95th Quantile"
//
> /
>                  1 "MPAD Tertile 1" 2 "MPAD Tertile 2" 3 "MPAD Tertile 3")
//
> /
>            pos(6) row(2)) ///
>     xtitle("Age (years)") ytitle("Pulmonary Artery Diameter (mm)", margin
(med
> large)) ///
>     title("Women") ///
>     xlabel(, labsize(medlarge)) ylabel(15(5)40, nogrid labsize(medlarge))
///
>  Adjusted Y-axis range
>     scheme(white_w3d) yscale(range(15 40)) ///
>     saving(female_qreg_plot, replace)
file female_qreg_plot.gph saved

.
.
. /* -----
> Combine the Two Plots into a Single Figure
> -----*/
. graph combine male_qreg_plot.gph female_qreg_plot.gph, ///
>     title("Sex-Stratified Quantile Regression of MPAD by Predicted Tertil
e")
> ///
>     ycommon xcommon ///
>     iscale(*1.2) ///
>     rows(1) ///
>     graphregion(margin(2 2 2 2)) ///
>     xsize(9) ysize(5)

.
```



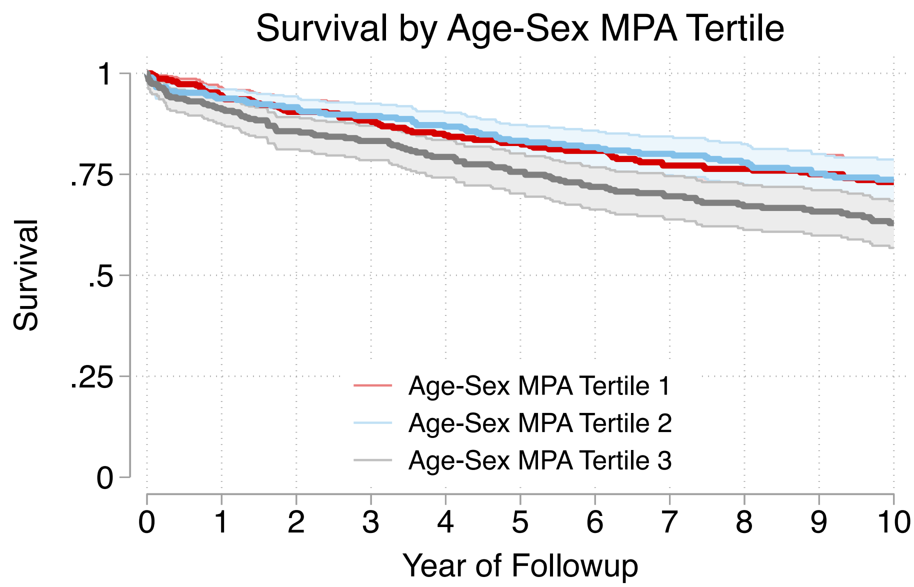
this graph shows:

- in colors = the overall tertile of PA size
- the lines = the age-sex specific thresholds (and the 95th percentile)

when the color change and line don't line up, those are patients who's PA size will be classified differently

In [25]:
```
%%stata

* Log-rank test for survival differences across MPAAA tertiles
sts test mpad_pred_tertile, logrank

* Kaplan-Meier Survival Curve by MPAAA Tertile
sts graph, by(mpad_pred_tertile) tmax(10) ci surv ///
 plotopts(lwidth(thick)) ///
 risktable(0(1)10, order(1 "Age-Sex MPA Tertile 1" 2 "Age-Sex MPA Tertile 2"
 xlabel(0(1)10, labsize(medlarge)) ///
 ylabel(,labsize(medlarge)) ///
 xtitle("Year of Followup", size(medlarge)) ///
 ytitle("Survival", size(medlarge)) ///
 legend(order(2 "Age-Sex MPA Tertile 1" 4 "Age-Sex MPA Tertile 2" 6 "Age-Sex
 title("Survival by Age-Sex MPA Tertile", size(large))
```

```
.
. * Log-rank test for survival differences across MPAAA tertiles
. sts test mpad_pred_tertile, logrank

        Failure _d: death==1
  Analysis time _t: _t

Equality of survivor functions
Log-rank test

                    |  Observed       Expected
mpad_pred_~e        |    events         events
--------------------+-------------------------
 MPA Pred Tertile 1 |        77          89.80
MPAD Pred Tertile 2 |        77          87.70
MPAD Pred Tertile 3 |       109          85.50
--------------------+-------------------------
              Total |       263         263.00

                            chi2(2) =   9.59
                            Pr>chi2 = 0.0083

.
. * Kaplan-Meier Survival Curve by MPAAA Tertile
. sts graph, by(mpad_pred_tertile) tmax(10) ci surv ///
>  plotopts(lwidth(thick)) ///
>  risktable(0(1)10, order(1 "Age-Sex MPA Tertile 1" 2 "Age-Sex MPA Tertile
2"
> 3 "Age-Sex MPA Tertile 3") title("Number Eligible", size(medium)) size(med
sma
> ll)) ///
>  xlabel(0(1)10, labsize(medlarge)) ///
>  ylabel(,labsize(medlarge)) ///
>  xtitle("Year of Followup", size(medlarge)) ///
>  ytitle("Survival", size(medlarge)) ///
>  legend(order(2 "Age-Sex MPA Tertile 1" 4 "Age-Sex MPA Tertile 2" 6 "Age-S
ex
> MPA Tertile 3") position(6) ring(0) rows(3) size(medsmall)) ///
>  title("Survival by Age-Sex MPA Tertile", size(large))

        Failure _d: death==1
  Analysis time _t: _t
(note:  named style i not found in class symbol, default attributes used)

.
```



Number Eligible											
Age-Sex MPA Tertile 1	301	278	258	246	225	215	201	188	178	168	139
Age-Sex MPA Tertile 2	302	265	254	242	224	212	201	191	177	159	131
Age-Sex MPA Tertile 3	309	272	253	234	218	203	189	176	158	146	122

interestingly, it works WORSE as a predictor.

hmmm...

In [26]:
```
%%stata

stcox i.mpad_pred_tertile
estat concordance

/* -----
Brier Score for MPAAA Tertile Model
-----*/
stbrier i.mpad_tertile, bt(12.4819)
```

```
.
. stcox i.mpad_pred_tertile

        Failure _d: death==1
  Analysis time _t: _t

Iteration 0:  Log likelihood = -1693.3646
Iteration 1:  Log likelihood = -1688.7719
Iteration 2:  Log likelihood = -1688.7557
Iteration 3:  Log likelihood = -1688.7557
Refining estimates:
Iteration 0:  Log likelihood = -1688.7557

Cox regression with Breslow method for ties

No. of subjects =        912                           Number of obs =    9
12
No. of failures =        263
Time at risk    = 6,981.3588
                                                        LR chi2(2)    =   9.
22
Log likelihood = -1688.7557                             Prob > chi2   = 0.01
00

----------------------------------------------------------------------------
--
          _t | Haz. ratio   Std. err.      z    P>|z|     [95% conf. interva
l]
-------------+--------------------------------------------------------------
--
mpad_pred_~e |
MPAD Pred..  |   1.023961   .1650353     0.15   0.883     .7466072    1.4043
47
MPAD Pred..  |   1.486917   .2213858     2.66   0.008     1.110586    1.9907
72
----------------------------------------------------------------------------
--

. estat concordance

        Failure _d: death==1
  Analysis time _t: _t

Harrell's C concordance statistic

  Number of subjects (N)             =      912
  Number of comparison pairs (P)     =   174623
  Number of orderings as expected (E) =    66451
  Number of tied predictions (T)     =    58044

          Harrell's C = (E + T/2) / P =   0.5467
                            Somers' D =   0.0935

.
. /* -----
> Brier Score for MPAAA Tertile Model
```

```
> -----*/
. stbrier i.mpad_tertile, bt(12.4819)

Mean estimation                          Number of obs   =        990

--------------------------------------------------------------
               |       Mean    Std. Err.     [95% Conf. Interval]
---------------+----------------------------------------------
   Brier score |   4.696547     .5729393     3.572231    5.820863
--------------------------------------------------------------

.
```

And if we add age and sex as predictors to the model:

In [27]:
```
%%stata

stcox i.mpad_pred_tertile c.age i.male
```

```
.
. stcox i.mpad_pred_tertile c.age i.male

        Failure _d: death==1
  Analysis time _t: _t

Iteration 0:  Log likelihood = -1693.3646
Iteration 1:  Log likelihood =  -1568.304
Iteration 2:  Log likelihood = -1567.7946
Iteration 3:  Log likelihood = -1567.7944
Refining estimates:
Iteration 0:  Log likelihood = -1567.7944

Cox regression with Breslow method for ties

No. of subjects =        912                           Number of obs =    9
12
No. of failures =        263
Time at risk    = 6,981.3588
                                                        LR chi2(4)    = 251.
14
Log likelihood = -1567.7944                             Prob > chi2   = 0.00
00

----------------------------------------------------------------------------
--
          _t | Haz. ratio   Std. err.      z    P>|z|     [95% conf. interva
l]
-------------+--------------------------------------------------------------
--
mpad_pred_~e |
MPAD Pred..  |   1.085267   .1750404     0.51   0.612     .7911304    1.4887
61
MPAD Pred..  |   1.683553   .2517543     3.48   0.000     1.255857    2.2569
05
             |
         age |   1.055469   .0040082    14.22   0.000     1.047642    1.0633
55
             |
        male |
        Male  |   1.483481   .1868292     3.13   0.002     1.158996    1.8988
12
----------------------------------------------------------------------------
--

.
```

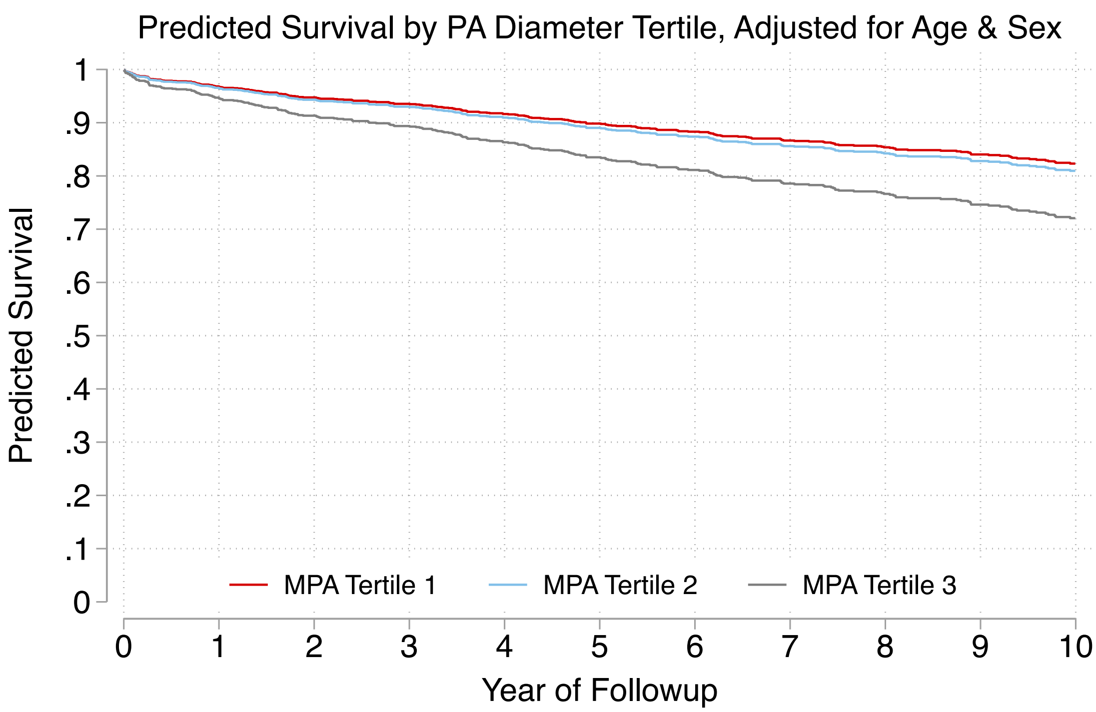
In [28]:

```
%%stata

stcox i.mpad_pred_tertile c.age male
stcurve, surv at(mpad_pred_tertile=1 male=0.3737 age=52) /// at mean values
          at(mpad_pred_tertile=2 male=0.3737 age=52) ///
          at(mpad_pred_tertile=3 male=0.3737 age=52) ///
          range(0 10) ///
          xlabel(0(1)10, labsize(medlarge)) ///
          ylabel(0(.1)1,labsize(medlarge)) ///
```

```
         xtitle("Year of Followup", size(medlarge)) ///
         ytitle("Predicted Survival", size(medlarge)) ///
         title("Predicted Survival by PA Diameter Tertile, Adjusted for Age
         legend(order(1 "MPA Tertile 1" 2 "MPA Tertile 2" 3 "MPA Tertile 3"
                position(6) ring(0) rows(1) size(medsmall))
```

```
.
. stcox i.mpad_pred_tertile c.age male

        Failure _d: death==1
  Analysis time _t: _t

Iteration 0:  Log likelihood = -1693.3646
Iteration 1:  Log likelihood =  -1568.304
Iteration 2:  Log likelihood = -1567.7946
Iteration 3:  Log likelihood = -1567.7944
Refining estimates:
Iteration 0:  Log likelihood = -1567.7944

Cox regression with Breslow method for ties

No. of subjects =        912                           Number of obs =    9
12
No. of failures =        263
Time at risk    = 6,981.3588
                                                       LR chi2(4)    = 251.
14
Log likelihood = -1567.7944                            Prob > chi2   = 0.00
00

----------------------------------------------------------------------------
--
          _t | Haz. ratio   Std. err.      z    P>|z|     [95% conf. interva
l]
-------------+--------------------------------------------------------------
--
mpad_pred_~e |
MPAD Pred..  |   1.085267   .1750404     0.51   0.612     .7911304    1.4887
61
MPAD Pred..  |   1.683553   .2517543     3.48   0.000     1.255857    2.2569
05
             |
         age |   1.055469   .0040082    14.22   0.000     1.047642    1.0633
55
        male |   1.483481   .1868292     3.13   0.002     1.158996    1.8988
12
----------------------------------------------------------------------------
--

. stcurve, surv at(mpad_pred_tertile=1 male=0.3737 age=52) /// at mean value
s
>           at(mpad_pred_tertile=2 male=0.3737 age=52) ///
>           at(mpad_pred_tertile=3 male=0.3737 age=52) ///
>           range(0 10) ///
>           xlabel(0(1)10, labsize(medlarge)) ///
>           ylabel(0(.1)1,labsize(medlarge)) ///
>           xtitle("Year of Followup", size(medlarge)) ///
>           ytitle("Predicted Survival", size(medlarge)) ///
>           title("Predicted Survival by PA Diameter Tertile, Adjusted for A
ge
> & Sex", size(medlarge)) ///
>           legend(order(1 "MPA Tertile 1" 2 "MPA Tertile 2" 3 "MPA Tertile
```

```
3")
>  ///
>                    position(6) ring(0) rows(1) size(medsmall))
note: function evaluated at specified values of selected covariates and
      overall means of other covariates (if any).

.
```



In [29]:
```
%%stata

stcox i.mpad_pred_tertile c.age i.male
estat concordance

/* -----
Brier Score for MPAD Tertile Model
-----*/
stbrier i.mpad_tertile c.age i.male, bt(12.4819)
```

```
.
. stcox i.mpad_pred_tertile c.age i.male

        Failure _d: death==1
  Analysis time _t: _t

Iteration 0:  Log likelihood = -1693.3646
Iteration 1:  Log likelihood =  -1568.304
Iteration 2:  Log likelihood = -1567.7946
Iteration 3:  Log likelihood = -1567.7944
Refining estimates:
Iteration 0:  Log likelihood = -1567.7944

Cox regression with Breslow method for ties

No. of subjects =        912                           Number of obs =    9
12
No. of failures =        263
Time at risk    = 6,981.3588
                                                       LR chi2(4)    = 251.
14
Log likelihood = -1567.7944                            Prob > chi2   = 0.00
00

-----------------------------------------------------------------------------
--
          _t | Haz. ratio   Std. err.      z    P>|z|     [95% conf. interva
l]
-------------+---------------------------------------------------------------
--
mpad_pred_~e |
MPAD Pred..  |   1.085267   .1750404     0.51   0.612     .7911304    1.4887
61
MPAD Pred..  |   1.683553   .2517543     3.48   0.000     1.255857    2.2569
05
             |
         age |   1.055469   .0040082    14.22   0.000     1.047642    1.0633
55
             |
        male |
       Male  |   1.483481   .1868292     3.13   0.002     1.158996    1.8988
12
-----------------------------------------------------------------------------
--

. estat concordance

        Failure _d: death==1
  Analysis time _t: _t

Harrell's C concordance statistic

  Number of subjects (N)             =      912
  Number of comparison pairs (P)     =   174623
  Number of orderings as expected (E) =   131795
  Number of tied predictions (T)     =      351
```

```
          Harrell's C = (E + T/2) / P =   0.7557
                             Somers' D =   0.5115

.
. /* -----
> Brier Score for MPAD Tertile Model
> -----*/
. stbrier i.mpad_tertile c.age i.male, bt(12.4819)

Mean estimation                     Number of obs   =       990

--------------------------------------------------------------
               |       Mean   Std. Err.     [95% Conf. Interval]
---------------+----------------------------------------------
   Brier score |   5.741934    .9364765    3.904224    7.579643
--------------------------------------------------------------

.
```

Overall - this is pretty similar in performance to the non-standardized version

Takeaway: not much in the way of accuracy boost from age-sex standardizing PA size.

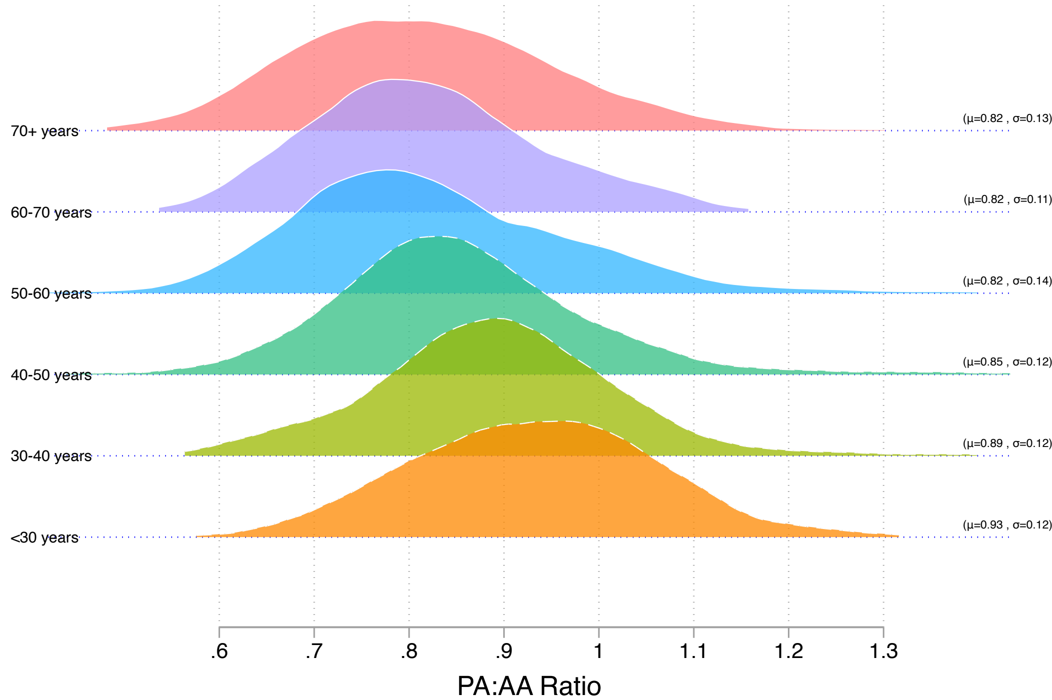
PA to AA ratio

Considering all the same methods with PA:AA gives us...

In [30]:

```
%%stata

/* MPA:AA */
ridgeline mpaaa, by(age_decade) yline ylw(0.2) overlap(1.7) ylc(blue) ylp(dc
        labpos(right) bwid(0.05) laboffset(-0.05) showstats xlabel(0.6(.1)1.
        palette(CET C6) alpha(75) xtitle("PA:AA Ratio")
```

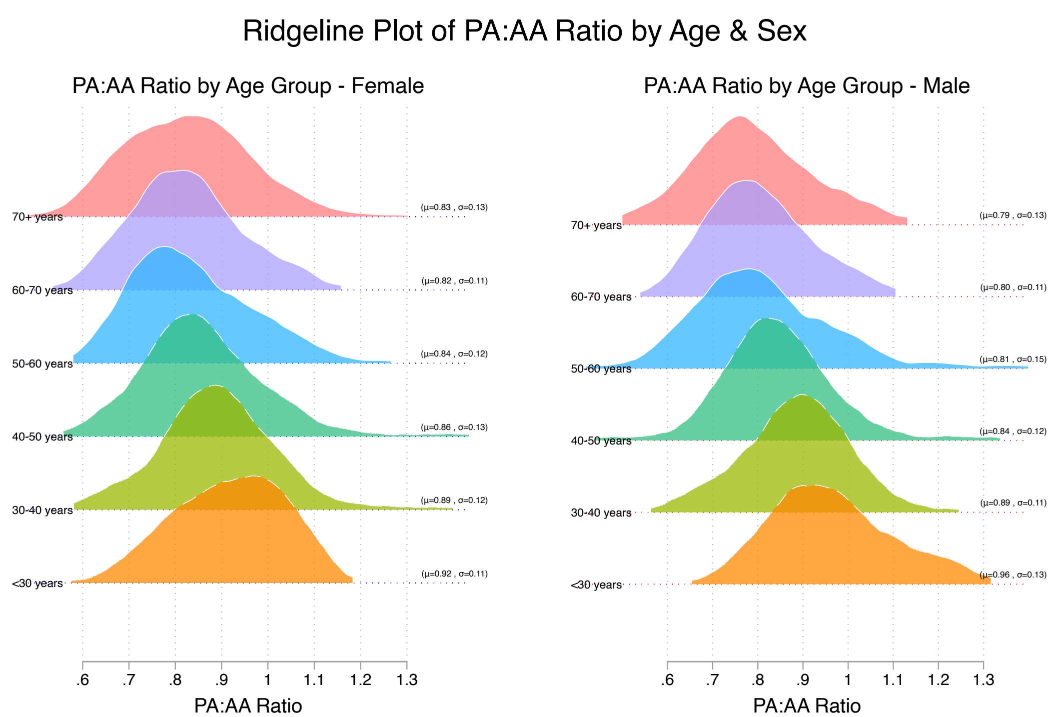
```
.
. /* MPA:AA */
. ridgeline mpaaa, by(age_decade) yline ylw(0.2) overlap(1.7) ylc(blue) ylp
(dot
> ) ////
>          labpos(right) bwid(0.05) laboffset(-0.05) showstats xlabel(0.6(.1)
1.3
> ) ///
>          palette(CET C6) alpha(75) xtitle("PA:AA Ratio")

.
```



In [31]:

```
%%stata

* Ridgeline plot for Females (male == 0) - PA:AA Ratio
ridgeline mpaaa if male == 0, by(age_decade) yline ylw(0.2) overlap(1.7) ///
    ylc(blue) ylp(dot) labpos(right) bwid(0.05) laboffset(-0.05) ///
    showstats xlabel(0.6(.1)1.3) palette(CET C6) alpha(75) ///
    xtitle("PA:AA Ratio") ///
    title("PA:AA Ratio by Age Group - Female") ///
    saving(ridgeline_mpaaa_female, replace)

* Ridgeline plot for Males (male == 1) - PA:AA Ratio
ridgeline mpaaa if male == 1, by(age_decade) yline ylw(0.2) overlap(1.7) ///
    ylc(red) ylp(dot) labpos(right) bwid(0.05) laboffset(-0.05) ///
    showstats xlabel(0.6(.1)1.3) palette(CET C6) alpha(75) ///
    xtitle("PA:AA Ratio") ///
    title("PA:AA Ratio by Age Group - Male") ///
    saving(ridgeline_mpaaa_male, replace)

graph combine ridgeline_mpaaa_female.gph ridgeline_mpaaa_male.gph, ///
    title("Ridgeline Plot of PA:AA Ratio by Age & Sex")
```

```
.
. * Ridgeline plot for Females (male == 0) - PA:AA Ratio
. ridgeline mpaaa if male == 0, by(age_decade) yline ylw(0.2) overlap(1.7)
///
>     ylc(blue) ylp(dot) labpos(right) bwid(0.05) laboffset(-0.05) ///
>     showstats xlabel(0.6(.1)1.3) palette(CET C6) alpha(75) ///
>     xtitle("PA:AA Ratio") ///
>     title("PA:AA Ratio by Age Group - Female") ///
>     saving(ridgeline_mpaaa_female, replace)

.
. * Ridgeline plot for Males (male == 1) - PA:AA Ratio
. ridgeline mpaaa if male == 1, by(age_decade) yline ylw(0.2) overlap(1.7)
///
>     ylc(red) ylp(dot) labpos(right) bwid(0.05) laboffset(-0.05) ///
>     showstats xlabel(0.6(.1)1.3) palette(CET C6) alpha(75) ///
>     xtitle("PA:AA Ratio") ///
>     title("PA:AA Ratio by Age Group - Male") ///
>     saving(ridgeline_mpaaa_male, replace)

.
. graph combine ridgeline_mpaaa_female.gph ridgeline_mpaaa_male.gph, ///
>     title("Ridgeline Plot of PA:AA Ratio by Age & Sex")
(note:  named style i not found in class symbol, default attributes used)
(note:  named style i not found in class symbol, default attributes used)

.
.
```



In [32]:
```
%%stata

/* Run Bootstrap Quantile Regressions for the Full Population */
bsqreg mpaaa c.age, quantile(50) reps(500) // 50th Percentile
```

```
bsqreg mpaaa c.age, quantile(90) reps(500) // 90th Percentile
bsqreg mpaaa c.age, quantile(95) reps(500) // 95th Percentile
```

```
.
. /* Run Bootstrap Quantile Regressions for the Full Population */
. bsqreg mpaaa c.age, quantile(50) reps(500) // 50th Percentile
(fitting base model)

Bootstrap replications (500): .........10.........20.........30.........4
0.....
> ....50.........60.........70.........80.........90.........100.........11
0...
> ......120.........130.........140.........150.........160.........17
0.......
> .180.........190.........200.........210.........220.........230.........2
40.
> ........250.........260.........270.........280.........290.........30
0......
> ...310.........320.........330.........340.........350.........36
0.........37
> 0.........380.........390.........400.........410.........420.........43
0....
> .....440.........450.........460.........470.........480.........49
0.........
> 500 done

Median regression, bootstrap(500) SEs              Number of obs =        9
90
  Raw sum of deviations  50.6267 (about .84615385)
  Min sum of deviations 47.96687                   Pseudo R2     =     0.05
25

----------------------------------------------------------------------------
--
       mpaaa | Coefficient  Std. err.      t    P>|t|     [95% conf. interva
l]
-------------+--------------------------------------------------------------
--
         age |  -.0021378   .0002924    -7.31   0.000    -.0027116   -.00156
41
       _cons |     .95458    .015935    59.90   0.000     .9233097    .98585
03
----------------------------------------------------------------------------
--

. bsqreg mpaaa c.age, quantile(90) reps(500) // 90th Percentile
(fitting base model)

Bootstrap replications (500): .........10.........20.........30.........4
0.....
> ....50.........60.........70.........80.........90.........100.........11
0...
> ......120.........130.........140.........150.........160.........17
0.......
> .180.........190.........200.........210.........220.........230.........2
40.
> ........250.........260.........270.........280.........290.........30
0......
> ...310.........320.........330.........340.........350.........36
```

```
0........37
> 0........380........390........400........410........420........43
0....
> .....440........450........460........470........480........49
0........
> 500 done

.9 Quantile regression, bootstrap(500) SEs          Number of obs =        9
90
  Raw sum of deviations 23.77614 (about 1.024735)
  Min sum of deviations 23.14079                    Pseudo R2     =     0.02
67

----------------------------------------------------------------------------
--
       mpaaa | Coefficient  Std. err.      t    P>|t|     [95% conf. interva
l]
-------------+--------------------------------------------------------------
--
         age |  -.0011581   .0002729    -4.24   0.000    -.0016936   -.00062
25
       _cons |   1.078691   .0132536    81.39   0.000     1.052683      1.10
47
----------------------------------------------------------------------------
--

. bsqreg mpaaa c.age, quantile(95) reps(500) // 95th Percentile
(fitting base model)

Bootstrap replications (500): .........10........20........30........4
0.....
> ....50........60........70........80........90........100........11
0...
> ......120........130........140........150........160........17
0.......
> .180........190........200........210........220........230........2
40.
> ........250........260........270........280........290........30
0......
> ...310........320........330........340........350........36
0........37
> 0........380........390........400........410........420........43
0....
> .....440........450........460........470........480........49
0........
> 500 done

.95 Quantile regression, bootstrap(500) SEs         Number of obs =        9
90
  Raw sum of deviations 14.43796 (about 1.0669856)
  Min sum of deviations 14.22666                    Pseudo R2     =     0.01
46

----------------------------------------------------------------------------
--
       mpaaa | Coefficient  Std. err.      t    P>|t|     [95% conf. interva
```

```
l]
-------------+----------------------------------------------------------------
--
         age |   -.001436   .0005365    -2.68   0.008    -.0024889   -.00038
32
       _cons |   1.132314   .0321086    35.27   0.000     1.069305    1.1953
23
------------------------------------------------------------------------------
--

.
```

In [33]:

```
%%stata


/* Extract Colors from Spectral Palette */
colorpalette Spectral, n(6) nograph
local color6 `"`r(p1)'"' // Reverse order
local color5 `"`r(p2)'"'
local color4 `"`r(p3)'"'
local color3 `"`r(p4)'"'
local color2 `"`r(p5)'"'
local color1 `"`r(p6)'"'

/* Run Quantile Regressions for the Full Population */
qreg mpaaa c.age, quantile(50) // 50th Percentile
predict mpaaa_q50_all, xb

qreg mpaaa c.age, quantile(90) // 90th Percentile
predict mpaaa_q90_all, xb

qreg mpaaa c.age, quantile(95) // 95th Percentile
predict mpaaa_q95_all, xb

/* Scatter Plot with Both Genders + Quantile Regression Lines */
twoway ///
    (scatter mpaaa age if age_decade == 0 & male == 0, mcolor("`color1'") ms
    (scatter mpaaa age if age_decade == 0 & male == 1, mcolor("`color1'") ms
    (scatter mpaaa age if age_decade == 1 & male == 0, mcolor("`color2'") ms
    (scatter mpaaa age if age_decade == 1 & male == 1, mcolor("`color2'") ms
    (scatter mpaaa age if age_decade == 2 & male == 0, mcolor("`color3'") ms
    (scatter mpaaa age if age_decade == 2 & male == 1, mcolor("`color3'") ms
    (scatter mpaaa age if age_decade == 3 & male == 0, mcolor("`color4'") ms
    (scatter mpaaa age if age_decade == 3 & male == 1, mcolor("`color4'") ms
    (scatter mpaaa age if age_decade == 4 & male == 0, mcolor("`color5'") ms
    (scatter mpaaa age if age_decade == 4 & male == 1, mcolor("`color5'") ms
    (scatter mpaaa age if age_decade == 5 & male == 0, mcolor("`color6'") ms
    (scatter mpaaa age if age_decade == 5 & male == 1, mcolor("`color6'") ms
    (line mpaaa_q50_all age, lcolor(black) lpattern(solid) lwidth(medthick))
    (line mpaaa_q90_all age, lcolor(gs8) lpattern(solid) lwidth(medthick)) |
    (line mpaaa_q95_all age, lcolor(gs12) lpattern(solid) lwidth(medthick)),
    legend(order(13 "50th Quantile" 14 "90th Quantile" 15 "95th Quantile"))
    xtitle("Age (years)") ytitle("PA:AA Ratio") ///
    title("Quantile Regression of PA:AA Ratio by Age (Both Sexes)") ///
    xlabel(, labsize(medlarge)) ylabel(0.6(0.1)1.3, nogrid labsize(medlarge)
```

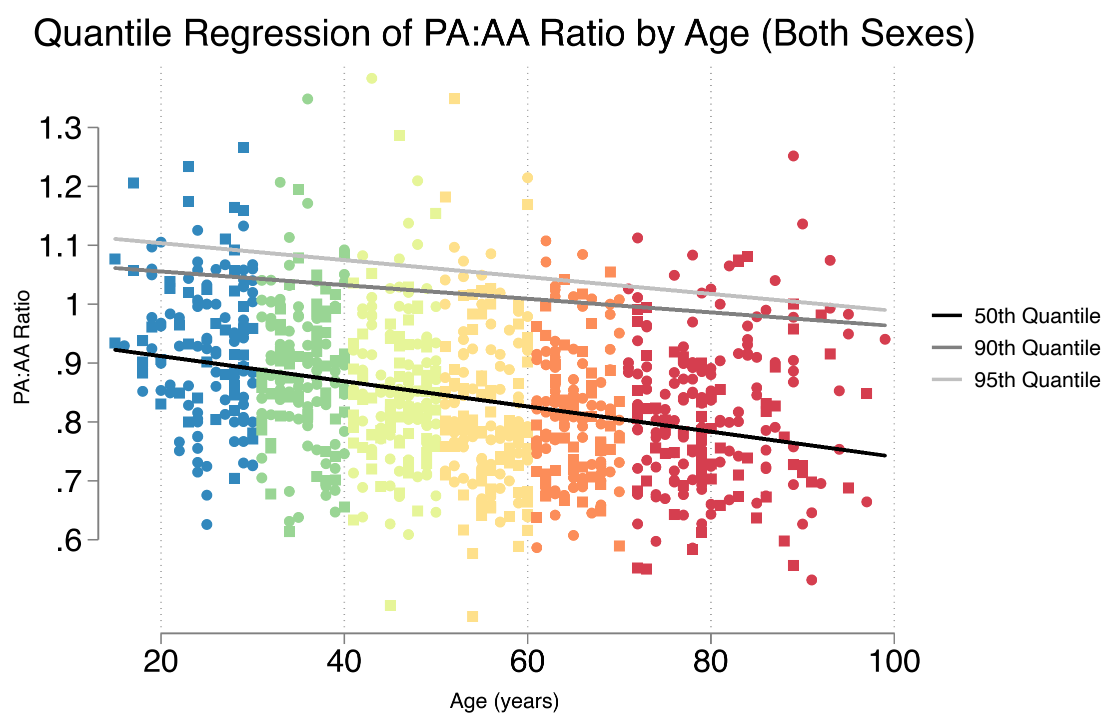
```
    scheme(white_w3d)
```

```
.
.
. /* Extract Colors from Spectral Palette */
. colorpalette Spectral, n(6) nograph

. local color6 `"`r(p1)'"' // Reverse order

. local color5 `"`r(p2)'"'

. local color4 `"`r(p3)'"'

. local color3 `"`r(p4)'"'

. local color2 `"`r(p5)'"'

. local color1 `"`r(p6)'"'

.
. /* Run Quantile Regressions for the Full Population */
. qreg mpaaa c.age, quantile(50) // 50th Percentile

Iteration 1:  WLS sum of weighted deviations =   48.15387

Iteration 1:  Sum of abs. weighted deviations =  48.165025
Iteration 2:  Sum of abs. weighted deviations =  48.068058
Iteration 3:  Sum of abs. weighted deviations =  47.966871

Median regression                                 Number of obs =       9
90
  Raw sum of deviations  50.6267 (about .84615385)
  Min sum of deviations 47.96687                  Pseudo R2     =     0.05
25

----------------------------------------------------------------------------
--
       mpaaa | Coefficient  Std. err.      t    P>|t|     [95% conf. interva
l]
-------------+--------------------------------------------------------------
--
         age |  -.0021378   .0002296    -9.31   0.000    -.0025883   -.00168
73
       _cons |     .95458   .0128506    74.28   0.000     .9293625    .97979
75
----------------------------------------------------------------------------
--

. predict mpaaa_q50_all, xb

.
. qreg mpaaa c.age, quantile(90) // 90th Percentile

Iteration 1:  WLS sum of weighted deviations =  46.411226

Iteration 1:  Sum of abs. weighted deviations =  46.443596
Iteration 2:  Sum of abs. weighted deviations =  35.234124
Iteration 3:  Sum of abs. weighted deviations =   23.17094
```

```
Iteration 4:  Sum of abs. weighted deviations =   23.14185
Iteration 5:  Sum of abs. weighted deviations =   23.14079

.9 Quantile regression                              Number of obs =        9
90
  Raw sum of deviations 23.77614 (about 1.024735)
  Min sum of deviations 23.14079                    Pseudo R2     =     0.02
67

------------------------------------------------------------------------------
--
       mpaaa | Coefficient  Std. err.      t    P>|t|     [95% conf. interva
l]
-------------+----------------------------------------------------------------
--
         age |  -.0011581   .0003108    -3.73   0.000    -.0017679   -.00054
82
       _cons |   1.078691    .017396    62.01   0.000     1.044554    1.1128
29
------------------------------------------------------------------------------
--

. predict mpaaa_q90_all, xb

.
. qreg mpaaa c.age, quantile(95) // 95th Percentile

Iteration 1:  WLS sum of weighted deviations =  45.915238

Iteration 1:  Sum of abs. weighted deviations =   46.19575
Iteration 2:  Sum of abs. weighted deviations =  28.313518
Iteration 3:  Sum of abs. weighted deviations =   14.43733
Iteration 4:  Sum of abs. weighted deviations =  14.229348
Iteration 5:  Sum of abs. weighted deviations =   14.22666

.95 Quantile regression                             Number of obs =        9
90
  Raw sum of deviations 14.43796 (about 1.0669856)
  Min sum of deviations 14.22666                    Pseudo R2     =     0.01
46

------------------------------------------------------------------------------
--
       mpaaa | Coefficient  Std. err.      t    P>|t|     [95% conf. interva
l]
-------------+----------------------------------------------------------------
--
         age |   -.001436   .0005878    -2.44   0.015    -.0025895   -.00028
26
       _cons |   1.132314   .0329022    34.41   0.000     1.067748     1.196
88
------------------------------------------------------------------------------
--

. predict mpaaa_q95_all, xb
```

```
.
. /* Scatter Plot with Both Genders + Quantile Regression Lines */
. twoway ///
>     (scatter mpaaa age if age_decade == 0 & male == 0, mcolor("`color1'")
msy
> mbol(O)) || /// <30 years, Female
>     (scatter mpaaa age if age_decade == 0 & male == 1, mcolor("`color1'")
msy
> mbol(S)) || /// <30 years, Male
>     (scatter mpaaa age if age_decade == 1 & male == 0, mcolor("`color2'")
msy
> mbol(O)) || /// 30-40 years, Female
>     (scatter mpaaa age if age_decade == 1 & male == 1, mcolor("`color2'")
msy
> mbol(S)) || /// 30-40 years, Male
>     (scatter mpaaa age if age_decade == 2 & male == 0, mcolor("`color3'")
msy
> mbol(O)) || /// 40-50 years, Female
>     (scatter mpaaa age if age_decade == 2 & male == 1, mcolor("`color3'")
msy
> mbol(S)) || /// 40-50 years, Male
>     (scatter mpaaa age if age_decade == 3 & male == 0, mcolor("`color4'")
msy
> mbol(O)) || /// 50-60 years, Female
>     (scatter mpaaa age if age_decade == 3 & male == 1, mcolor("`color4'")
msy
> mbol(S)) || /// 50-60 years, Male
>     (scatter mpaaa age if age_decade == 4 & male == 0, mcolor("`color5'")
msy
> mbol(O)) || /// 60-70 years, Female
>     (scatter mpaaa age if age_decade == 4 & male == 1, mcolor("`color5'")
msy
> mbol(S)) || /// 60-70 years, Male
>     (scatter mpaaa age if age_decade == 5 & male == 0, mcolor("`color6'")
msy
> mbol(O)) || /// 70+ years, Female
>     (scatter mpaaa age if age_decade == 5 & male == 1, mcolor("`color6'")
msy
> mbol(S)) || /// 70+ years, Male
>     (line mpaaa_q50_all age, lcolor(black) lpattern(solid) lwidth(medthic
k))
> || /// 50th quantile
>     (line mpaaa_q90_all age, lcolor(gs8) lpattern(solid) lwidth(medthick))
||
>  /// 90th quantile
>     (line mpaaa_q95_all age, lcolor(gs12) lpattern(solid) lwidth(medthic
k)),
> /// 95th quantile
>     legend(order(13 "50th Quantile" 14 "90th Quantile" 15 "95th Quantil
e")) /
> //
>     xtitle("Age (years)") ytitle("PA:AA Ratio") ///
>     title("Quantile Regression of PA:AA Ratio by Age (Both Sexes)") ///
>     xlabel(, labsize(medlarge)) ylabel(0.6(0.1)1.3, nogrid labsize(medlarg
e))
>  ///
```

```
>     scheme(white_w3d)

.
.
```



In [34]:
```
%%stata



/* -----
Sex-Stratified Quantile Regression Scatter Plots for PA:AA Ratio
-----*/

/* Run Bootstrap Quantile Regressions for the Men */
bsqreg mpaaa c.age if male == 1, quantile(50) reps(500) // 50th Percentile
bsqreg mpaaa c.age if male == 1, quantile(90) reps(500) // 90th Percentile
bsqreg mpaaa c.age if male == 1, quantile(95) reps(500) // 95th Percentile

/* Extract Colors from Spectral Palette */
colorpalette Spectral, n(6) nograph
local color6 `"`r(p1)'"' // Reverse order
local color5 `"`r(p2)'"'
local color4 `"`r(p3)'"'
local color3 `"`r(p4)'"'
local color2 `"`r(p5)'"'
local color1 `"`r(p6)'"'

/* -----
Generate Quantile Regression Predictions for Males
-----*/
qreg mpaaa c.age if male == 1, quantile(50) // Male
predict mpaaa_q50_male if male == 1, xb
qreg mpaaa c.age if male == 1, quantile(90) // Male
predict mpaaa_q90_male if male == 1, xb
```

```
qreg mpaaa c.age if male == 1, quantile(95) // Male
predict mpaaa_q95_male if male == 1, xb

/* -----
Male Scatter Plot
-----*/
twoway ///
    (scatter mpaaa age if age_decade == 0 & male == 1, mcolor("`color1'") ms
    (scatter mpaaa age if age_decade == 1 & male == 1, mcolor("`color2'") ms
    (scatter mpaaa age if age_decade == 2 & male == 1, mcolor("`color3'") ms
    (scatter mpaaa age if age_decade == 3 & male == 1, mcolor("`color4'") ms
    (scatter mpaaa age if age_decade == 4 & male == 1, mcolor("`color5'") ms
    (scatter mpaaa age if age_decade == 5 & male == 1, mcolor("`color6'") ms
    (line mpaaa_q50_male age, lcolor(black) lpattern(solid) lwidth(medthick)
    (line mpaaa_q90_male age, lcolor(gs8) lpattern(solid) lwidth(medthick))
    (line mpaaa_q95_male age, lcolor(gs12) lpattern(solid) lwidth(medthick))
    ///
    legend(order(7 "50th Quantile" 8 "90th Quantile" 9 "95th Quantile") pos(
    xtitle("Age (years)") ytitle("PA:AA Ratio", margin(medlarge)) ///
    title("Men") ///
    xlabel(, labsize(medlarge)) ylabel(0.6(0.1)1.3, nogrid labsize(medlarge)
    scheme(white_w3d) yscale(range(0.6 1.3)) ///
    saving(male_qreg_plot, replace)


/* Run Bootstrap Quantile Regressions for the Women */
bsqreg mpaaa c.age if male == 0, quantile(50) reps(500) // 50th Percentile
bsqreg mpaaa c.age if male == 0, quantile(90) reps(500) // 90th Percentile
bsqreg mpaaa c.age if male == 0, quantile(95) reps(500) // 95th Percentile

/* -----
Generate Quantile Regression Predictions for Females
-----*/
qreg mpaaa c.age if male == 0, quantile(50) // Female
predict mpaaa_q50_female if male == 0, xb
qreg mpaaa c.age if male == 0, quantile(90) // Female
predict mpaaa_q90_female if male == 0, xb
qreg mpaaa c.age if male == 0, quantile(95) // Female
predict mpaaa_q95_female if male == 0, xb

/* -----
Female Scatter Plot
-----*/
twoway ///
    (scatter mpaaa age if age_decade == 0 & male == 0, mcolor("`color1'") ms
    (scatter mpaaa age if age_decade == 1 & male == 0, mcolor("`color2'") ms
    (scatter mpaaa age if age_decade == 2 & male == 0, mcolor("`color3'") ms
    (scatter mpaaa age if age_decade == 3 & male == 0, mcolor("`color4'") ms
    (scatter mpaaa age if age_decade == 4 & male == 0, mcolor("`color5'") ms
    (scatter mpaaa age if age_decade == 5 & male == 0, mcolor("`color6'") ms
    (line mpaaa_q50_female age, lcolor(black) lpattern(solid) lwidth(medthic
    (line mpaaa_q90_female age, lcolor(gs8) lpattern(solid) lwidth(medthick)
    (line mpaaa_q95_female age, lcolor(gs12) lpattern(solid) lwidth(medthick
    ///
    legend(order(7 "50th Quantile" 8 "90th Quantile" 9 "95th Quantile") pos(
    xtitle("Age (years)") ytitle("PA:AA Ratio", margin(medlarge)) ///
```

```
    title("Women") ///
    xlabel(, labsize(medlarge)) ylabel(0.6(0.1)1.3, nogrid labsize(medlarge)
    scheme(white_w3d) yscale(range(0.6 1.3)) ///
    saving(female_qreg_plot, replace)

/* -----
Combine the Two Plots into a Single Figure
-----*/
graph combine male_qreg_plot.gph female_qreg_plot.gph, ///
    title("Sex-Stratified Quantile Regression of PA:AA Ratio by Age") ///
    ycommon xcommon ///
    iscale(*1.2) ///
    rows(1) ///
    graphregion(margin(2 2 2 2)) ///
    xsize(9) ysize(5)
```

```
.
.
. /* -----
> Sex-Stratified Quantile Regression Scatter Plots for PA:AA Ratio
> -----*/
.
. /* Run Bootstrap Quantile Regressions for the Men */
. bsqreg mpaaa c.age if male == 1, quantile(50) reps(500) // 50th Percentile
(fitting base model)

Bootstrap replications (500): .........10.........20.........30.........4
0.....
> ....50.........60.........70.........80.........90.........100.........11
0...
> ......120.........130.........140.........150.........160.........17
0.......
> .180.........190.........200.........210.........220.........230.........2
40.
> ........250.........260.........270.........280.........290.........30
0......
> ...310.........320.........330.........340.........350.........36
0.........37
> 0.........380.........390.........400.........410.........420.........43
0....
> .....440.........450.........460.........470.........480.........49
0.........
> 500 done

Median regression, bootstrap(500) SEs              Number of obs =        3
70
  Raw sum of deviations 19.73065 (about .83475784)
  Min sum of deviations 18.00374                   Pseudo R2     =     0.08
75

----------------------------------------------------------------------------
--
       mpaaa | Coefficient  Std. err.      t    P>|t|     [95% conf. interva
l]
-------------+--------------------------------------------------------------
--
         age |  -.0028723   .0003666    -7.84   0.000    -.0035931   -.00215
15
       _cons |   .9800146   .0224854    43.58   0.000     .9357987     1.024
23
----------------------------------------------------------------------------
--

. bsqreg mpaaa c.age if male == 1, quantile(90) reps(500) // 90th Percentile
(fitting base model)

Bootstrap replications (500): .........10.........20.........30.........4
0.....
> ....50.........60.........70.........80.........90.........100.........11
0...
> ......120.........130.........140.........150.........160.........17
0.......
```

```
> .180.........190.........200.........210.........220.........230.........2
40.
> .........250.........260.........270.........280.........290.........30
0......
> ...310.........320.........330.........340.........350.........36
0.........37
> 0.........380.........390.........400.........410.........420.........43
0....
> .....440.........450.........460.........470.........480.........49
0.........
> 500 done

.9 Quantile regression, bootstrap(500) SEs          Number of obs =        3
70
  Raw sum of deviations 9.633967 (about 1.0172414)
  Min sum of deviations 9.206198                    Pseudo R2     =     0.04
44

------------------------------------------------------------------------------
--
       mpaaa | Coefficient  Std. err.      t    P>|t|     [95% conf. interva
l]
-------------+----------------------------------------------------------------
--
         age |  -.0020999   .0008742    -2.40   0.017     -.003819   -.00038
08
       _cons |   1.121952   .0473363    23.70   0.000     1.028868    1.2150
35
------------------------------------------------------------------------------
--

. bsqreg mpaaa c.age if male == 1, quantile(95) reps(500) // 95th Percentile
(fitting base model)

Bootstrap replications (500): .........10.........20.........30.........4
0.....
> ....50.........60.........70.........80.........90.........100.........11
0...
> ......120.........130.........140.........150.........160.........17
0........
> .180.........190.........200.........210.........220.........230.........2
40.
> .........250.........260.........270.........280.........290.........30
0......
> ...310.........320.........330.........340.........350.........36
0.........37
> 0.........380.........390.........400.........410.........420.........43
0....
> .....440.........450.........460.........470.........480.........49
0.........
> 500 done

.95 Quantile regression, bootstrap(500) SEs         Number of obs =        3
70
  Raw sum of deviations 6.000896 (about 1.0749186)
  Min sum of deviations 5.726216                    Pseudo R2     =     0.04
```

```
58

------------------------------------------------------------------------------
--
       mpaaa | Coefficient  Std. err.      t    P>|t|     [95% conf. interva
l]
-------------+----------------------------------------------------------------
--
         age |  -.0032398    .000991    -3.27   0.001    -.0051885   -.00129
11
       _cons |   1.249011   .0636665    19.62   0.000     1.123815    1.3742
07
------------------------------------------------------------------------------
--

.
. /* Extract Colors from Spectral Palette */
. colorpalette Spectral, n(6) nograph

. local color6 `"`r(p1)'"' // Reverse order

. local color5 `"`r(p2)'"'

. local color4 `"`r(p3)'"'

. local color3 `"`r(p4)'"'

. local color2 `"`r(p5)'"'

. local color1 `"`r(p6)'"'

.
. /* -----
> Generate Quantile Regression Predictions for Males
> -----*/
. qreg mpaaa c.age if male == 1, quantile(50) // Male

Iteration 1:  WLS sum of weighted deviations =  18.072056

Iteration 1:  Sum of abs. weighted deviations =  18.079947
Iteration 2:  Sum of abs. weighted deviations =  18.009275
Iteration 3:  Sum of abs. weighted deviations =  18.006029
Iteration 4:  Sum of abs. weighted deviations =  18.003737

Median regression                                  Number of obs =        3
70
  Raw sum of deviations 19.73065 (about .83475784)
  Min sum of deviations 18.00374                   Pseudo R2     =     0.08
75

------------------------------------------------------------------------------
--
       mpaaa | Coefficient  Std. err.      t    P>|t|     [95% conf. interva
l]
-------------+----------------------------------------------------------------
--
```

```
         age |  -.0028723   .0003572    -8.04   0.000    -.0035748   -.00216
99
       _cons |   .9800146   .0199446    49.14   0.000     .9407948    1.0192
34
------------------------------------------------------------------------------
--

. predict mpaaa_q50_male if male == 1, xb
(620 missing values generated)

. qreg mpaaa c.age if male == 1, quantile(90) // Male

Iteration 1:  WLS sum of weighted deviations =  16.516374

Iteration 1:  Sum of abs. weighted deviations =  16.648165
Iteration 2:  Sum of abs. weighted deviations =  12.965204
Iteration 3:  Sum of abs. weighted deviations =  9.5208214
Iteration 4:  Sum of abs. weighted deviations =  9.2905708
Iteration 5:  Sum of abs. weighted deviations =  9.2359831
Iteration 6:  Sum of abs. weighted deviations =  9.2071549
Iteration 7:  Sum of abs. weighted deviations =  9.2061984

.9 Quantile regression                              Number of obs =        3
70
  Raw sum of deviations 9.633967 (about 1.0172414)
  Min sum of deviations 9.206198                    Pseudo R2     =     0.04
44

------------------------------------------------------------------------------
--
       mpaaa | Coefficient  Std. err.      t    P>|t|     [95% conf. interva
l]
-------------+----------------------------------------------------------------
--
         age |  -.0020999   .0007879    -2.67   0.008    -.0036491   -.00055
06
       _cons |   1.121952   .0439879    25.51   0.000     1.035452    1.2084
51
------------------------------------------------------------------------------
--

. predict mpaaa_q90_male if male == 1, xb
(620 missing values generated)

. qreg mpaaa c.age if male == 1, quantile(95) // Male

Iteration 1:  WLS sum of weighted deviations =   16.06492

Iteration 1:  Sum of abs. weighted deviations =  16.066351
Iteration 2:  Sum of abs. weighted deviations =  11.129776
Iteration 3:  Sum of abs. weighted deviations =   5.726216

.95 Quantile regression                             Number of obs =        3
70
  Raw sum of deviations 6.000896 (about 1.0749186)
  Min sum of deviations 5.726216                    Pseudo R2     =     0.04
```

```
58

------------------------------------------------------------------------------
--
       mpaaa | Coefficient  Std. err.      t    P>|t|     [95% conf. interva
l]
-------------+----------------------------------------------------------------
--
         age |  -.0032398   .0013303    -2.44   0.015    -.0058557   -.00062
38
       _cons |   1.249011   .0742749    16.82   0.000     1.102955    1.3950
67
------------------------------------------------------------------------------
--

. predict mpaaa_q95_male if male == 1, xb
(620 missing values generated)

.
. /* -----
> Male Scatter Plot
> -----*/
. twoway ///
>     (scatter mpaaa age if age_decade == 0 & male == 1, mcolor("`color1'")
msy
> mbol(S)) || /// <30 years
>     (scatter mpaaa age if age_decade == 1 & male == 1, mcolor("`color2'")
msy
> mbol(S)) || /// 30-40 years
>     (scatter mpaaa age if age_decade == 2 & male == 1, mcolor("`color3'")
msy
> mbol(S)) || /// 40-50 years
>     (scatter mpaaa age if age_decade == 3 & male == 1, mcolor("`color4'")
msy
> mbol(S)) || /// 50-60 years
>     (scatter mpaaa age if age_decade == 4 & male == 1, mcolor("`color5'")
msy
> mbol(S)) || /// 60-70 years
>     (scatter mpaaa age if age_decade == 5 & male == 1, mcolor("`color6'")
msy
> mbol(S)) || /// 70+ years
>     (line mpaaa_q50_male age, lcolor(black) lpattern(solid) lwidth(medthic
k))
>  || /// 50th percentile line
>     (line mpaaa_q90_male age, lcolor(gs8) lpattern(solid) lwidth(medthic
k)) |
> | /// 90th percentile line
>     (line mpaaa_q95_male age, lcolor(gs12) lpattern(solid) lwidth(medthic
k)),
>  /// 95th percentile line
>     ///
>     legend(order(7 "50th Quantile" 8 "90th Quantile" 9 "95th Quantile") po
s(6
> ) row(1)) /// Only quantile labels in legend
>     xtitle("Age (years)") ytitle("PA:AA Ratio", margin(medlarge)) ///
>     title("Men") ///
```

```
>     xlabel(, labsize(medlarge)) ylabel(0.6(0.1)1.3, nogrid labsize(medlarg
e))
>  /// Adjusted Y-axis range
>     scheme(white_w3d) yscale(range(0.6 1.3)) ///
>     saving(male_qreg_plot, replace)
file male_qreg_plot.gph saved

.
.
. /* Run Bootstrap Quantile Regressions for the Women */
. bsqreg mpaaa c.age if male == 0, quantile(50) reps(500) // 50th Percentile
(fitting base model)

Bootstrap replications (500): .........10.........20.........30.........4
0.....
> ....50.........60.........70.........80.........90.........100.........11
0...
> ......120.........130.........140.........150.........160.........17
0........
> .180.........190.........200.........210.........220.........230.........2
40.
> ........250.........260.........270.........280.........290.........30
0......
> ...310.........320.........330.........340.........350.........36
0.........37
> 0.........380.........390.........400.........410.........420.........43
0....
> .....440.........450.........460.........470.........480.........49
0.........
> 500 done

Median regression, bootstrap(500) SEs              Number of obs =        6
20
  Raw sum of deviations 30.79985 (about .85060976)
  Min sum of deviations 29.65024                   Pseudo R2     =     0.03
73

----------------------------------------------------------------------------
--
       mpaaa | Coefficient  Std. err.      t    P>|t|     [95% conf. interva
l]
-------------+--------------------------------------------------------------
--
         age |  -.0016429   .0003034    -5.42   0.000    -.0022386   -.00104
71
       _cons |   .9338699   .0169172    55.20   0.000     .9006477    .96709
21
----------------------------------------------------------------------------
--

. bsqreg mpaaa c.age if male == 0, quantile(90) reps(500) // 90th Percentile
(fitting base model)

Bootstrap replications (500): .........10.........20.........30.........4
0.....
> ....50.........60.........70.........80.........90.........100.........11
```

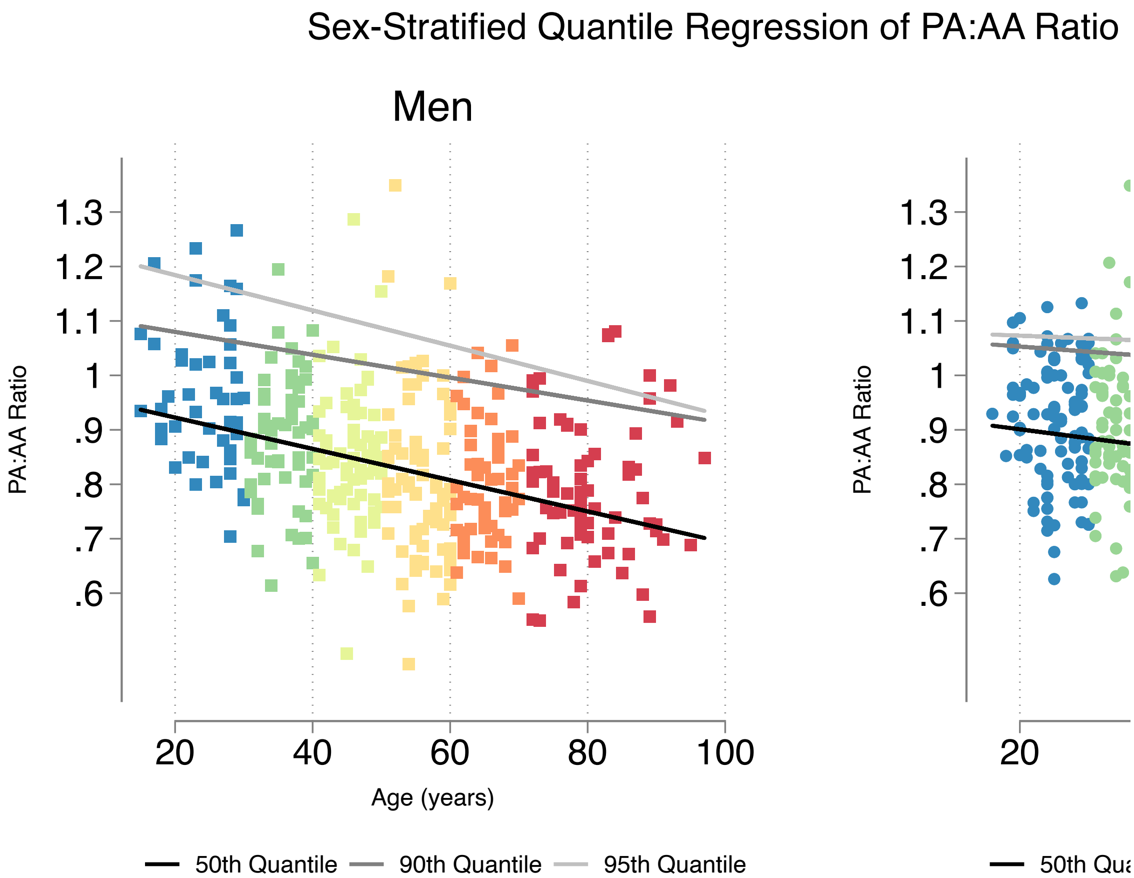
```
0...
> ......120........130........140........150........160........17
0.......
> .180........190........200........210........220........230........2
40.
> ........250........260........270........280........290........30
0......
> ...310........320........330........340........350........36
0........37
> 0........380........390........400........410........420........43
0....
> .....440........450........460........470........480........49
0........
> 500 done

.9 Quantile regression, bootstrap(500) SEs         Number of obs =        6
20
  Raw sum of deviations 14.12762 (about 1.0265781)
  Min sum of deviations 13.82977                   Pseudo R2     =     0.02
11

----------------------------------------------------------------------------
--
       mpaaa | Coefficient  Std. err.      t    P>|t|     [95% conf. interva
l]
-------------+--------------------------------------------------------------
--
         age |  -.0009342   .0002829    -3.30   0.001    -.0014898   -.00037
86
       _cons |   1.071655    .013371    80.15   0.000     1.045396    1.0979
13
----------------------------------------------------------------------------
--

. bsqreg mpaaa c.age if male == 0, quantile(95) reps(500) // 95th Percentile
(fitting base model)

Bootstrap replications (500): .........10........20........30........4
0.....
> ....50........60........70........80........90........100........11
0...
> ......120........130........140........150........160........17
0.......
> .180........190........200........210........220........230........2
40.
> ........250........260........270........280........290........30
0......
> ...310........320........330........340........350........36
0........37
> 0........380........390........400........410........420........43
0....
> .....440........450........460........470........480........49
0........
> 500 done

.95 Quantile regression, bootstrap(500) SEs        Number of obs =        6
```

```
20
  Raw sum of deviations 8.426463 (about 1.0649819)
  Min sum of deviations  8.39009                   Pseudo R2     =     0.00
43

---------------------------------------------------------------------------
--
       mpaaa | Coefficient  Std. err.      t    P>|t|     [95% conf. interva
l]
-------------+-------------------------------------------------------------
--
         age |  -.0004906   .0005492    -0.89   0.372    -.0015691    .00058
78
       _cons |   1.082819   .0278234    38.92   0.000     1.028179    1.1374
59
---------------------------------------------------------------------------
--


.
. /* -----
> Generate Quantile Regression Predictions for Females
> -----*/
. qreg mpaaa c.age if male == 0, quantile(50) // Female

Iteration 1:  WLS sum of weighted deviations =  29.811813

Iteration 1:  Sum of abs. weighted deviations =  29.823386
Iteration 2:  Sum of abs. weighted deviations =  29.742514
Iteration 3:  Sum of abs. weighted deviations =  29.652217
Iteration 4:  Sum of abs. weighted deviations =  29.650236

Median regression                                   Number of obs =        6
20
  Raw sum of deviations 30.79985 (about .85060976)
  Min sum of deviations 29.65024                   Pseudo R2     =     0.03
73

---------------------------------------------------------------------------
--
       mpaaa | Coefficient  Std. err.      t    P>|t|     [95% conf. interva
l]
-------------+-------------------------------------------------------------
--
         age |  -.0016429   .0002963    -5.54   0.000    -.0022247   -.0010
61
       _cons |   .9338699   .0166099    56.22   0.000     .9012511    .96648
86
---------------------------------------------------------------------------
--

. predict mpaaa_q50_female if male == 0, xb
(370 missing values generated)

. qreg mpaaa c.age if male == 0, quantile(90) // Female

Iteration 1:  WLS sum of weighted deviations =  28.799449
```

```
Iteration 1:  Sum of abs. weighted deviations =  28.846315
Iteration 2:  Sum of abs. weighted deviations =  24.601914
Iteration 3:  Sum of abs. weighted deviations =  13.847022
Iteration 4:  Sum of abs. weighted deviations =  13.829769

.9 Quantile regression                             Number of obs =        6
20
  Raw sum of deviations 14.12762 (about 1.0265781)
  Min sum of deviations 13.82977                   Pseudo R2     =     0.02
11

-----------------------------------------------------------------------------
--
       mpaaa | Coefficient  Std. err.      t    P>|t|     [95% conf. interva
l]
-------------+---------------------------------------------------------------
--
         age |  -.0009342   .0003384    -2.76   0.006    -.0015988   -.00026
95
       _cons |   1.071655   .0189738    56.48   0.000     1.034394    1.1089
15
-----------------------------------------------------------------------------
--

. predict mpaaa_q90_female if male == 0, xb
(370 missing values generated)

. qreg mpaaa c.age if male == 0, quantile(95) // Female

Iteration 1:  WLS sum of weighted deviations =  28.494836

Iteration 1:  Sum of abs. weighted deviations =  28.708416
Iteration 2:  Sum of abs. weighted deviations =    23.0951
Iteration 3:  Sum of abs. weighted deviations =  8.7738135
Iteration 4:  Sum of abs. weighted deviations =  8.3928053
Iteration 5:  Sum of abs. weighted deviations =  8.3900904

.95 Quantile regression                            Number of obs =        6
20
  Raw sum of deviations 8.426463 (about 1.0649819)
  Min sum of deviations  8.39009                   Pseudo R2     =     0.00
43

-----------------------------------------------------------------------------
--
       mpaaa | Coefficient  Std. err.      t    P>|t|     [95% conf. interva
l]
-------------+---------------------------------------------------------------
--
         age |  -.0004906    .000586    -0.84   0.403    -.0016415    .00066
02
       _cons |   1.082819   .0328534    32.96   0.000     1.018301    1.1473
37
-----------------------------------------------------------------------------
--
```

```
. predict mpaaa_q95_female if male == 0, xb
(370 missing values generated)

.
. /* -----
> Female Scatter Plot
> -----*/
. twoway ///
>     (scatter mpaaa age if age_decade == 0 & male == 0, mcolor("`color1'")
msy
> mbol(O)) || /// <30 years
>     (scatter mpaaa age if age_decade == 1 & male == 0, mcolor("`color2'")
msy
> mbol(O)) || /// 30-40 years
>     (scatter mpaaa age if age_decade == 2 & male == 0, mcolor("`color3'")
msy
> mbol(O)) || /// 40-50 years
>     (scatter mpaaa age if age_decade == 3 & male == 0, mcolor("`color4'")
msy
> mbol(O)) || /// 50-60 years
>     (scatter mpaaa age if age_decade == 4 & male == 0, mcolor("`color5'")
msy
> mbol(O)) || /// 60-70 years
>     (scatter mpaaa age if age_decade == 5 & male == 0, mcolor("`color6'")
msy
> mbol(O)) || /// 70+ years
>     (line mpaaa_q50_female age, lcolor(black) lpattern(solid) lwidth(medth
ick
> )) || /// 50th percentile line
>     (line mpaaa_q90_female age, lcolor(gs8) lpattern(solid) lwidth(medthic
k))
>  || /// 90th percentile line
>     (line mpaaa_q95_female age, lcolor(gs12) lpattern(solid) lwidth(medthi
ck)
> ), /// 95th percentile line
>     ///
>     legend(order(7 "50th Quantile" 8 "90th Quantile" 9 "95th Quantile") po
s(6
> ) row(1)) /// Only quantile labels in legend
>     xtitle("Age (years)") ytitle("PA:AA Ratio", margin(medlarge)) ///
>     title("Women") ///
>     xlabel(, labsize(medlarge)) ylabel(0.6(0.1)1.3, nogrid labsize(medlarg
e))
>  /// Adjusted Y-axis range
>     scheme(white_w3d) yscale(range(0.6 1.3)) ///
>     saving(female_qreg_plot, replace)
file female_qreg_plot.gph saved

.
. /* -----
> Combine the Two Plots into a Single Figure
> -----*/
. graph combine male_qreg_plot.gph female_qreg_plot.gph, ///
>     title("Sex-Stratified Quantile Regression of PA:AA Ratio by Age") ///
>     ycommon xcommon ///
```

```
>     iscale(*1.2) ///
>     rows(1) ///
>     graphregion(margin(2 2 2 2)) ///
>     xsize(9) ysize(5)

.
```



overall - we can see the PA:AA is expected to get smaller with age, and that there is a less pronounced sex difference than in PA_d

Unadjusted Survival Analysis

In [35]:
```
%%stata

/* -----
PA:AA Ratio (MPAAA) Tertiles and Survival Analysis
-----*/

* Generate tertile cutoffs for MPAAA
xtile mpaaa_tertile = mpaaa, n(3)  // Divides into 3 equal-sized groups

* Label the tertile groups
label define mpaaa_tertile_lbl 1 "MPAAA Tertile 1" 2 "MPAAA Tertile 2" 3 "MP
label values mpaaa_tertile mpaaa_tertile_lbl

* Verify the tertile distribution
```

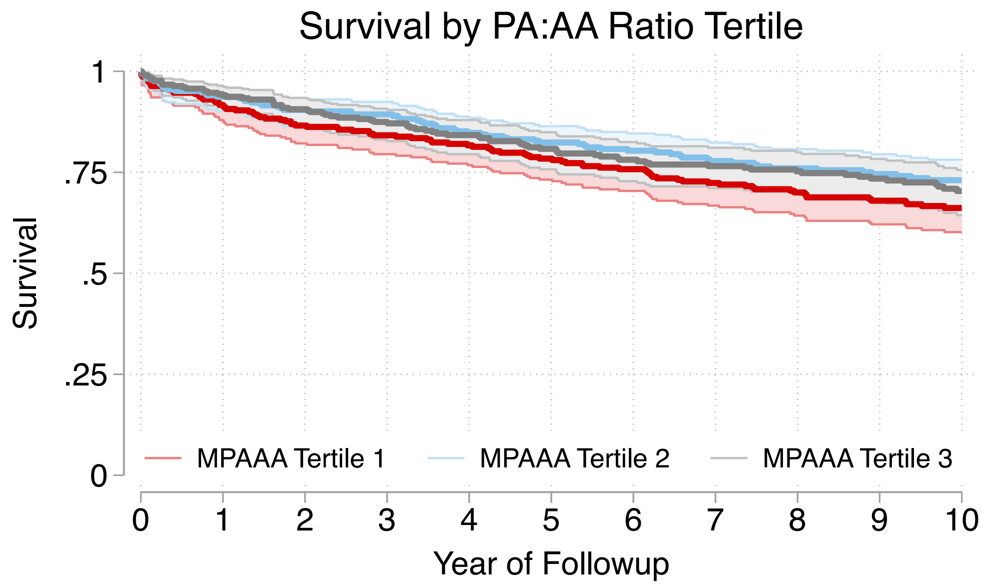
```
tab mpaaa_tertile

/* -----
Survival Analysis by MPAAA Tertile
-----*/

* Log-rank test for survival differences across MPAAA tertiles
sts test mpaaa_tertile, logrank
```

```
.
. /* -----
> PA:AA Ratio (MPAAA) Tertiles and Survival Analysis
> -----*/
.
. * Generate tertile cutoffs for MPAAA
. xtile mpaaa_tertile = mpaaa, n(3)  // Divides into 3 equal-sized groups

.
. * Label the tertile groups
. label define mpaaa_tertile_lbl 1 "MPAAA Tertile 1" 2 "MPAAA Tertile 2" 3
"MPA
> AA Tertile 3"

. label values mpaaa_tertile mpaaa_tertile_lbl

.
. * Verify the tertile distribution
. tab mpaaa_tertile

 3 quantiles of |
          mpaaa |      Freq.     Percent        Cum.
----------------+-----------------------------------
MPAAA Tertile 1 |        330       33.33       33.33
MPAAA Tertile 2 |        330       33.33       66.67
MPAAA Tertile 3 |        330       33.33      100.00
----------------+-----------------------------------
          Total |        990      100.00

.
. /* -----
> Survival Analysis by MPAAA Tertile
> -----*/
.
. * Log-rank test for survival differences across MPAAA tertiles
. sts test mpaaa_tertile, logrank

        Failure _d: death==1
  Analysis time _t: _t

Equality of survivor functions
Log-rank test

                |  Observed       Expected
mpaaa_tert~e    |    events         events
----------------+-------------------------
MPAAA Tertile 1 |       101          88.47
MPAAA Tertile 2 |        76          86.14
MPAAA Tertile 3 |        86          88.39
----------------+-------------------------
          Total |       263         263.00

                          chi2(2) =   3.03
                          Pr>chi2 = 0.2195

.
```

not a good mortality risk predictor at all - I suspect this confounding by age runs the other way on this one (larger PA:AA is worse, but the normal range drops with age while general mortality risk increases)

In [36]:
```
%%stata

* Kaplan-Meier Survival Curve by MPAAA Tertile
sts graph, by(mpaaa_tertile) tmax(10) ci surv ///
 plotopts(lwidth(thick)) ///
 risktable(0(1)10, order(1 "MPAAA Tertile 1" 2 "MPAAA Tertile 2" 3 "MPAAA Te
 xlabel(0(1)10, labsize(medlarge)) ///
 ylabel(,labsize(medlarge)) ///
 xtitle("Year of Followup", size(medlarge)) ///
 ytitle("Survival", size(medlarge)) ///
 legend(order(2 "MPAAA Tertile 1" 4 "MPAAA Tertile 2" 6 "MPAAA Tertile 3") p
 title("Survival by PA:AA Ratio Tertile", size(large))
```

```
.
. * Kaplan-Meier Survival Curve by MPAAA Tertile
. sts graph, by(mpaaa_tertile) tmax(10) ci surv ///
>  plotopts(lwidth(thick)) ///
>  risktable(0(1)10, order(1 "MPAAA Tertile 1" 2 "MPAAA Tertile 2" 3 "MPAAA
Ter
> tile 3") title("Number Eligible", size(medium)) size(medsmall)) ///
>  xlabel(0(1)10, labsize(medlarge)) ///
>  ylabel(,labsize(medlarge)) ///
>  xtitle("Year of Followup", size(medlarge)) ///
>  ytitle("Survival", size(medlarge)) ///
>  legend(order(2 "MPAAA Tertile 1" 4 "MPAAA Tertile 2" 6 "MPAAA Tertile 3")
po
> sition(6) ring(0) rows(1) size(medsmall)) ///
>  title("Survival by PA:AA Ratio Tertile", size(large))

        Failure _d: death==1
  Analysis time _t: _t
(note:  named style i not found in class symbol, default attributes used)

.
```



Number Eligible

MPAAA Tertile 1	307	273	253	241	225	213	200	189	174	163	131
MPAAA Tertile 2	295	262	249	236	218	206	192	179	166	153	133
MPAAA Tertile 3	310	280	263	245	224	211	199	187	173	157	128

From the original analysis (and repeated, though not included here), the gist is:

PA:AA falls with age because

AA increases with age faster than PA (though both increase with age)

In [37]:
```
%%stata

/* -----
Cox Proportional Hazards Model by MPAAA Tertile
-----*/
stcox i.mpaaa_tertile
estat concordance

/* -----
Brier Score for MPAAA Tertile Model
-----*/
stbrier i.mpaaa_tertile, bt(12.4819)
```

```
.
. /* -----
> Cox Proportional Hazards Model by MPAAA Tertile
> -----*/
. stcox i.mpaaa_tertile

        Failure _d: death==1
  Analysis time _t: _t

Iteration 0:  Log likelihood = -1693.3646
Iteration 1:  Log likelihood =  -1691.864
Iteration 2:  Log likelihood =  -1691.862
Refining estimates:
Iteration 0:  Log likelihood =  -1691.862

Cox regression with Breslow method for ties

No. of subjects =        912                            Number of obs =    9
12
No. of failures =        263
Time at risk    = 6,981.3588
                                                        LR chi2(2)    =   3.
01
Log likelihood = -1691.862                              Prob > chi2   = 0.22
26

----------------------------------------------------------------------------
--
          _t | Haz. ratio   Std. err.      z    P>|z|     [95% conf. interva
l]
-------------+--------------------------------------------------------------
--
mpaaa_tert~e |
MPAAA Ter..  |   .7728009   .1173687    -1.70   0.090     .5738415    1.0407
43
MPAAA Ter..  |   .8522742   .1250772    -1.09   0.276     .6392336    1.1363
16
----------------------------------------------------------------------------
--

. estat concordance

        Failure _d: death==1
  Analysis time _t: _t

Harrell's C concordance statistic

  Number of subjects (N)             =      912
  Number of comparison pairs (P)     =   174623
  Number of orderings as expected (E) =    63638
  Number of tied predictions (T)     =    58188

          Harrell's C = (E + T/2) / P =   0.5310
                             Somers' D =   0.0621

.
```

```
. /* -----
> Brier Score for MPAAA Tertile Model
> -----*/
. stbrier i.mpaaa_tertile, bt(12.4819)

Mean estimation                          Number of obs   =        990

---------------------------------------------------------------
               |       Mean    Std. Err.    [95% Conf. Interval]
---------------+-----------------------------------------------
   Brier score |   4.535835     .4818537    3.590262    5.481408
---------------------------------------------------------------

.
```

In [38]:

```
%%stata

stcox i.mpaaa_tertile c.age i.male
estat concordance

/* -----
Brier Score for MPAAA Tertile Model
-----*/
stbrier i.mpaaa_tertile c.age i.male, bt(12.4819)
```

```
.
. stcox i.mpaaa_tertile c.age i.male

        Failure _d: death==1
  Analysis time _t: _t

Iteration 0:  Log likelihood = -1693.3646
Iteration 1:  Log likelihood = -1570.7149
Iteration 2:  Log likelihood = -1569.3077
Iteration 3:  Log likelihood = -1569.3064
Refining estimates:
Iteration 0:  Log likelihood = -1569.3064

Cox regression with Breslow method for ties

No. of subjects =        912                           Number of obs =    9
12
No. of failures =        263
Time at risk    = 6,981.3588
                                                        LR chi2(4)    = 248.
12
Log likelihood = -1569.3064                             Prob > chi2   = 0.00
00

----------------------------------------------------------------------------
--
          _t | Haz. ratio   Std. err.      z    P>|z|     [95% conf. interva
l]
-------------+--------------------------------------------------------------
--
mpaaa_tert~e |
MPAAA Ter..  |   1.199892   .1848864     1.18   0.237     .8871237    1.6229
32
MPAAA Ter..  |    1.66826   .2515077     3.39   0.001      1.24147    2.2417
72
             |
         age |   1.056287   .0039128    14.78   0.000     1.048646    1.0639
84
             |
        male |
       Male  |   1.553325   .1988822     3.44   0.001     1.208586    1.9963
98
----------------------------------------------------------------------------
--

. estat concordance

        Failure _d: death==1
  Analysis time _t: _t

Harrell's C concordance statistic

  Number of subjects (N)             =      912
  Number of comparison pairs (P)     =   174623
  Number of orderings as expected (E) =   131276
  Number of tied predictions (T)     =      395
```

```
          Harrell's C = (E + T/2) / P =   0.7529
                           Somers' D =   0.5058

.
. /* -----
> Brier Score for MPAAA Tertile Model
> -----*/
. stbrier i.mpaaa_tertile c.age i.male, bt(12.4819)

Mean estimation                       Number of obs   =       990

----------------------------------------------------------------
               |       Mean    Std. Err.     [95% Conf. Interval]
---------------+------------------------------------------------
   Brier score |       5.93     .9602664     4.045606    7.814393
----------------------------------------------------------------

.
```

add age and sex to the model makes it pretty good again - and highest tertile is associated with mortality once controlling for age

Age, Sex adjusted MPA:AA

In [39]:

```
%%stata

/* -----
Define Colors for MPAAA Predicted Tertiles
-----*/

colorpalette Spectral, n(3) nograph
local color3 `"`r(p1)'"'  // MPAAA Tertile 3 (High)
local color2 `"`r(p2)'"'  // MPAAA Tertile 2 (Middle)
local color1 `"`r(p3)'"'  // MPAAA Tertile 1 (Low)

/* -----
Generate Quantile Regression Predictions for Males
-----*/

qreg mpaaa c.age if male == 1, quantile(0.333) // Male
predict mpaaa_33_male if male == 1, xb
qreg mpaaa c.age if male == 1, quantile(0.666) // Male
predict mpaaa_66_male if male == 1, xb
qreg mpaaa c.age if male == 1, quantile(0.95) // Male
predict mpaaa_95_male if male == 1, xb

/* -----
Male Scatter Plot (Color by MPAAA Predicted Tertile)
-----*/
twoway ///
    (scatter mpaaa age if male == 1 & mpaaa_tertile == 1, mcolor("`color1'")
    (scatter mpaaa age if male == 1 & mpaaa_tertile == 2, mcolor("`color2'")
    (scatter mpaaa age if male == 1 & mpaaa_tertile == 3, mcolor("`color3'")
```

```
    (line mpaaa_33_male age, lcolor(black) lpattern(solid) lwidth(medthick))
    (line mpaaa_66_male age, lcolor(gs8) lpattern(dash) lwidth(medthick)) ||
    (line mpaaa_95_male age, lcolor(gs12) lpattern(longdash) lwidth(medthick
    ///
    legend(order(4 "1st-2nd Age-Sex Tertile" 5 "2nd-3rd Age-Sex Tertile" 6 "
                 1 "MPAAA Tertile 1" 2 "MPAAA Tertile 2" 3 "MPAAA Tertile 3"
           pos(6) row(2)) ///
    xtitle("Age (years)") ytitle("PA:AA Ratio", margin(medlarge)) ///
    title("Men") ///
    xlabel(, labsize(medlarge)) ylabel(0.6(0.1)1.3, nogrid labsize(medlarge)
    scheme(white_w3d) yscale(range(0.6 1.3)) ///
    saving(male_qreg_plot, replace)


/* -----
Generate Quantile Regression Predictions for Females
-----*/

qreg mpaaa c.age if male == 0, quantile(0.333) // Female
predict mpaaa_33_female if male == 0, xb
qreg mpaaa c.age if male == 0, quantile(0.666) // Female
predict mpaaa_66_female if male == 0, xb
qreg mpaaa c.age if male == 0, quantile(0.95) // Female
predict mpaaa_95_female if male == 0, xb

/* -----
Female Scatter Plot (Color by MPAAA Predicted Tertile)
-----*/
twoway ///
    (scatter mpaaa age if male == 0 & mpaaa_tertile == 1, mcolor("`color1'")
    (scatter mpaaa age if male == 0 & mpaaa_tertile == 2, mcolor("`color2'")
    (scatter mpaaa age if male == 0 & mpaaa_tertile == 3, mcolor("`color3'")
    (line mpaaa_33_female age, lcolor(black) lpattern(solid) lwidth(medthick
    (line mpaaa_66_female age, lcolor(gs8) lpattern(dash) lwidth(medthick))
    (line mpaaa_95_female age, lcolor(gs12) lpattern(longdash) lwidth(medthi
    ///
    legend(order(4 "1st-2nd Age-Sex Tertile" 5 "2nd-3rd Age-Sex Tertile" 6 "
                 1 "MPAAA Tertile 1" 2 "MPAAA Tertile 2" 3 "MPAAA Tertile 3"
           pos(6) row(2)) ///
    xtitle("Age (years)") ytitle("PA:AA Ratio", margin(medlarge)) ///
    title("Women") ///
    xlabel(, labsize(medlarge)) ylabel(0.6(0.1)1.3, nogrid labsize(medlarge)
    scheme(white_w3d) yscale(range(0.6 1.3)) ///
    saving(female_qreg_plot, replace)


/* -----
Combine the Two Plots into a Single Figure
-----*/
graph combine male_qreg_plot.gph female_qreg_plot.gph, ///
    title("Sex-Stratified Quantile Regression of PA:AA Ratio by Predicted Te
    ycommon xcommon ///
    iscale(*1.2) ///
    rows(1) ///
    graphregion(margin(2 2 2 2)) ///
    xsize(9) ysize(5)
```

```
.
. /* -----
> Define Colors for MPAAA Predicted Tertiles
> -----*/
.
. colorpalette Spectral, n(3) nograph

. local color3 `"`r(p1)'"'  // MPAAA Tertile 3 (High)

. local color2 `"`r(p2)'"'  // MPAAA Tertile 2 (Middle)

. local color1 `"`r(p3)'"'  // MPAAA Tertile 1 (Low)

.
. /* -----
> Generate Quantile Regression Predictions for Males
> -----*/
.
. qreg mpaaa c.age if male == 1, quantile(0.333) // Male

Iteration 1:  WLS sum of weighted deviations =  17.631278

Iteration 1:  Sum of abs. weighted deviations =  17.821262
Iteration 2:  Sum of abs. weighted deviations =  16.957048
Iteration 3:  Sum of abs. weighted deviations =    15.8715
Iteration 4:  Sum of abs. weighted deviations =  15.867128

.333 Quantile regression                           Number of obs =        3
70
  Raw sum of deviations 17.43487 (about .77903683)
  Min sum of deviations 15.86713                   Pseudo R2     =     0.08
99

----------------------------------------------------------------------------
--
       mpaaa | Coefficient  Std. err.      t    P>|t|     [95% conf. interva
l]
-------------+--------------------------------------------------------------
--
         age |  -.0027043   .0003659    -7.39   0.000    -.0034239   -.00198
47
       _cons |   .9340505    .020431    45.72   0.000     .8938743    .97422
67
----------------------------------------------------------------------------
--

. predict mpaaa_33_male if male == 1, xb
(620 missing values generated)

. qreg mpaaa c.age if male == 1, quantile(0.666) // Male

Iteration 1:  WLS sum of weighted deviations =  17.866733

Iteration 1:  Sum of abs. weighted deviations =  17.881113
Iteration 2:  Sum of abs. weighted deviations =  17.574345
Iteration 3:  Sum of abs. weighted deviations =  17.260269
```

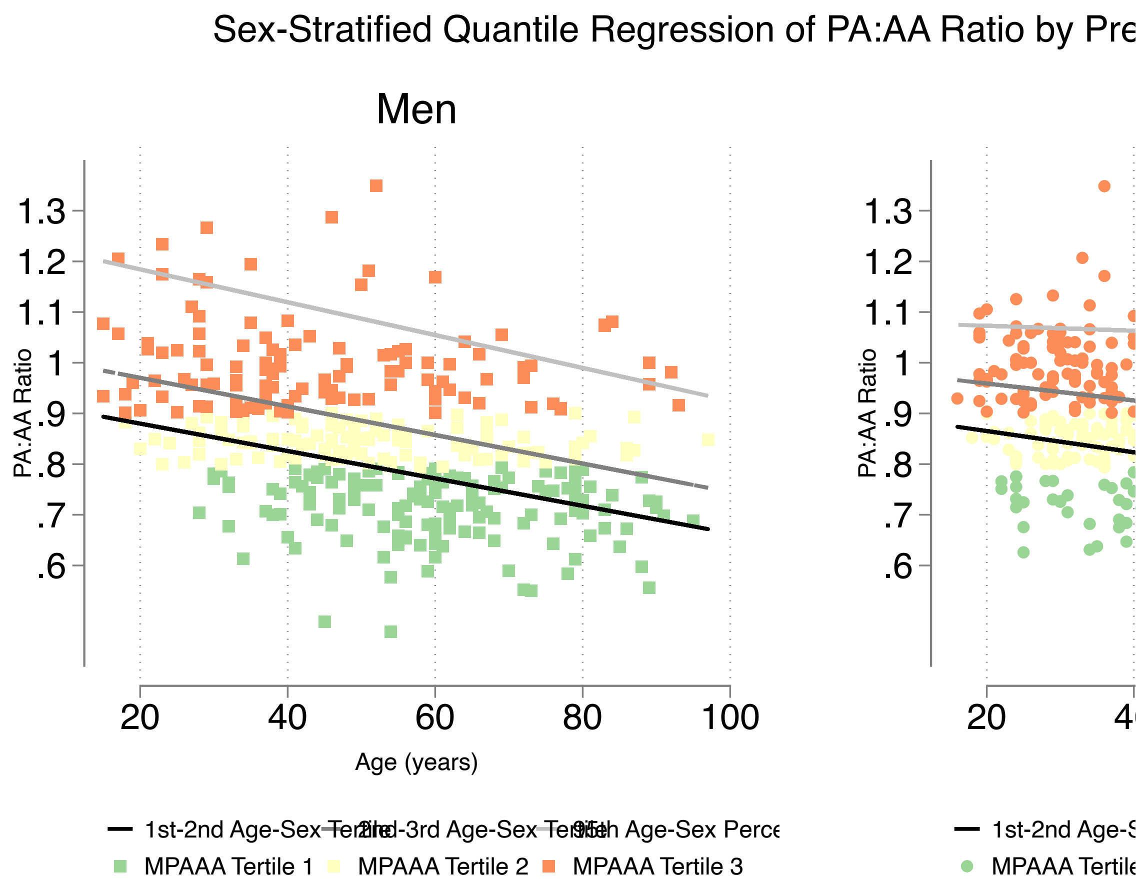
```
Iteration 4:  Sum of abs. weighted deviations =  17.259819

.666 Quantile regression                            Number of obs =        3
70
  Raw sum of deviations 18.66995 (about .89296636)
  Min sum of deviations 17.25982                    Pseudo R2     =     0.07
55

----------------------------------------------------------------------------
--
       mpaaa | Coefficient  Std. err.      t    P>|t|     [95% conf. interva
l]
-------------+--------------------------------------------------------------
--
         age |  -.0028184   .0004891    -5.76   0.000    -.0037802   -.00185
66
       _cons |   1.026636   .0273079    37.59   0.000     .9729365    1.0803
35
----------------------------------------------------------------------------
--

. predict mpaaa_66_male if male == 1, xb
(620 missing values generated)

. qreg mpaaa c.age if male == 1, quantile(0.95) // Male

Iteration 1:  WLS sum of weighted deviations =   16.06492

Iteration 1:  Sum of abs. weighted deviations =  16.066351
Iteration 2:  Sum of abs. weighted deviations =  11.129776
Iteration 3:  Sum of abs. weighted deviations =   5.726216

.95 Quantile regression                             Number of obs =        3
70
  Raw sum of deviations 6.000896 (about 1.0749186)
  Min sum of deviations 5.726216                    Pseudo R2     =     0.04
58

----------------------------------------------------------------------------
--
       mpaaa | Coefficient  Std. err.      t    P>|t|     [95% conf. interva
l]
-------------+--------------------------------------------------------------
--
         age |  -.0032398   .0013303    -2.44   0.015    -.0058557   -.00062
38
       _cons |   1.249011   .0742749    16.82   0.000     1.102955    1.3950
67
----------------------------------------------------------------------------
--

. predict mpaaa_95_male if male == 1, xb
(620 missing values generated)

.
. /* -----
```

```
> Male Scatter Plot (Color by MPAAA Predicted Tertile)
> -----*/
. twoway ///
>     (scatter mpaaa age if male == 1 & mpaaa_tertile == 1, mcolor("`color
1'")
> msymbol(S)) || /// MPAAA Tertile 1 (Low)
>     (scatter mpaaa age if male == 1 & mpaaa_tertile == 2, mcolor("`color
2'")
> msymbol(S)) || /// MPAAA Tertile 2 (Middle)
>     (scatter mpaaa age if male == 1 & mpaaa_tertile == 3, mcolor("`color
3'")
> msymbol(S)) || /// MPAAA Tertile 3 (High)
>     (line mpaaa_33_male age, lcolor(black) lpattern(solid) lwidth(medthic
k))
> || /// 33rd percentile line
>     (line mpaaa_66_male age, lcolor(gs8) lpattern(dash) lwidth(medthick))
||
> /// 66th percentile line
>     (line mpaaa_95_male age, lcolor(gs12) lpattern(longdash) lwidth(medthi
ck)
> ), /// 95th percentile line
>     ///
>     legend(order(4 "1st-2nd Age-Sex Tertile" 5 "2nd-3rd Age-Sex Tertile" 6
"9
> 5th Age-Sex Percentile" ///
>                   1 "MPAAA Tertile 1" 2 "MPAAA Tertile 2" 3 "MPAAA Tertile
3")
>  ///
>            pos(6) row(2)) ///
>     xtitle("Age (years)") ytitle("PA:AA Ratio", margin(medlarge)) ///
>     title("Men") ///
>     xlabel(, labsize(medlarge)) ylabel(0.6(0.1)1.3, nogrid labsize(medlarg
e))
>  /// Adjusted Y-axis range
>     scheme(white_w3d) yscale(range(0.6 1.3)) ///
>     saving(male_qreg_plot, replace)
file male_qreg_plot.gph saved

.
.
. /* -----
> Generate Quantile Regression Predictions for Females
> -----*/
.
. qreg mpaaa c.age if male == 0, quantile(0.333) // Female

Iteration 1:  WLS sum of weighted deviations =  29.481779

Iteration 1:  Sum of abs. weighted deviations =   29.54164
Iteration 2:  Sum of abs. weighted deviations =  28.828928
Iteration 3:  Sum of abs. weighted deviations =  26.392896
Iteration 4:  Sum of abs. weighted deviations =  26.197821
Iteration 5:  Sum of abs. weighted deviations =  26.189561
Iteration 6:  Sum of abs. weighted deviations =  26.185366
Iteration 7:  Sum of abs. weighted deviations =  26.183595
```

```
.333 Quantile regression                          Number of obs =        6
20
  Raw sum of deviations 27.48991 (about .80064309)
  Min sum of deviations  26.1836                  Pseudo R2     =     0.04
75

------------------------------------------------------------------------------
--
       mpaaa | Coefficient  Std. err.      t    P>|t|     [95% conf. interva
l]
-------------+----------------------------------------------------------------
--
         age |  -.0020983   .0002977    -7.05   0.000    -.0026828   -.00151
38
       _cons |   .9072726   .0166867    54.37   0.000     .8745031    .94004
21
------------------------------------------------------------------------------
--

. predict mpaaa_33_female if male == 0, xb
(370 missing values generated)

. qreg mpaaa c.age if male == 0, quantile(0.666) // Female

Iteration 1:  WLS sum of weighted deviations =  29.701315

Iteration 1:  Sum of abs. weighted deviations =   29.77867
Iteration 2:  Sum of abs. weighted deviations =  29.448708
Iteration 3:  Sum of abs. weighted deviations =  28.095946
Iteration 4:  Sum of abs. weighted deviations =  28.049182
Iteration 5:  Sum of abs. weighted deviations =  28.045446

.666 Quantile regression                          Number of obs =        6
20
  Raw sum of deviations   29.003 (about .90365449)
  Min sum of deviations 28.04545                  Pseudo R2     =     0.03
30

------------------------------------------------------------------------------
--
       mpaaa | Coefficient  Std. err.      t    P>|t|     [95% conf. interva
l]
-------------+----------------------------------------------------------------
--
         age |  -.0016714   .0003616    -4.62   0.000    -.0023816   -.00096
13
       _cons |   .9924711   .0202725    48.96   0.000     .9526598    1.0322
82
------------------------------------------------------------------------------
--

. predict mpaaa_66_female if male == 0, xb
(370 missing values generated)

. qreg mpaaa c.age if male == 0, quantile(0.95) // Female
```

```
Iteration 1:  WLS sum of weighted deviations =  28.494836

Iteration 1:  Sum of abs. weighted deviations =  28.708416
Iteration 2:  Sum of abs. weighted deviations =    23.0951
Iteration 3:  Sum of abs. weighted deviations =  8.7738135
Iteration 4:  Sum of abs. weighted deviations =  8.3928053
Iteration 5:  Sum of abs. weighted deviations =  8.3900904

.95 Quantile regression                             Number of obs =       6
20
  Raw sum of deviations 8.426463 (about 1.0649819)
  Min sum of deviations  8.39009                    Pseudo R2     =    0.00
43

----------------------------------------------------------------------------
--
       mpaaa | Coefficient  Std. err.      t    P>|t|     [95% conf. interva
l]
-------------+--------------------------------------------------------------
--
         age |  -.0004906    .000586    -0.84   0.403    -.0016415    .00066
02
       _cons |   1.082819   .0328534    32.96   0.000     1.018301    1.1473
37
----------------------------------------------------------------------------
--

. predict mpaaa_95_female if male == 0, xb
(370 missing values generated)

.
. /* -----
> Female Scatter Plot (Color by MPAAA Predicted Tertile)
> -----*/
. twoway ///
>     (scatter mpaaa age if male == 0 & mpaaa_tertile == 1, mcolor("`color
1'")
> msymbol(O)) || /// MPAAA Tertile 1 (Low)
>     (scatter mpaaa age if male == 0 & mpaaa_tertile == 2, mcolor("`color
2'")
> msymbol(O)) || /// MPAAA Tertile 2 (Middle)
>     (scatter mpaaa age if male == 0 & mpaaa_tertile == 3, mcolor("`color
3'")
> msymbol(O)) || /// MPAAA Tertile 3 (High)
>     (line mpaaa_33_female age, lcolor(black) lpattern(solid) lwidth(medthi
ck)
> ) || /// 33rd percentile line
>     (line mpaaa_66_female age, lcolor(gs8) lpattern(dash) lwidth(medthic
k)) |
> | /// 66th percentile line
>     (line mpaaa_95_female age, lcolor(gs12) lpattern(longdash) lwidth(medt
hic
> k)), /// 95th percentile line
>     ///
>     legend(order(4 "1st-2nd Age-Sex Tertile" 5 "2nd-3rd Age-Sex Tertile" 6
"9
```

```
> 5th Age-sex Quantile" ///
>                 1 "MPAAA Tertile 1" 2 "MPAAA Tertile 2" 3 "MPAAA Tertile
3")
>  ///
>            pos(6) row(2)) ///
>     xtitle("Age (years)") ytitle("PA:AA Ratio", margin(medlarge)) ///
>     title("Women") ///
>     xlabel(, labsize(medlarge)) ylabel(0.6(0.1)1.3, nogrid labsize(medlarg
e))
>  /// Adjusted Y-axis range
>     scheme(white_w3d) yscale(range(0.6 1.3)) ///
>     saving(female_qreg_plot, replace)
file female_qreg_plot.gph saved

.
.
. /* -----
> Combine the Two Plots into a Single Figure
> -----*/
. graph combine male_qreg_plot.gph female_qreg_plot.gph, ///
>     title("Sex-Stratified Quantile Regression of PA:AA Ratio by Predicted
Ter
> tile") ///
>     ycommon xcommon ///
>     iscale(*1.2) ///
>     rows(1) ///
>     graphregion(margin(2 2 2 2)) ///
>     xsize(9) ysize(5)

.
```



In [40]:

```
%%stata


* Run Quantile Regression Separately for Males
qreg mpaaa c.age if male == 1, quantile(0.333)
predict mpaaa_q33_male if male == 1, xb

qreg mpaaa c.age if male == 1, quantile(0.666)
predict mpaaa_q66_male if male == 1, xb

* Run Quantile Regression Separately for Females
qreg mpaaa c.age if male == 0, quantile(0.333)
predict mpaaa_q33_female if male == 0, xb

qreg mpaaa c.age if male == 0, quantile(0.666)
predict mpaaa_q66_female if male == 0, xb

/* -----
Step 2: Categorize Individuals Into Predicted Tertiles for MPAAA
-----*/

* Create an empty variable for tertile classification
gen mpaaa_pred_tertile = .

* Assign individuals into tertiles based on their predicted quantile cutoffs
replace mpaaa_pred_tertile = 1 if male == 1 & mpaaa < mpaaa_q33_male
replace mpaaa_pred_tertile = 2 if male == 1 & mpaaa >= mpaaa_q33_male & mpaa
```

```
replace mpaaa_pred_tertile = 3 if male == 1 & mpaaa >= mpaaa_q66_male

replace mpaaa_pred_tertile = 1 if male == 0 & mpaaa < mpaaa_q33_female
replace mpaaa_pred_tertile = 2 if male == 0 & mpaaa >= mpaaa_q33_female & mp
replace mpaaa_pred_tertile = 3 if male == 0 & mpaaa >= mpaaa_q66_female

/* -----
Step 3: Label the Categories for MPAAA Predicted Tertiles
-----*/

label define mpaaa_pred_tertile_lbl 1 "PA:AA Pred Tertile 1" 2 "PA:AA Pred T
label values mpaaa_pred_tertile mpaaa_pred_tertile_lbl

* Verify the tertile distribution
tab mpaaa_pred_tertile mpaaa_tertile
```

```
.
.
. * Run Quantile Regression Separately for Males
. qreg mpaaa c.age if male == 1, quantile(0.333)

Iteration 1:  WLS sum of weighted deviations =  17.631278

Iteration 1:  Sum of abs. weighted deviations =  17.821262
Iteration 2:  Sum of abs. weighted deviations =  16.957048
Iteration 3:  Sum of abs. weighted deviations =    15.8715
Iteration 4:  Sum of abs. weighted deviations =  15.867128

.333 Quantile regression                           Number of obs =       3
70
  Raw sum of deviations 17.43487 (about .77903683)
  Min sum of deviations 15.86713                   Pseudo R2     =    0.08
99

------------------------------------------------------------------------------
--
       mpaaa | Coefficient  Std. err.      t    P>|t|     [95% conf. interva
l]
-------------+----------------------------------------------------------------
--
         age |  -.0027043   .0003659    -7.39   0.000    -.0034239   -.00198
47
       _cons |   .9340505    .020431    45.72   0.000     .8938743    .97422
67
------------------------------------------------------------------------------
--

. predict mpaaa_q33_male if male == 1, xb
(620 missing values generated)

.
. qreg mpaaa c.age if male == 1, quantile(0.666)

Iteration 1:  WLS sum of weighted deviations =  17.866733

Iteration 1:  Sum of abs. weighted deviations =  17.881113
Iteration 2:  Sum of abs. weighted deviations =  17.574345
Iteration 3:  Sum of abs. weighted deviations =  17.260269
Iteration 4:  Sum of abs. weighted deviations =  17.259819

.666 Quantile regression                           Number of obs =       3
70
  Raw sum of deviations 18.66995 (about .89296636)
  Min sum of deviations 17.25982                   Pseudo R2     =    0.07
55

------------------------------------------------------------------------------
--
       mpaaa | Coefficient  Std. err.      t    P>|t|     [95% conf. interva
l]
-------------+----------------------------------------------------------------
--
```

```
         age |  -.0028184   .0004891    -5.76   0.000    -.0037802   -.00185
66
       _cons |   1.026636   .0273079    37.59   0.000     .9729365    1.0803
35
----------------------------------------------------------------------------
--

. predict mpaaa_q66_male if male == 1, xb
(620 missing values generated)

.
. * Run Quantile Regression Separately for Females
. qreg mpaaa c.age if male == 0, quantile(0.333)

Iteration 1:  WLS sum of weighted deviations =  29.481779

Iteration 1:  Sum of abs. weighted deviations =   29.54164
Iteration 2:  Sum of abs. weighted deviations =  28.828928
Iteration 3:  Sum of abs. weighted deviations =  26.392896
Iteration 4:  Sum of abs. weighted deviations =  26.197821
Iteration 5:  Sum of abs. weighted deviations =  26.189561
Iteration 6:  Sum of abs. weighted deviations =  26.185366
Iteration 7:  Sum of abs. weighted deviations =  26.183595

.333 Quantile regression                           Number of obs =        6
20
  Raw sum of deviations 27.48991 (about .80064309)
  Min sum of deviations  26.1836                   Pseudo R2     =     0.04
75

----------------------------------------------------------------------------
--
       mpaaa | Coefficient  Std. err.      t    P>|t|     [95% conf. interva
l]
-------------+--------------------------------------------------------------
--
         age |  -.0020983   .0002977    -7.05   0.000    -.0026828   -.00151
38
       _cons |   .9072726   .0166867    54.37   0.000     .8745031    .94004
21
----------------------------------------------------------------------------
--

. predict mpaaa_q33_female if male == 0, xb
(370 missing values generated)

.
. qreg mpaaa c.age if male == 0, quantile(0.666)

Iteration 1:  WLS sum of weighted deviations =  29.701315

Iteration 1:  Sum of abs. weighted deviations =   29.77867
Iteration 2:  Sum of abs. weighted deviations =  29.448708
Iteration 3:  Sum of abs. weighted deviations =  28.095946
Iteration 4:  Sum of abs. weighted deviations =  28.049182
Iteration 5:  Sum of abs. weighted deviations =  28.045446
```

```
.666 Quantile regression                           Number of obs =       6
20
  Raw sum of deviations   29.003 (about .90365449)
  Min sum of deviations 28.04545                   Pseudo R2     =    0.03
30

----------------------------------------------------------------------------
--
       mpaaa | Coefficient  Std. err.      t    P>|t|     [95% conf. interva
l]
-------------+--------------------------------------------------------------
--
         age |  -.0016714   .0003616    -4.62   0.000    -.0023816   -.00096
13
       _cons |   .9924711   .0202725    48.96   0.000     .9526598    1.0322
82
----------------------------------------------------------------------------
--

. predict mpaaa_q66_female if male == 0, xb
(370 missing values generated)

.
. /* -----
> Step 2: Categorize Individuals Into Predicted Tertiles for MPAAA
> -----*/
.
. * Create an empty variable for tertile classification
. gen mpaaa_pred_tertile = .
(990 missing values generated)

.
. * Assign individuals into tertiles based on their predicted quantile cutof
fs
. replace mpaaa_pred_tertile = 1 if male == 1 & mpaaa < mpaaa_q33_male
(124 real changes made)

. replace mpaaa_pred_tertile = 2 if male == 1 & mpaaa >= mpaaa_q33_male & mp
aaa
>  < mpaaa_q66_male
(123 real changes made)

. replace mpaaa_pred_tertile = 3 if male == 1 & mpaaa >= mpaaa_q66_male
(123 real changes made)

.
. replace mpaaa_pred_tertile = 1 if male == 0 & mpaaa < mpaaa_q33_female
(208 real changes made)

. replace mpaaa_pred_tertile = 2 if male == 0 & mpaaa >= mpaaa_q33_female &
mpa
> aa < mpaaa_q66_female
(205 real changes made)

. replace mpaaa_pred_tertile = 3 if male == 0 & mpaaa >= mpaaa_q66_female
```

```
(207 real changes made)

.
. /* -----
> Step 3: Label the Categories for MPAAA Predicted Tertiles
> -----*/
.
. label define mpaaa_pred_tertile_lbl 1 "PA:AA Pred Tertile 1" 2 "PA:AA Pred
Te
> rtile 2" 3 "PA:AA Pred Tertile 3"

. label values mpaaa_pred_tertile mpaaa_pred_tertile_lbl

.
. * Verify the tertile distribution
. tab mpaaa_pred_tertile mpaaa_tertile

                     |       3 quantiles of mpaaa
  mpaaa_pred_tertile | MPAAA Ter  MPAAA Ter  MPAAA Ter |     Total
---------------------+---------------------------------+----------
PA:AA Pred Tertile 1 |       272         60          0 |       332 
PA:AA Pred Tertile 2 |        58        221         49 |       328 
PA:AA Pred Tertile 3 |         0         49        281 |       330 
---------------------+---------------------------------+----------
               Total |       330        330        330 |       990 

.
.
.
```

In [41]:

```
%%stata

kappaetc mpaaa_pred_tertile mpaaa_tertile
```

```
.
. kappaetc mpaaa_pred_tertile mpaaa_tertile

Interrater agreement                             Number of subjects =     99
0
                                                Ratings per subject =
2
                                        Number of rating categories =
3
----------------------------------------------------------------------------
--
                     |   Coef.  Std. Err.    t    P>|t|   [95% Conf. Interva
l]
---------------------+------------------------------------------------------
--
   Percent Agreement |  0.7818    0.0131  59.53   0.000     0.7560     0.807
6
Brennan and Prediger |  0.6727    0.0197  34.15   0.000     0.6341     0.711
4
Cohen/Conger's Kappa |  0.6727    0.0197  34.16   0.000     0.6341     0.711
4
    Scott/Fleiss' Pi |  0.6727    0.0197  34.16   0.000     0.6341     0.711
4
           Gwet's AC |  0.6727    0.0197  34.15   0.000     0.6341     0.711
4
Krippendorff's Alpha |  0.6729    0.0197  34.16   0.000     0.6342     0.711
5
----------------------------------------------------------------------------
--

.
```

interestingly, even lower agreement (= more reclassification) than in the MPA diameter case

In [42]:
```
%%stata

tab3way mpaaa_pred_tertile mpaaa_tertile age_decade
```

```
.
. tab3way mpaaa_pred_tertile mpaaa_tertile age_decade
```

Table entries are cell frequencies
Missing categories ignored

	Age (by decade) and 3 quantiles of mpaaa		
	<30 years		
mpaaa_pred_tertile	MPAAA Tertile 1	MPAAA Tertile 2	MPAAA Tertile 3
PA:AA Pred Tertile 1	18	25	
PA:AA Pred Tertile 2		15	24
PA:AA Pred Tertile 3			65

	Age (by decade) and 3 quantiles of mpaaa		
	30-40 years		
mpaaa_pred_tertile	MPAAA Tertile 1	MPAAA Tertile 2	MPAAA Tertile 3
PA:AA Pred Tertile 1	27	19	
PA:AA Pred Tertile 2		45	20
PA:AA Pred Tertile 3			54

	Age (by decade) and 3 quantiles of mpaaa		
	40-50 years		
mpaaa_pred_tertile	MPAAA Tertile 1	MPAAA Tertile 2	MPAAA Tertile 3
PA:AA Pred Tertile 1	49	15	
PA:AA Pred Tertile 2		58	5
PA:AA Pred Tertile 3		1	41

	Age (by decade) and 3 quantiles of mpaaa		
	50-60 years		
mpaaa_pred_tertile	MPAAA Tertile 1	MPAAA Tertile 2	MPAAA Tertile 3
PA:AA Pred Tertile 1	73	1	
PA:AA Pred Tertile 2	6	35	
PA:AA Pred Tertile 3		3	41

	Age (by decade) and 3 quantiles of mpaaa		
	60-70 years		
mpaaa_pred_tertile	MPAAA Tertile 1	MPAAA Tertile 2	MPAAA Tertile 3
PA:AA Pred Tertile 1	48		
PA:AA Pred Tertile 2	15	39	
PA:AA Pred Tertile 3		12	31

```
------------------------------------------------------------------------
                     |     Age (by decade) and 3 quantiles of mpaaa
                     | ------------------- 70+ years -------------------
  mpaaa_pred_tertile | MPAAA Tertile 1  MPAAA Tertile 2  MPAAA Tertile 3
---------------------+--------------------------------------------------
PA:AA Pred Tertile 1 |              57
PA:AA Pred Tertile 2 |              37               29
PA:AA Pred Tertile 3 |                               33               49
------------------------------------------------------------------------

.
```

In [43]:

```
%%stata

/* -----
Log-Rank Test for Survival Differences Across PA:AA Predicted Tertiles
-----*/
sts test mpaaa_pred_tertile, logrank
```

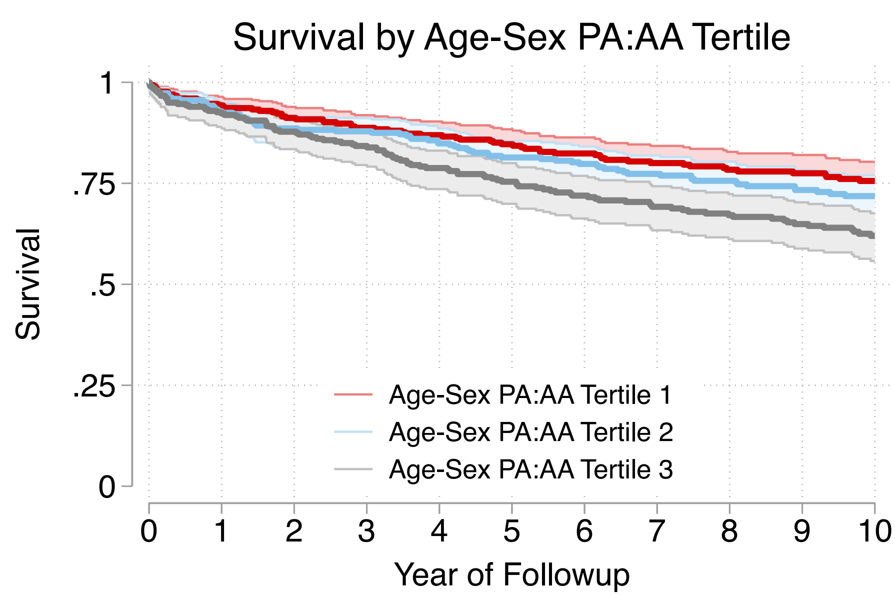
```
.
. /* -----
> Log-Rank Test for Survival Differences Across PA:AA Predicted Tertiles
> -----*/
. sts test mpaaa_pred_tertile, logrank

        Failure _d: death==1
  Analysis time _t: _t

Equality of survivor functions
Log-rank test

                     |  Observed       Expected
mpaaa_pred~e         |    events         events
---------------------+-------------------------
PA:AA Pred Tertile 1 |        74          92.17
PA:AA Pred Tertile 2 |        79          86.99
PA:AA Pred Tertile 3 |       110          83.84
---------------------+-------------------------
               Total |       263         263.00

                               chi2(2) =  12.49
                               Pr>chi2 = 0.0019

.
```

In [44]:

```
%%stata

/* -----
Kaplan-Meier Survival Curve by PA:AA Predicted Tertile
-----*/
sts graph, by(mpaaa_pred_tertile) tmax(10) ci surv ///
 plotopts(lwidth(thick)) ///
 risktable(0(1)10, order(1 "Age-Sex PA:AA Tertile 1" 2 "Age-Sex PA:AA Tertil
 xlabel(0(1)10, labsize(medlarge)) ///
 ylabel(,labsize(medlarge)) ///
```

```
xtitle("Year of Followup", size(medlarge)) ///
ytitle("Survival", size(medlarge)) ///
legend(order(2 "Age-Sex PA:AA Tertile 1" 4 "Age-Sex PA:AA Tertile 2" 6 "Age
title("Survival by Age-Sex PA:AA Tertile", size(large))
```

```
.
. /* -----
> Kaplan-Meier Survival Curve by PA:AA Predicted Tertile
> -----*/
. sts graph, by(mpaaa_pred_tertile) tmax(10) ci surv ///
>  plotopts(lwidth(thick)) ///
>  risktable(0(1)10, order(1 "Age-Sex PA:AA Tertile 1" 2 "Age-Sex PA:AA Tert
ile
>  2" 3 "Age-Sex PA:AA Tertile 3") title("Number Eligible", size(medium)) si
ze(
> medsmall)) ///
>  xlabel(0(1)10, labsize(medlarge)) ///
>  ylabel(,labsize(medlarge)) ///
>  xtitle("Year of Followup", size(medlarge)) ///
>  ytitle("Survival", size(medlarge)) ///
>  legend(order(2 "Age-Sex PA:AA Tertile 1" 4 "Age-Sex PA:AA Tertile 2" 6 "A
ge-
> Sex PA:AA Tertile 3") position(6) ring(0) rows(3) size(medsmall)) ///
>  title("Survival by Age-Sex PA:AA Tertile", size(large))

        Failure _d: death==1
  Analysis time _t: _t
(note:  named style i not found in class symbol, default attributes used)

.
```



Number Eligible

	0	1	2	3	4	5	6	7	8	9	10
Age-Sex PA:AA Tertile 1	306	277	263	249	234	224	209	200	185	175	143
Age-Sex PA:AA Tertile 2	301	267	250	240	222	208	197	184	170	153	131
Age-Sex PA:AA Tertile 3	305	271	252	233	211	198	185	171	158	145	118

definitely a better predictor than the unadjusted version - though not amazingly so

In [45]:
```
%%stata

/* -----
Cox Proportional Hazards Model by PA:AA Predicted Tertile
-----*/
        stcox i.mpaaa_pred_tertile
        estat concordance

/* -----
Brier Score for PA:AA Tertile Model
-----*/
stbrier i.mpaaa_tertile, bt(12.4819)
```

```
.
. /* -----
> Cox Proportional Hazards Model by PA:AA Predicted Tertile
> -----*/
.         stcox i.mpaaa_pred_tertile

        Failure _d: death==1
  Analysis time _t: _t

Iteration 0:  Log likelihood = -1693.3646
Iteration 1:  Log likelihood = -1687.3764
Iteration 2:  Log likelihood = -1687.3465
Iteration 3:  Log likelihood = -1687.3465
Refining estimates:
Iteration 0:  Log likelihood = -1687.3465

Cox regression with Breslow method for ties

No. of subjects =        912                           Number of obs =    9
12
No. of failures =        263
Time at risk    = 6,981.3588
                                                        LR chi2(2)    =  12.
04
Log likelihood = -1687.3465                             Prob > chi2   = 0.00
24

----------------------------------------------------------------------------
--
          _t | Haz. ratio   Std. err.      z    P>|z|     [95% conf. interva
l]
-------------+--------------------------------------------------------------
--
mpaaa_pred~e |
PA:AA Pre..  |   1.131214   .1830167     0.76   0.446      .823817    1.5533
13
PA:AA Pre..  |   1.634908   .2459271     3.27   0.001     1.217457    2.1954
99
----------------------------------------------------------------------------
--

.         estat concordance

        Failure _d: death==1
  Analysis time _t: _t

Harrell's C concordance statistic

  Number of subjects (N)             =      912
  Number of comparison pairs (P)     =   174623
  Number of orderings as expected (E) =    68108
  Number of tied predictions (T)     =    57825

          Harrell's C = (E + T/2) / P =   0.5556
                            Somers' D =   0.1112
```

```
.
. /* -----
> Brier Score for PA:AA Tertile Model
> -----*/
. stbrier i.mpaaa_tertile, bt(12.4819)

Mean estimation                          Number of obs   =        990

--------------------------------------------------------------
               |       Mean    Std. Err.     [95% Conf. Interval]
---------------+----------------------------------------------
   Brier score |   4.535835     .4818537     3.590262    5.481408
--------------------------------------------------------------

.
```

In [46]:

```
%%stata

/* -----
Cox Proportional Hazards Model by PA:AA Predicted Tertile with Age and Sex A
-----*/
stcox i.mpaaa_pred_tertile c.age i.male
estat concordance

/* -----
Brier Score for PA:AA Tertile Model
-----*/
stbrier i.mpaaa_tertile, bt(12.4819)
```

```
.
. /* -----
> Cox Proportional Hazards Model by PA:AA Predicted Tertile with Age and Sex
Ad
> justment
> -----*/
. stcox i.mpaaa_pred_tertile c.age i.male

        Failure _d: death==1
  Analysis time _t: _t

Iteration 0:  Log likelihood = -1693.3646
Iteration 1:  Log likelihood = -1570.1557
Iteration 2:  Log likelihood = -1569.1491
Iteration 3:  Log likelihood = -1569.1491
Refining estimates:
Iteration 0:  Log likelihood = -1569.1491

Cox regression with Breslow method for ties

No. of subjects =        912                            Number of obs =    9
12
No. of failures =        263
Time at risk    = 6,981.3588
                                                        LR chi2(4)    = 248.
43
Log likelihood = -1569.1491                             Prob > chi2   = 0.00
00

----------------------------------------------------------------------------
--
          _t | Haz. ratio   Std. err.      z    P>|z|     [95% conf. interva
l]
-------------+--------------------------------------------------------------
--
mpaaa_pred~e |
PA:AA Pre..  |   1.171174   .1895864     0.98   0.329      .852769    1.6084
65
PA:AA Pre..  |   1.646089   .2498688     3.28   0.001     1.222489     2.216
47
             |
         age |   1.053179   .0038795    14.07   0.000     1.045602     1.060
81
             |
        male |
       Male  |    1.48923   .1877327     3.16   0.002     1.163213    1.9066
21
----------------------------------------------------------------------------
--

. estat concordance

        Failure _d: death==1
  Analysis time _t: _t

Harrell's C concordance statistic
```

```
  Number of subjects (N)              =      912
  Number of comparison pairs (P)      =   174623
  Number of orderings as expected (E) =   131661
  Number of tied predictions (T)      =      361

          Harrell's C = (E + T/2) / P =   0.7550
                            Somers' D =   0.5100

.
. /* -----
> Brier Score for PA:AA Tertile Model
> -----*/
. stbrier i.mpaaa_tertile, bt(12.4819)

Mean estimation                    Number of obs   =       990

----------------------------------------------------------------
               |       Mean    Std. Err.    [95% Conf. Interval]
---------------+------------------------------------------------
   Brier score |   4.535835     .4818537    3.590262    5.481408
----------------------------------------------------------------

.
.
```

overall - not super accurate as a predictor

Summary and Next steps

- In comparison to the chest paper, we have a much broader spectrum of patients - with both pulmonary vascular diasease and not
- like the TTE CTPA paper, we find that PA size increases with age by quantile regression (and other methods)
- however, AA also increases with age even faster, so expected MPA:AA drops with age. - not reported in TTE paper or evaluated in Chest PH Paper

Re-creating the analysis as they did doesn't work as well because the tertile's blend together a lot of the exposure risk - especially in our group where moderate enlargement isn't as important as extreme enlargmenent.

Thus, appraoches could be:

- we could ask them to do MPA:AA - is it more predictive?
- we could ask them about the age confounding issue that we see in the more general population (e.g. is increased mortality risk because MPA enlarges as you get older? seems like probably not, since no univariate association in their paper)
- we could ask if they think age-correcting would be additionally predictive in their cohort or not?

- we could cook up more models and see if there is a difference in predictiveness (we can't do Reveal, but could adjust for comorbidities, etc.)

My TODO:

- play around with using pulm artery size and aorta, independently, as predictors - (this may be better)
- construct a variety of models with all the covariates we used in the original manuscript and see if additional info more predictive or not.
- create a continuous versions of the models on the z-score and see if prediction improve in a more rigorous way